Non-equilibrium and spin transport in hybrids of superconductors and magnets

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Preface

The work presented in this dissertation has been carried out during my Ph.D. under the supervision of Prof. Dr. Wolfgang Belzig between July 2016 and December 2019 in the Quantum Transport Group of the Department of Physics, University of Konstanz. The result of chapter 3 [1] is submitted and the works presented in chapters 4 [2] and 5 [3] are already published in refereed journals. Except where the references are made, this thesis is entirely the outcome of multiple scientific collaborations established by Prof. Belzig and thereby contains a lot of valuable inputs from his side. All the numerics implemented to obtain the results of this dissertation are exclusively produced for this reason and have been developed entirely by the author; nevertheless, Dr. Akashdeep Kamra and Dr. Robert Hussein are greatly acknowledged for all the fruitful discussions and the technical support whenever needed. Moreover, the hospitality of QuSpin (NTNU) and mainly Dr. Akashdeep Kamra, during my visit to Trondheim in 2018 are appreciated, where some steps towards the initial results of chapter 3 developed. To properly understand the idea behind the numerical results throughout the different chapters, there are two analytically solvable examples presented in the appendix A. An outline of the main theoretical techniques used in this thesis is presented in chapter 2, where the detailed steps towards derivating of the formulas are not included but could be found in the references of the chapter.

[1] **Ali Rezaei**, Robert Hussein, Akashdeep Kamra, and Wolfgang Belzig, arXiv:1908.09610 (2019).
[2] **Ali Rezaei**, A. Kamra, P. Machon, and W. Belzig, New Journal of Physics **20**, 073034 (2018).
[3] Akashdeep Kamra, **Ali Rezaei**, and Wolfgang Belzig, Phys. Rev. Lett. **121**, 247702 (2018).

Contents

Abstract

BCS superconductivity and ferromagnetism rarely coexist in bulk materials due to their mutually antagonistic spin orderings. Many exciting new phenomena which are not present in each system alone, however, could occur in proximity-coupled superconductor/ferromagnet (S/F) heterostructures. The goal of this dissertation is to theoretically present a variety of fascinating but still not experimentally reported phenomena occurring in magnetic superconducting hybrid devices, which in addition to serving as a platform for novel fundamental spin physics, could become essential building blocks in superconducting spintronics.

One of the key concepts in quantum transport is the controlled creation and interaction of charge and spin in nanoscale hybrid structures. Conventional S is composed of singlet Cooper pairs that carry no net spin. Achieving exotic superconducting phases as well as exploiting zero-dissipation supercurrents for switching magnetic memories, for example, requires equal-spin triplet Cooper pairs which are challenging to obtain due to their elusive nature. We will demonstrate a minimal and experimentally achievable setup of a four-terminal S/F hybrid that allows for the generation, control, and detection of these pairs. In particular, we propose a detection scheme based on charge current measurements and show that equal-spin triplet correlations establish a tunable spin current flow into the magnets.

A very large thermopower is not directly obtainable in conventional S due to the presence of an almost perfect spin-dependent electron-hole symmetry around the Fermi energy. However, a thin S with a spin-split density of states, could break this symmetry and thereby provides a giant thermoelectric effect. We will show that spin-flip scattering could strongly enhance the thermoelectric response of an S/F bilayer via creating a sizeable figure of merit at low values of temperature and spin-splitting. The scattering rate shifts the maximum of the thermopower from a higher to a weaker magnetic field, which turns out to be of crucial importance for reducing the necessary spin-splitting in these structures and thereby avoiding additional detrimental effects

related to external magnetic fields.

Inducing a spin-splitting field in an adjacent S usually has been achieved via employing ferromagnets or externally applied magnetic fields. However, using ferromagnets in these designs is marred with numerous detrimental and parasitic effects which lower the practical feasibility of the proposed models. Deriving the interfacial self-energy, we will confirm that an insulating antiferromagnet in proximity to a thin S layer, can indeed more conveniently, be used to induce a sizeable, and disorder-resistant spin-splitting in the S than ferromagnets or applied magnetic fields. This prediction is a promising step towards eliminating the limiting impacts of ferromagnets employed to this end and allows realizing several concepts involving spin-split superconductors.

Zusammenfassung

BCS-Supraleitung und Ferromagnetismus koexistieren in Volumenmaterialien aufgrund ihrer entgegengesetzten Spinordnung nur selten. Viele spannende neue Phänomene, die nicht in jedem einzelnen System vorkommen, könnten jedoch in nahtlos gekoppelten Supraleiter/Ferromagneten (S/F) Heterostrukturen auftreten. Ziel dieser Dissertation ist es, in theoretischer Arbeit eine Vielzahl von faszinierenden, aber noch nicht bekannten Phänomenen in magnetisch supraleitenden Hybridbauelementen zu präsentieren, die nicht nur als Plattform für eine neuartige fundamentale Spinphysik dienen, sondern auch zu wesentlichen Bausteinen der supraleitenden Spintronik werden könnten.

Eines der Schlüsselkonzepte im Quantentransport ist die kontrollierte Erzeugung und Wechselwirkung von Ladung und Spin in Nanohybridstrukturen. Konventionelle Supraleitung hat ihren Ursprung in der Bildung von singlet Cooper-Paaren, welche keinen Nettospin aufweisen. Zum Beispiel, das Erreichen exotischer supraleitender Phasen sowie die Nutzung von dissipationsfreien Superströmen zum Schalten magnetischer Speicher erfordert gleichspinige Triplett-Cooper-Paare, die aufgrund ihrer flüchtigen Natur schwierig zu erhalten sind. Wir werden einen minimalen und experimentell realisierbaren Aufbau eines vierpoligen S/F-Hybrids demonstrieren, der die Erzeugung, Kontrolle und Erkennung dieser Paare ermöglicht. Insbesondere schlagen wir ein Detektionsschema vor, welches auf Ladestrommessungen basiert und zeigen, dass Gleichspin-Triplettkorrelationen einen regelbaren Spinstromfluss in den Magnete erzeugen.

Eine sehr große Thermokraft ist in konventionellen S nicht direkt erreichbar, da eine nahezu perfekte spinabhängige Elektronen-Loch-Symmetrie um die Fermi-Energie vorliegt. Ein dünner S mit einer Spin-aufgespaltenen Zustandsdichte könnte diese Symmetrie jedoch brechen und damit einen riesigen thermoelektrischen Effekt erzeugen. Wir werden zeigen, dass Spinumkehrstreuung die thermoelektrische Reaktion einer S/F-Doppelschicht stark verbessern könnte, indem sie bei niedrigen Temperatur-

werten und Spin-Splitting eine beachtliche Leistungszahl erzeugt. Die Streurate verschiebt das Maximum der Thermokraft von einem höheren zu einem schwächeren magnetischen Feld, was von entscheidender Bedeutung ist, um die notwendige Spinspaltung in diesen Strukturen zu reduzieren und dadurch zusätzliche abträgliche Auswirkungen externer Magnetfelder zu vermeiden.

Das Induzieren eines Zeeman-Feldes in einem benachbarten S wurde in der Regel durch den Einsatz von Ferromagneten oder extern angelegten Magnetfeldern erreicht. Die Verwendung von Ferromagneten in diesen Designs ist jedoch mit zahlreichen nachteiligen und parasitären Effekten behaftet, welche die praktische Umsetzbarkeit der vorgeschlagenen Modelle beeinträchtigen. Mit der Herleitung der Selbstenergie an der Grenzfläche weisen wir nach, dass ein isolierender Antiferromagnet in der Nähe einer dünnen S-Schicht tatsächlich in geeigneterer Weise eingesetzt werden kann, um eine erhebliche und Störstellen resistentere Spinaufspaltung im S, im Vergleich zu Ferromagneten oder angelegten Magnetfeldern, zu induzieren. Diese Prognose ist ein vielversprechender Schritt zur Beseitigung der limitierenden Auswirkungen der zu diesem Zweck eingesetzten Ferromagnete und ermöglicht die Realisierung mehrerer Konzepte mit Spin-Split-Supraleitern.

Chapter 1

Introduction

The dissipationless flow of currents is one of the most intriguing phenomena in nature. For instance, it can occur in metals such as mercury, aluminum, and gallium, when cooled below a critical temperature of about few Kelvin. These materials belong to the class of type-I superconductors, which exclude applied weak magnetic fields from their bodies. In particular, we will investigate the transport properties of hybrid (anti)ferromagnetic structures arising due to the proximity with such low-temperature superconductors. After a brief historical outline in the next section, we will elaborate on the proximity effect in Sec. 1.2 and conclude with the organization of the thesis in Sec 1.3.

1.1 Historical outline of superconductivity

In the year 1908, liquefaction of helium—the last permanent gas remained to be liquefied at the time, gave Heike Kamerlingh Onnes the required refrigeration procedure to reach temperatures of a few Kelvin, and thereby a new chapter in the field of low-temperature physics started in his laboratory at the University of Leiden. In 1911, Onnes was performing experiments to measure the resistivities of several metals at low temperatures and came across the result that no detectable resistance for the mercury (Hg) bar existed when the temperature was lowered below 4.20 K [1, 2]. This breakthrough was the unexpected discovery of superconductivity.

Superconductivity is a unique property of many metals, which remains to be one of the most exciting phases of matter, even a century after its first observation. The exactly zero dc electrical resistance as the temperature is decreased below a particular value—that depends on the elements, e.g., 4.15 K for mercury, is the first governing property of superconductivity. The specific temperature below which the resistance of material disappears, i.e., the phase transition temperature T_c, varies from milli-Kelvin (or micro-Kelvin) to values above 100 K [3]. The zero resistivity, or equivalently, the perfect conductivity, is also of vital importance for most potential applications, such as high-current transmission lines and high-field magnets [4]. A considerable amount of time through the 1920s and 1930s, while the magnetic properties of superconductors were the center of attention, went by with extensive discussions trying to distinguish the superconducting state via perfect conductivity as the only prerequisite characteristic and establish a way to explain the phenomenon [3, 5].

It didn't take long before physicists became aware of the fact that for a material to be recognized as a superconductor (S), it must own another characteristic hallmark—namely, the perfect diamagnetism, referred to as the "Meissner effect" [6]. Meissner and Ochsenfeld managed to observe in 1933 that when a sphere is cooled below its critical temperature T_c in a magnetic field below a particular threshold value, it completely excludes the magnetic flux from its body—meaning that the magnetic flux cannot exist inside the (type-I) S when it is in the superconducting phase [3]. If a material goes resistance-less at low temperatures, but it does not show perfect diamagnetism simultaneously—as might seem to be explained by perfect conductivity—then it is only a great conductor and not an S. On the other hand, it is solely a perfect diamagnet and not an S if at low temperatures it starts to become perfectly diamagnetic, but the resistance remains at a measurable value down to the absolute zero [2]. In 1935, two years after observation of the phenomenon by Meissner, the London brothers (Fritz and Heinz) proposed a simple theory to explain the effect, and trying to predict the penetration range of a static external magnetic field in superconducting materials [3].

The next significant theoretical advancement came in 1950 when Ginzburg and Landau developed a macroscopic theory describing superconductivity in terms of an order parameter and presented a derivation of the London equations. Even though the Ginzburg-Landau theory of superconductivity [1] is phenomenological, it has had remarkable success in explaining many of the principal superconducting properties. The main limitation, however, is that it does not explain the microscopic origins of superconductivity. That same year, H. Fröhlich theoretically predicted the "isotope effect"—supports the electron-phonon interaction of superconductivity [3]—explaining the mechanism behind decreasing of the transition temperature in superconducting material via increasing average isotopic mass. The experimental observation of the

[1] It concentrates on the superconducting electrons rather than on excitations [4, 7].

effect was confirmed and reported immediately in the same year by Maxwell [8] and Reynolds [9].

1957 was the year when a remarkably comprehensive and satisfactory step towards a theoretical understanding of classic superconductivity got developed. Owning to Bardeen, Cooper, and Schrieffer a beautiful microscopic theory explaining the interesting properties of conventional (low-temperatures) superconductors from first principles, known as the BCS theory [10] of superconductivity, got proposed which still after more than six decades of its existence gives a surprisingly good interpretation of many properties of superconductors and thereby has established itself as one of the fascinating theories in condensed matter physics. According to it, superconductivity is a macroscopic quantum effect that is induced due to the pairwise condensation of bound electrons, so-called Cooper pairs into the superconducting ground state. The attractive interaction among the electrons, which is mediated by the electron-phonon coupling, gives rise to the most symmetric formation of Cooper pairs [2]. The theory shows that Cooper pairs carry the supercurrent, and moreover, a part of the electrons continue to stay unpaired at finite temperatures, where there is an energy gap that separates these unpaired "quasiparticles" (normal) from the Cooper pairs (superconducting state). In 1959, Gor'kov [11] showed that Ginzburg-Landau theory could be derived as a limiting case, valid near T_c, from the BCS formalism and thereby made substantial progress towards the acceptance of its validity. Shortly after, the physical properties of inhomogeneous superconductors were predicted by Josephson in 1962 [12], and forthwith verified by experimental observations.

Elemental metals back in Kammerlingh Onnes' era, e.g., mercury, lead, bismuth, become superconducting only at the low temperatures of liquid helium, thus being described as conventional (or low-temperature) superconductors. In 1986, after a long time with no progress on setting a new record of the elevated superconducting transition temperature, J. G. Bednorz and K. A. Müller [13] overturned the situation and initiated the era of high-temperature superconductivity by the discovery of a new class of superconductors based on the copper oxide perovskite compounds—obeying the same general phenomenology as the classic superconductors, but with a much higher T_c compared to them. When the article appeared worldwide, it was faced with initial skepticism. The problem was resolved when the groups of Tanaka [14] and Chu [15], respectively, from Japan and the United States, reproduced the results of Bednorz and Müller and thereby making them accepted. Soon, many other researchers

[2] Vanishing relative angular momentum (called s-wave nowadays), and spin-singlets arrangements (antiparallel alignment of spins of the two electrons).

started to search for a higher transition temperature actively, and the recorded transition temperature began a rapid rise, e.g., T_c above 90 K in a mixed-phase Y-Ba-Cu-O compound system [16] was observed.

1.2 Proximity effect

The proximity effect (PE), also known as the Holm-Meissner effect, is the phenomena occurring when an S is brought into good electrical contact with a non-S, resulting in the appearance of superconducting-like properties in the latter due to leakage of the Cooper pairs [17]. For a normal-metal (N) near an S, superconducting correlations could spread over a substantial length scale from the interface leading to a modification of the electronic density of states (DOS) in N, which has been resolved experimentally using tunneling probes [18, 19]. The local DOS in the N exhibit a mini-gap in the diffusive limit—i.e., it's smaller than the superconducting gap, and vanishes within an energy window about the Fermi level—with a magnitude of the order of the Thouless energy or "correlation energy" defined as $\epsilon_{\mathrm{Th}} \equiv \hbar D/L_n^2$ [20, 21], where $\hbar$ is Planck's constant, D is the electronic diffusion constant, and L_n is the sample length. As the relation for the characteristic energy scale implies, the size of the mini-gap would decrease via the increasing thickness of the N [20].

Although the interplay between a proximity-coupled S and an N (or insulator) is investigated thoroughly during the last decades, the case when a ferromagnet (F) is utilized instead of N remains open. To this end, the next section discusses the PE involving S/F-heterostructure where the main objective is to review the phenomenon that arises due to the interplay of these two different ordering parameters.

1.2.1 State of the art

With the improvement of the microscopic theory of superconductivity, the amount of attention in S/F hybrids has enormously increased due to theoretical predictions, as well as considerable advances in experimental fabrications [22–26]. At first sight, owing to the different spin orientations between ferromagnetism and BCS superconductivity [3]; the coexistence of these two mutually counteractive orderings, except some rare cases, seems unlikely [28]. Nonetheless, due to the proximity phenomena and exchanging electrons from one region to the other trying to accommodate the different order parameters, one can expect stimulating effects near the interface of the

[3] The Pauli principle dictates a parallel arrangement of spins through the F exchange field, whereas spin-singlet Cooper pairs in (conventional) S require an antiparallel orientation [27].

contacting materials.

Fulde-Ferrel (FF) and Larkin-Ovchinnikov (LO) at the same time but independently, proposed the initial predictions towards the coexistence of superconductivity and ferromagnetism in bulk materials by showing the possibility to generate a non-conventional superconducting state with a spatially modulated order parameter, the so-called FFLO state [29, 30], in the presence of a strong exchange field [4].

When we talk about the BCS theory, one of the main ingredients is the pairing between electrons with opposite center-of-mass momenta and spins—matching between the Fermi energies of electrons in the up- and down-spin channel. The Fermi energies of up and down electrons become unequal in the phase where the system is partially polarized, leading to an exotic form of pairing, which was discovered separately by FF and LO. First, it was shown by FF that superconducting Cooper pairs have finite pairing momentum and spin-polarization under a strong enough magnetic field. Then, LO suggested that when the separation between Fermi surfaces is sufficiently large, the formation of electronic pairs with different momenta ($\boldsymbol{k}+\boldsymbol{Q}/2\uparrow$ and $-\boldsymbol{k}+\boldsymbol{Q}/2\downarrow$), is favored energetically over ordinary BCS pairs with opposite momenta ($\boldsymbol{k}\uparrow$ and $-\boldsymbol{k}\downarrow$) [31]. The Cooper pairs will contain a finite ($\boldsymbol{Q}\neq 0$) rather than zero total momentum as a result of the mismatched momenta, thereby spins' density and the superconducting gap become periodic functions of the spatial coordinates. The space modulation of the S order parameter leads to a nonuniform or anisotropic ground state, which breaks translational and rotational symmetries [32]. The FFLO state should be achievable if the Zeeman splitting of the conduction-electron bands distributes the Cooper pairs unfavorable, the system must be very clean [33], and additionally, superconductivity should not be destroyed by the orbital effect of the external magnetic field [34].

New attentions in the interplay of proximity-coupled S/F heterostructures at low temperatures were initiated via emerging the superconducting spintronics [35–37]. The underlying idea behind investigating this field was to utilize charge, and spin transports via creating a synergy between spin-polarized ordering and superconductivity [38], wishing that one could benefit from the manipulation of both currents separately for possible future applications. One of the essential features which imply a dramatic influence on the transport properties within low-power superconducting spintronics in S/F hybrids is the creation of the equal-spin triplet orderings. According to the Usadel theory, the length L_S at which the superconducting condensates suppress within a dirty ferromagnet scales $\sim \xi_F = \sqrt{\hbar D/2H_{ex}}$, where ξ_F is the coher-

[4] The electrons with different spins belong to different energy bands, and we could consider the energy shift of these bands as an exchange field acting on the electrons' spins [25].

ence length in the F, and H_{ex} is the exchange field. For most of standard ferromagnets and alloys, L_S is of the order of a few $(1 - 10)$ nanometers [39]; however, multiple observations verified that the surviving range for the superconducting correlations at S/F interfaces are much larger and actually analogous to $\xi_N \sim \sqrt{D/T}$, the coherence length of a N [40–44], where T stands for temperature. To understand the long-range PE mentioned above, one has to note that the (sufficiently strong) exchange-splitting field breaks the opposite-spin pairs of electrons apart—by attempting to align the spins of electrons of a Cooper pair parallelly—destroying the superconductivity, and thereby its condensate decays fast into F region, making the penetration depth of the conventional PE quite short-ranged. However, spin-flip processes at the interface of a conventional singlet pair S with certain inhomogeneous magnetic textures could lead to the creation of equal-spin triplet pairs [25, 45, 46] which are immune to the exchange interaction and can spread in F over two orders of magnitude larger than the penetration distance of the singlet pairs [47]. The provision and manipulation of equal-spin triplet pairs are notably tricky due to their elusive nature, to this end, chapter 3 is devoted to the timely issue of generation, control, and detection of them in proximity-coupled S/F systems.

Another remarkable discovery in the context of the superconducting spintronics properties of S/F hybrid-structures in recent years was the creation of a sizable spin-dependent thermoelectric (TE) effect [48–54] via lifting the spin degeneracy of the S DOS. In conventional solid-state materials, the charge conductance G and thermal conductance G_{th} obey the Wiedemann–Franz law, i.e., $G_{th}/G = LT$, where $L = (\pi^2 k_B^2/3e^2)$ is the proportionality constant or the Lorenz number. Moreover, according to the Mott formula, the Seebeck coefficient (or the thermopower) relates to the G as [55]

$$S = \frac{\pi^2}{3}\left(\frac{k_B^2 T}{e}\right)\left(\frac{\partial \ln G}{\partial \epsilon}\right). \tag{1.1}$$

These constraints together, result in a very poor TE performance [5] of conventional metallic structures since it is difficult to increase G and S without also increasing G_{th} (and vice versa) [56], and therefore semiconductors are used in the preparation of most traditional TE devices. Since there is an energy gap in the excitation spectrum of the semiconductors, their electronic structure looks similar to superconductors. The gap in semiconductors can be shifted respecting the average electron energy, however, since the energy gap in the S is symmetric—regarding positive and negative energies—

[5] It is evaluated via the figure of merit, $ZT = S^2GT/G_{th}$, as the benchmark of TE conversion efficiency.

TE effect from positive-energy charge carriers (electrons) cancels the effect from the negative ones (holes) [51]. As it has been shown recently, this spin-dependent electron-hole symmetry could be broken via putting the S in the proximity of a ferromagnetic insulator [50], or via applying an external magnetic field with a spin filter [51] which results in a spin-polarized conductance and thereby giant thermoelectricity through the junction [57] which we will further investigate in chapter 4.

So far, inducing a spin-splitting field in an adjacent S have usually been achieved via employing ferromagnets, both in experiment and theory. However, the use of ferromagnets in these designs is marred with several detrimental and parasitic effects—e.g., stray magnetic fields, low-energy magnonic excitations [58, 59], and the Chandrasekhar-Clogston (CC)-limited ultra-low-temperature degradation of the TE response [60]—which sharply limits the practical feasibility of the system. In chapter 4 we will discuss a promising prediction towards reducing the necessary spin-splitting in S/F heterostructures, which might also be useful to avoid the mentioned detrimental effects. Moreover, we will explain via a detailed derivation of the self-energy in chapter 5 that one could as well utilize an antiferromagnet with an uncompensated interface, i.e., containing finite surface magnetization [61–63], alternately to an F to induce spin-splitting in the adjacent S and overcome these effects [64].

1.3 Objective and thesis organization

In this dissertation, we want to present a variety of interesting and still not investigated effects that arise due to PE in structures combining superconductors with magnetic materials (ferro and antiferromagnets), named magnetic superconducting heterostructures. To help the reader grasp the primary motivation and results as efficiently as possible, we organize the thesis by the logical order of the published materials over the chronological order, as presented within chapters 3, 4, and 5.

The content of the dissertation is as follows.

- Chapter 2 introduces a brief review of the theoretical frameworks implemented throughout this thesis, which are necessary to describe the proximity-coupled hybrid structures containing superconductors.

- Remarkably, the spectroscopic evidence of equal-spin triplet correlations in an Al/EuS structure was recently reported by Diesch et al. [65]. In chapter 3, we go a step further, demonstrating that minimal setup of a four-terminal proximity-coupled S/F-hybrid, consisting of a central island interfaced by two superconductors and two ferromagnets in the diffusive limit, indeed, facilitate the generation, control, and detection of spin-triplet Cooper pairs [66]. Implementing the currents conservation within the finite-element semi-classical circuit theory, the chapter explains that as a result of inducing triplet pair correlations, the relative magnetization angle qualifies for voltages exceeding the superconducting gap as a control knob for spin and electron currents. In particular, we highlight the main feature of our proposed setup, which is introducing a detection scheme of equal-spin triplet correlations based on standard charge current measurements and show that they establish a tunable spin current flow into the ferromagnets.

- Chapter 4 is motivated and builds on the recently found sizable TE effects in magnetic superconducting heterostructures. Considering a tunnel structure formed by a thin insulating layer sandwiched between a spin-split superconducting film–that, on the other hand, is induced due to the spin-splitting field—and a ferromagnetic metal, we discuss that how the TE effects are modified in response to the coherent spin-flip scattering inside the S. Incorporating spin-flip and incoherent broadening self-consistently within the semi-classical Green function description of the S, via a numerical model, we find that the TE performance will indeed, drastically enhance in the experimentally relevant low-field and -temperature regime. Two main results of the chapter, which are essential for applications in the low-temperature regime; the maximum of Seebeck coefficient will exceed k_B/e by a factor of ≈ 5, and a sizeable figure of merit of $ZT \gtrsim 1$ is achievable [60].

- In chapter 5 we pay special attention to the feasibility of employing insulating antiferromagnets or ferrimagnets for a magnetic proximity effect in an adjacent (thin) S. The main result for the chapter is that antiferromagnet/superconducting systems can, indeed, be used to induce a spin-splitting in the S than ferromagnets or applied magnetic fields, even in the presence of the interfacial disorder. This allows us to eliminate the detrimental, and parasitic effects of ferromagnets employed to this end and constitutes a significant step towards realizing several concepts and devices involving spin-split superconductors [64]. In addition to their fundamental physics importance, the results presented in this chapter have high relevance for the recently invigorated fields of antiferromagnetic spintronics as well as spin-triplet superconductors.

- There are detailed summaries at the end of each chapter; however, we will end the dissertation with the main conclusion in chapter 6. In appendix A, there will be an extension to chapter 3 with two examples that could be solved analytically via the quantum circuit theory. First, Sec. A.1.1 addresses the results for a regime in a four-terminal S/F system, in which we can drive a straightforward expression for the local DOS. Second, Sec. A.1.2 will study the transport properties of a simple FNF system with arbitrary magnetizations alignment on the F leads. To complete the story regarding the TE performance of the system studied in chapter 4, in appendix B, we will discuss the effect of a larger inhomogeneous broadening on the achievable maximum Seebeck coefficient. Moreover, appendix C includes technical notes and a detailed derivation of the self-energy, and the conventions followed for expressing the Green functions in the frequency and momentum representation which complement the results of chapter 5.

Chapter 2

Theoretical fundamentals

One of the main reasons the BCS model grew into such a prominent theory in understanding superconductivity was due to its formulation in terms of the full microscopic Green functions [1] by Gor'kov [67], where he introduced a set of coupled equations for their time-evolution [68].

The Gor'kov's coupled equations are quite difficult to solve; however, it turned out they can get remarkably simplified via integrating the Green function over the energy within the so-called quasi-classical approximation. By relying on the fact that the superconducting phenomena deal with the energy scales much less than the Fermi energy and thereby focusing only on a low-energy regime (on mesoscopic length scale), Larkin-Ovchinnikov [69], and Eilenberger [70] were able to develop transport-like equations for a set of Green functions associated to Gor'kov's Green functions [67]. The quasiparticle motion is treated classically within the quasi-classical approach; however, the particle-hole and the spin degrees of freedom are addressed quantum mechanically [71].

Additionally, the method extended for the dirty case as a limiting case of the Eilenberger's equation [72]. Usadel noticed that in the diffusive limit, the Larkin-Ovchinnikov-Eilenberger Green functions are almost isotropic in space, and thereby he further simplified the general equations by utilizing this condition to explain the diffusive transport of the Cooper pairs [68].

A disadvantage of the quasi-classical approach is that in order to achieve a unique physical solution, one needs to supplement the transport-like equation for interfaces of heterostructures—a first-order matrix differential equation for the quasi-classical

[1] One of the well-established mathematical techniques for the theoretical study of problems in condensed matter and many-body physics.

propagator—with proper boundary conditions [71]. To lift the technical difficulties of the quasi-classical approximation [2] and thereby enhance its feasibility, the interfaces of arbitrarily complex structures in a discretized version of the Usadel equation, can be treated in a systematic way via simplified boundary conditions within the so-called "quantum circuit theory" [74–76] which builds on top of the quasi-classical Green function. In this theory, which is analogous to that of classic electric circuits, a mesoscopic layout is separated into easily characterisable units, e.g., terminals, connectors, and finite-element nodes, where the only difference between them is that we know a priori the equilibrium Green functions in the terminals and yet it has to be determined on the node [77]. This method has been developed individually for S/N [75], F/N [78], and subsequently applied to S/F heterostructures [79–83].

This chapter—which follows most closely textbooks by J. Rammer [84], Y. Nazarov, and Y. Blanter [85], a section by V. Chandrasekhar in [86], and Refs. [74, 75, 87, 88]—is meant to present a summary of the main theoretical concepts which are essential to understand and characterize the physics of the magnetic superconducting heterostructures investigated in the following chapters.

2.1 Green functions and the Gor'kov equations

The BCS Hamiltonian including a contact interaction for the attraction between electrons [89]

$$H = \int d^3r \left(\sum_{\sigma} \hat{\psi}^{\dagger}_{\sigma}(\boldsymbol{r}) h(\boldsymbol{r},t) \hat{\psi}_{\sigma}(\boldsymbol{r}) - \frac{g}{2} \sum_{\sigma \neq \sigma'} \hat{\psi}^{\dagger}_{\sigma}(\boldsymbol{r}) \hat{\psi}^{\dagger}_{\sigma'}(\boldsymbol{r}) \hat{\psi}_{\sigma'}(\boldsymbol{r}) \hat{\psi}_{\sigma}(\boldsymbol{r}) \right) \tag{2.1}$$

where σ (σ') is the spin index, $h(\boldsymbol{r},t) = (\partial_{\boldsymbol{r}}/i - |e|\boldsymbol{A}/c)^2/2m + u(\boldsymbol{r}) - \mu$ with $u(\boldsymbol{r})$ being the one-particle potential, g parametrizes the strength of Gorkov electron-phonon coupling, and we have set $\hbar = 1$. The energy relative to the chemical potential is measured using $-\mu$, and only in a region of the Debye energy of the Fermi surface, there is nonzero interaction among the electrons [89].

The Heisenberg field operators read

$$\psi_{\sigma}(\boldsymbol{r},t) = \exp(iHt)\, \psi_{\sigma}(\boldsymbol{r}) \exp(-iHt)\,, \tag{2.2}$$

$$\psi^{\dagger}_{\sigma}(\boldsymbol{r},t) = \exp(iHt)\, \psi^{\dagger}_{\sigma}(\boldsymbol{r}) \exp(-iHt)\,, \tag{2.3}$$

[2] Although the convenient method to handle the nonequilibrium transport in diffusive proximity nanostructures is the quasi-classical Keldysh Green function, it is technically a complicated task [73].

which obey the usual anticommutation relations

$$\left\{\psi_\sigma\left(\boldsymbol{r},t\right),\psi^\dagger_{\sigma'}\left(\boldsymbol{r}',t'\right)\right\}=\delta_{\sigma\sigma'}\delta\left(\boldsymbol{r}-\boldsymbol{r}'\right)\delta\left(t-t'\right), \tag{2.4}$$

$$\left\{\psi_\sigma\left(\boldsymbol{r},t\right),\psi_{\sigma'}\left(\boldsymbol{r}',t'\right)\right\}=\left\{\psi^\dagger_\sigma\left(\boldsymbol{r},t\right),\psi^\dagger_{\sigma'}\left(\boldsymbol{r}',t'\right)\right\}=0, \tag{2.5}$$

and equation of motions for $\psi_\alpha(\boldsymbol{r},t)$ via employing the time dependence relation as $i\partial_t\psi_\sigma(\boldsymbol{r},t)=[\psi_\sigma(\boldsymbol{r},t),H]$ read:

$$i\partial_t\psi_\sigma(\boldsymbol{r},t)=h(\boldsymbol{r},t)\psi_\sigma(\boldsymbol{r},t)-g\sum_{\sigma'}\psi^\dagger_{\sigma'}(\boldsymbol{r},t)\psi_{\sigma'}(\boldsymbol{r},t)\psi_\sigma(\boldsymbol{r},t), \tag{2.6}$$

Taking its complex conjugate will results in

$$i\partial_t\psi^\dagger_\sigma(\boldsymbol{r},t)=-h(\boldsymbol{r},t)\psi^\dagger_\sigma(\boldsymbol{r},t)+g\sum_{\sigma'}\psi^\dagger_\sigma(\boldsymbol{r},t)\psi^\dagger_{\sigma'}(\boldsymbol{r},t)\psi_{\sigma'}(\boldsymbol{r},t). \tag{2.7}$$

Using the pseudo-spinors in particle-hole (Nambu) space of the form $\Psi^\dagger=\left(\psi^\dagger,\psi\right)$, where $\psi^\dagger$ (ψ) is the electron creation (annihilation) field operator, the so-called contour-ordered Green function reads

$$\begin{aligned}&\underline{G}(\boldsymbol{r},t;\boldsymbol{r}',t')\equiv-i\left\langle\hat{T}_c\Psi(\boldsymbol{r},t)\Psi^\dagger(\boldsymbol{r}',t')\right\rangle\\&=-i\begin{pmatrix}\left\langle\hat{T}_c\psi_\uparrow(\boldsymbol{r},t)\psi^\dagger_\uparrow(\boldsymbol{r}',t')\right\rangle & \left\langle\hat{T}_c\psi_\uparrow(\boldsymbol{r},t)\psi_\downarrow(\boldsymbol{r}',t')\right\rangle\\ \left\langle\hat{T}_c\psi^\dagger_\downarrow(\boldsymbol{r},t)\psi^\dagger_\uparrow(\boldsymbol{r}',t')\right\rangle & \left\langle\hat{T}_c\psi^\dagger_\downarrow(\boldsymbol{r},t)\psi_\downarrow(\boldsymbol{r}',t')\right\rangle\end{pmatrix}=\begin{pmatrix}G(\boldsymbol{r},t;\boldsymbol{r}',t') & F(\boldsymbol{r},t;\boldsymbol{r}',t')\\ \tilde{F}(\boldsymbol{r},t;\boldsymbol{r}',t') & \tilde{G}(\boldsymbol{r},t;\boldsymbol{r}',t')\end{pmatrix}\end{aligned} \tag{2.8}$$

where $\langle\ldots\rangle$ stands for an average over a statistical ensemble [90]. Here, $\hat{T}_c$ is the contour-ordering operator, G is the normal Green function, and F is the anomalous part which measures pair correlations in the superconducting state; with $\tilde{F}(\boldsymbol{r},t;\boldsymbol{r}',t')=F^\dagger(\boldsymbol{r}',t';\boldsymbol{r},t)$.

We will employ $i\partial_t$ on each component of the Nambu Green function in Eq.(2.8) and to make it more convenient we define representers 1 and $1'$ instead of the coordinate sets $(\boldsymbol{r},t)$ and $(\boldsymbol{r}',t')$, respectively. As a result, the Gor'kov coupled equations are:

$$\begin{pmatrix}i\partial_t-h(1) & \Delta(1)\\ -\Delta^*(1) & -i\partial_t-h^*(1)\end{pmatrix}\underline{G}(1,1')=\delta(\boldsymbol{r}-\boldsymbol{r}')\delta(t-t')=\delta(1-1'), \tag{2.9}$$

where, the superconducting order parameter Δ (Δ^*) will get the following self-consistent form utilizing the anomalous Green function F ($F^\dagger$) :

$$\Delta(1) = g \langle \psi_\uparrow(1)\psi_\downarrow(1)\rangle = -ig \lim_{1'\to 1} F(1,1'), \tag{2.10}$$

$$\Delta^*(1) = ig \lim_{1'\to 1} F^\dagger(1,1'). \tag{2.11}$$

To this point, we have disregarded the effects of impurity scatterings in the derivation of the Gor'kov matrix Eqs. (2.9). In general, as we will show in the next section, the corresponding contributions could be taken into account within a self-energy term.

2.2 Perturbation theory: Self-energy

The self-energy term $\underline{\Sigma}$ in general, includes external fields [91] and possible impurity scattering processes—e.g., elastic spin-flip or spin-orbit and inelastic electron-electron and electron-phonon scattering [57]. Incorporating the diagram in Fiq. 2.1, we may treat the interactions in a perturbation picture, and thereby complement the free (non-interacting) Green function with the present self-energies in the system. As a result, the total Green function of the system can be expressed as:

$$\begin{aligned}\underline{G} = \underline{G}_0 &+ \underline{G}_0 \otimes \underline{\Sigma} \otimes \underline{G}_0 \\ &+ \underline{G}_0 \otimes \underline{\Sigma} \otimes \underline{G}_0 \otimes \underline{\Sigma} \otimes \underline{G}_0 \\ &+ \underline{G}_0 \otimes \underline{\Sigma} \otimes \underline{G}_0 \otimes \underline{\Sigma} \otimes \underline{G}_0 \otimes \underline{\Sigma} \otimes \underline{G}_0 \\ &+ ...,\end{aligned} \tag{2.12}$$

where, by some iteration, we obtain the following equation for $\underline{G}(1,1')$, namely the matrix Dyson equation

$$\underline{G} = \underline{G}_0 + \underline{G}_0 \otimes \underline{\Sigma} \otimes \underline{G}_0, \tag{2.13}$$

or

$$\underline{G}^{-1} = \underline{G}_0^{-1} - \underline{\Sigma}, \tag{2.14}$$

here, $\otimes$ implies matrix multiplication in the spatial and contour time variables [87]. Furthermore, we could iterate the Eq. (2.12) from the other side and write down equivalently the Eq. (2.13) as $\underline{G} = \underline{G}_0 + \underline{G} \otimes \underline{\Sigma} \otimes \underline{G}_0$. These two equations are repetitive in equilibrium since Fourier transformation turns the time convolutions into simple products for which the order of factors is irrelevant. However, they contain different

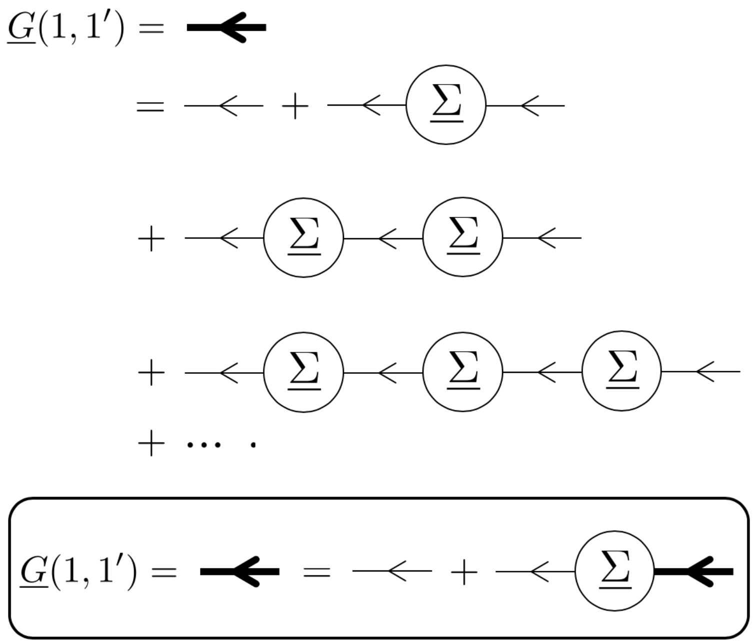

Figure 2.1: The diagram representation of the possible interactions of the electrons with their surroundings, namely the self-energy $\underline{\Sigma}$, adapted from Ref. [84]. The self-energy parts are joined with the rest via two free particle Green function $\underline{G}_0$ lines.

information in a non-equilibrium state, and thereby subtracting them is a useful way of expressing the out of equilibrium dynamics [87].

$\underline{G}_0$ obeys the equation of motion

$$(\underline{G}_0^{-1} \otimes \underline{G}_0)(1,1') = \delta(1-1') = (\underline{G}_0 \otimes \underline{G}_0^{-1})(1,1'), \tag{2.15}$$

where, the inverse free Green function $\underline{G}_0^{-1}(1,1') = \underline{G}_0^{-1}(1)\delta(1-1')$ and $\underline{G}_0^{-1}(1) = i\underline{\tau}_3\partial_t - h(1)$. Operating the inversed free Green function from left-side on the Dyson equation (Eq. (2.12)), we obtain

$$(\underline{G}_0^{-1} - \underline{\Delta} - \underline{\Sigma}) \otimes \underline{G} = \delta(1-1'), \tag{2.16}$$

likewise, if we do the same from the right side, the conjugated Gor'kov-Dyson equation reads

$$\underline{G} \otimes (\underline{G}_0^{-1} - \underline{\Delta} - \underline{\Sigma}) = \delta(1-1'), \tag{2.17}$$

and, the pair potential is given by exploiting the part of electron-phonon self-energy which is responsible for superconductivity [88]

$$\underline{\Delta}(1,1') = \left[\frac{\big(\Delta(1) - \Delta^*(1)\big)}{2}\underline{\tau}_x + \frac{i\big(\Delta(1) + \Delta^*(1)\big)}{2}\underline{\tau}_y\right]\delta(1-1'), \tag{2.18}$$

here, $\underline{\tau}_i$ represents the i-th 2×2 Pauli matrix in particle-hole space.

2.2.1 Keldysh Green function

To generalize the Gor'kov-Dyson Eq. (2.16) and its conjugate (2.16), we will implement the Keldysh technique which allows us to describe many-body systems out of thermal equilibrium [88]. We take the contour-ordered Nambu Green function in Eq. (2.8) and label it from here on as $\underline{G}\left(\boldsymbol{r},t;\boldsymbol{r}',t'\right) \to \underline{G}^{11}\left(1,1'\right) = -i\left\langle \hat{T}\Psi(1)\Psi^\dagger(1')\right\rangle$, where $\hat{T}$ is the forward (ordinary) time-ordering operator, and $\underline{G}^{11}\left(1,1'\right)$ denotes the case when both t and t' are along the forward-in-time cotour [87] (upper branch of the Keldysh contour)—indicated via the superscript 1. The other components in terms of average values of Heisenberg operators are the greater Green function

$$\underline{G}^{12}(1,1') = i\left\langle \Psi^\dagger(1')\Psi(1)\right\rangle = G^>(1,1'), \tag{2.19}$$

the lesser $\underline{G}$

$$\underline{G}^{21}(1,1') = -i\left\langle \Psi(1)\Psi^\dagger(1')\right\rangle = G^<(1,1'), \tag{2.20}$$

and

$$\underline{G}^{22}(1,1') = -i\left\langle \hat{\tilde{T}}\Psi(1)\Psi^\dagger(1')\right\rangle. \tag{2.21}$$

Both time coordinates t and t' are on the backward-in-time contour (lower branch) [3] for $\underline{G}^{22}$, and $\hat{\tilde{T}}$ indicates backward (anti-) time-ordering operator [87]. The greater and lesser Green functions are related through the retarded, advanced

$$\begin{aligned}\underline{G}^R(1,1') &= \theta(t-t')\left[\underline{G}^<(1,1') - \underline{G}^>(1,1')\right],\\ \underline{G}^A(1,1') &= -\theta(t'-t)\left[\underline{G}^<(1,1') - \underline{G}^>(1,1')\right],\end{aligned} \tag{2.22}$$

[3] labeld as superscript 2

and the Keldysh Green functions

$$\underline{G}^K(1,1') = \underline{G}^<(1,1') + \underline{G}^>(1,1'). \tag{2.23}$$

The four components of the full Green function in keldysh spcae

$$\hat{G}(1,1') = \begin{pmatrix} \underline{G}^{11}(1,1') & \underline{G}^{12}(1,1') \\ \underline{G}^{21}(1,1') & \underline{G}^{22}(1,1') \end{pmatrix}, \tag{2.24}$$

are linearly related via equations of the form $\underline{G}^{11} + \underline{G}^{22} = \underline{G}^{12} + \underline{G}^{21}$. One can therefore perform a transformation to set one of the matrix components to zero [86]; employing a transformation in Keldysh space $\hat{G} \equiv \hat{\tau}_z \hat{G}$, a rotation as $\hat{G} \to \hat{L}\hat{G}\hat{L}^\dagger$ where $\hat{L} = (\mathbb{1}_4 - i\hat{\tau}_y)/\sqrt{2}$, and finally implementing [87]

$$\underline{G}^R(1,1') = \underline{G}^{11}(1,1') - \underline{G}^{12}(1,1') = \underline{G}^{21}(1,1') - \underline{G}^{22}(1,1'), \tag{2.25}$$

$$\underline{G}^A(1,1') = \underline{G}^{11}(1,1') - \underline{G}^{21}(1,1') = \underline{G}^{12}(1,1') - \underline{G}^{22}(1,1'), \tag{2.26}$$

$$\underline{G}^K(1,1') = \underline{G}^{21}(1,1') + \underline{G}^{12}(1,1') = \underline{G}^{11}(1,1') + \underline{G}^{22}(1,1'), \tag{2.27}$$

the resulting 4×4 Nambu-Keldysh matrices will have the forms

$$\hat{G} = \begin{pmatrix} \underline{G}^R & \underline{G}^K \\ 0_2 & \underline{G}^A \end{pmatrix}, \qquad \hat{\Sigma} = \begin{pmatrix} \underline{\Sigma}^R & \underline{\Sigma}^K \\ 0_2 & \underline{\Sigma}^A \end{pmatrix}, \tag{2.28}$$

where, the Green function in terms of its spin matrix elements G and F of the quasi-classical Green function, and their associated particle-hole conjugates ($\tilde{G}$, $\tilde{F}$) takes the form [92]

$$\underline{G}^{R,A} = \begin{pmatrix} G^{R,A} & F^{R,A} \\ -\tilde{F}^{R,A} & -\tilde{G}^{R,A} \end{pmatrix}, \qquad \underline{G}^K = \begin{pmatrix} G^K & F^K \\ \tilde{F}^K & \tilde{G}^K \end{pmatrix}. \tag{2.29}$$

Lastly, using Eq. (2.28) and defining block-diagonal matrices $\hat{\tau}_i = \operatorname{diag}(\underline{\tau}_i, \underline{\tau}_i)$, $\hat{\Delta} = \operatorname{diag}(\underline{\Delta}, \underline{\Delta})$, and $\hat{G}_0^{-1} = \operatorname{diag}\left(\underline{G}_0^{-1}, \underline{G}_0^{-1}\right)$ we could derive the same Gor'kov equation as Eq. (2.16) ((2.17)) in the non-equilibrium Keldysh formalism. Note that, since overall this thesis we are mostly interested in the properties of S/F-hybrids, we must replace the G and F elements in the Eq. (2.29) with 2×2 spin matrices:

$$\underline{g} = \begin{pmatrix} g_{\uparrow\uparrow} & g_{\uparrow\downarrow} \\ g_{\downarrow\uparrow} & g_{\downarrow\downarrow} \end{pmatrix}, \tag{2.30}$$

enabling us to parametrize the S correlations inside the F [92]. This process results in $\hat{G} \to \check{G}$, now corresponding to an 8×8 Keldysh⊗Nambu⊗spin matrix.

2.3 Quasi-classical Green functions

In this section, we will introduce a powerful tool to treat non-equilibrium superconducting systems. Substracting the Gor'kov-Dyson equation of motion (Eq. (2.16)) and its conjugate (Eq. (2.17)) in the non-equilibrium Keldysh formalism, we obtain

$$\left[\hat{G}_0^{-1} - \hat{\Delta} - \hat{\Sigma}, \hat{G}\right] = 0_4. \tag{2.31}$$

Since this equation depends on two coordinates, it is quite tricky to solve it. On the other hand, the full $\hat{G}$ oscillates quickly as a function of $|\boldsymbol{r} - \boldsymbol{r}'|$ [88] on the scale of Fermi wavelength and thereby holds information at length scales much shorter than those we are interested [93]. The way around this is switching from the coordinates $(\boldsymbol{r}, t)$ and $(\boldsymbol{r}', t')$ to the centeral $(\boldsymbol{R}, T)$ and relative coordinates $(\boldsymbol{r}, t)$ defined as

$$\begin{aligned} \boldsymbol{r} &= \boldsymbol{R} + \boldsymbol{r}/2, & \boldsymbol{r}' &= \boldsymbol{R} - \boldsymbol{r}/2, \\ t &= T + t/2, & t' &= T - t/2, \end{aligned} \tag{2.32}$$

where,

$$\hat{G}(1, 1') \to \hat{G}(\boldsymbol{R} + \boldsymbol{r}/2, \boldsymbol{R} - \boldsymbol{r}/2, T + t/2, T - t/2) \tag{2.33}$$

and then Fourier transformation of it with respect to $\boldsymbol{r}$ and t, in Wigner representation will be

$$\hat{G}(\boldsymbol{R}, T; \boldsymbol{r}, t) = \int d\boldsymbol{p} d\epsilon \exp(-i\epsilon t) \exp(i\boldsymbol{p} \cdot \boldsymbol{r}) \hat{G}(\boldsymbol{R}, T; \boldsymbol{p}, \epsilon), \tag{2.34}$$

with the quasiparticle energy ϵ. The quasi-classical Green functions are then defined via integrating overt the Gor'kov Green function as

$$\hat{g}(\boldsymbol{R}, T; p, \epsilon) = \frac{i}{\pi} \int_{-\infty}^{\infty} d\xi \hat{G}(\boldsymbol{R}, T; \boldsymbol{p}, \epsilon), \tag{2.35}$$

with a normalization constraint for $\hat{g}(\boldsymbol{R}, T; \boldsymbol{p}, \epsilon)$ as $\hat{g}^2(\boldsymbol{R}, T; \boldsymbol{p}, \epsilon) = \mathbb{1}_4$[4]. In order to capture the low-energy behavior of the electron Green functions, we have employed

[4]The symbol 0_n ($\mathbb{1}_n$) labels the zero (identity) matrix in $n \times n$ dimensions.

the ξ-integration to exploit its pronounced peak around the Fermi momentum p_F, where $\xi = \boldsymbol{v}_{\mathrm{F}}(\boldsymbol{p} - \boldsymbol{p}_{\mathrm{F}})$ is the energy measured respecting the Fermi level [93] with $\boldsymbol{v}_{\mathrm{F}}$ being the Fermi velocity. With the assumption that the self-energy depends weakly on ξ and thereby one could set $\xi = 0$ in it [94].

The Eilenberger equation of motion for the quasi-classical Green functions [70] read

$$i\partial_T\{\hat{\tau}_z, \hat{g}\} + i\boldsymbol{v}_{\mathrm{F}} \cdot \partial_{\boldsymbol{R}}\, \hat{g} + \epsilon\left[\hat{\tau}_z - \hat{\Delta}, \hat{g}\right] - \left[\hat{\Sigma}, \hat{g}\right] = 0_4, \tag{2.36}$$

where, $\hat{\Sigma}$ denotes the elastic scattering self-energy. If we are interested only in stationary problems, Eq. (2.36) then will be simplified to the following form

$$\left[\epsilon\hat{\tau}_z - \hat{\Delta} - \hat{\Sigma}, \hat{g}\right] + i\boldsymbol{v}_{\mathrm{F}} \cdot \partial_{\boldsymbol{R}}\, \hat{g} = 0_4. \tag{2.37}$$

Now that we have derived the Eilenberger quasi-classical Keldysh Green function, the physical observables which we are interested in, e.g., the electric and thermal current densities will have the form [86]

$$\boldsymbol{j}(\boldsymbol{R}, T) = -\frac{eN_0}{4}\int_{-\infty}^{\infty} \mathrm{d}\epsilon \int \frac{d\Omega_p}{4\pi}\boldsymbol{v}_F \,\mathrm{tr}\left[\underline{\tau}_z \underline{g}^K(\boldsymbol{R}, T; \boldsymbol{p}, \epsilon)\right], \tag{2.38}$$

$$\boldsymbol{j}_{\mathrm{th}}(\boldsymbol{R}, T) = -\frac{N_0}{4}\int_{-\infty}^{\infty} \mathrm{d}\epsilon \int \frac{d\Omega_p}{4\pi}\epsilon\; \boldsymbol{v}_F \,\mathrm{tr}\left[\mathbb{1}_2 \underline{g}^K(\boldsymbol{R}, T; \boldsymbol{p}, \epsilon)\right], \tag{2.39}$$

with $d\Omega_p$ solid angle in momentum space spanning the Fermi surface, and $N_0 = m^2|\boldsymbol{v}_F|/2\pi^2$ is the single-particle DOS at the Fermi energy.

2.3.1 Usadel equation

The Usadel equation assumes that the Fermi energy is considerably higher than any other energy scale involved in the system, and fermions govern the essential physics at Fermi level [95]. Moreover, the superconducting material usually has strong impurity scattering—the impurity scattering rate $(1/\tau)$ is much larger than any other energy (ϵ, Δ) [86] except for the Fermi energy. In this limit, referred to as the diffusive "dirty" limit, due to the strong elastic scattering, the transport of quasiparticles is diffusive rather than ballistic; thus the Green functions can be treated as nearly isotropic, and we can expand them linearly in spherical harmonics as [86, 88]

$$\hat{g} = \hat{g}_s + \hat{\boldsymbol{p}} \cdot \hat{\boldsymbol{g}}_{\boldsymbol{p}} \tag{2.40}$$

the subscripts indicate s- and p-wave components, and $\hat{\boldsymbol{g}}_{\boldsymbol{p}} \ll \hat{g}_s$. Similarily, for the self-energy

$$\hat{\Sigma} = \hat{\Sigma}_s + \hat{\boldsymbol{p}} \cdot \hat{\boldsymbol{\Sigma}}_{\boldsymbol{p}}, \tag{2.41}$$

where, $\hat{\boldsymbol{p}}$ is a vector of unit magnitude in the direction of momentum, however, both isotropic and correlation terms $\hat{g}_s$, $\hat{\boldsymbol{g}}_{\boldsymbol{p}}$ (and $\hat{\Sigma}_s$, $\hat{\boldsymbol{\Sigma}}_{\boldsymbol{p}}$) do not depend on its direction. Implementing the normalization constraints $\hat{g}_s^2 = \mathbb{1}_4$ and thereby $\hat{g}_s\hat{\boldsymbol{g}}_{\boldsymbol{p}} + \hat{\boldsymbol{g}}_{\boldsymbol{p}}\hat{g}_s = 0$ [86], inserting Eqs. (2.40) and (2.41) into the Eilenberger equation ((2.37)), and replacing $\boldsymbol{v}_{\rm F}$ with $v_{\rm F}\hat{\boldsymbol{p}}$, we obtain

$$\begin{aligned} &\left[\epsilon\hat{\tau}_z - \hat{\Delta} - \hat{\Sigma}_s, \hat{g}_s\right] + iv_{\rm F}(\hat{\boldsymbol{p}} \cdot \hat{\boldsymbol{p}})\partial_{\boldsymbol{R}}\, \hat{\boldsymbol{g}}_{\boldsymbol{p}} \\ &+ \hat{\boldsymbol{p}}\left[\epsilon\hat{\tau}_z - \hat{\Delta} - \hat{\Sigma}_s, \hat{\boldsymbol{g}}_{\boldsymbol{p}}\right] - \hat{\boldsymbol{p}}\left[\hat{\boldsymbol{\Sigma}}_{\boldsymbol{p}}, \hat{g}_s\right] + iv_{\rm F}\hat{\boldsymbol{p}} \cdot \partial_{\boldsymbol{R}}\, \hat{g}_s = 0_4. \end{aligned} \tag{2.42}$$

Collecting all the terms which are even in $\hat{\boldsymbol{p}}$ (terms in the first line) and averaging them in all directions of momentum

$$\left[\epsilon\hat{\tau}_z - \hat{\Delta}, \hat{g}_s\right] + i\frac{v_{\rm F}}{3}\partial_{\boldsymbol{R}}\, \hat{\boldsymbol{g}}_{\boldsymbol{p}} = 0_4, \tag{2.43}$$

and all the other terms which are odd in $\hat{\boldsymbol{p}}$ (terms proportional to it, the second line in Eq. (2.42)) together

$$\left[\epsilon\hat{\tau}_z - \hat{\Delta}, \hat{\boldsymbol{g}}_{\boldsymbol{p}}\right] - \frac{i}{2\tau_{tr}}\left[\hat{\boldsymbol{g}}_{\boldsymbol{p}}, \hat{g}_s\right] + iv_{\rm F}\partial_{\boldsymbol{R}}\, \hat{g}_s = 0_4, \tag{2.44}$$

where, the spin-dependent scattering is neglected [86]. The s- and p-wave components of the self-energy matrix are $\hat{\Sigma}_s = -(i/2\tau)\hat{g}_s - (i/2\tau_{sf})\hat{\tau}_z\hat{g}_s\hat{\tau}_z$ [5], and $\hat{\boldsymbol{\Sigma}}_{\boldsymbol{p}} = -(i/2)(1/\tau - 1/\tau_{tr})\hat{\boldsymbol{g}}_{\boldsymbol{p}}$ [6], respectively. Here, τ_{tr} is the effective transport time, and $1/\tau_{sf}$ is the spin-flip scattering rate. Assuming strong elastic scattering, we can neglect the first term in Eq. (2.44) respecting the third one, and thereby the second term will simplify to

$$\frac{i}{\tau_{tr}}\hat{g}_s\hat{\boldsymbol{g}}_{\boldsymbol{p}} + iv_{\rm F}\partial_{\boldsymbol{R}}\, \hat{g}_s = 0_4, \tag{2.45}$$

In order to express $\hat{\boldsymbol{g}}_{\boldsymbol{p}}$ in terms of $\hat{g}_s$, we will first introduce the elastic mean free path as $l = v_{\rm F}\tau_{tr}$, then multiply Eq. (2.45) from the left-side by $\hat{g}_s$ while exploiting the normalization relation which leads to $\hat{\boldsymbol{g}}_{\boldsymbol{p}} = -l\hat{g}_s\partial_{\boldsymbol{R}}\, \hat{g}_s$. Finally, inserting $\hat{\boldsymbol{g}}_{\boldsymbol{p}}$ into

[5] with spin-flip scattering included

[6] Under the assumption that non-spin-flip impurity scattering is much stronger than the spin-flip one, we consider only the contribution of the former for the p-wave component [86, 87].

Eq. (2.43) results in a diffusion-like expression for $\hat{g}_s$, also known as the so-called Usadel equation for dirty bulk-superconductors [72]

$$i\partial_T\{\hat{\tau}_z, \hat{g}_s\} + \left[\epsilon\hat{\tau}_z - \hat{\Delta}, \hat{g}_s\right] - iD\partial_{\boldsymbol{R}}(\hat{g}_s\partial_{\boldsymbol{R}}\,\hat{g}_s) = 0_4, \tag{2.46}$$

with $D = v_{\mathrm{F}}l/3$ the diffusion coefficient. The fact that elastic scattering self-energy is no more present in the equation of motion indicates the diffusive nature of the Usadel equation.

The equilibrium properties could be described employing the diagonal components of $\hat{g}_s$ ($\underline{g}_s^{R,A}$), however, the kinetic equation will be derived via the off-diagonal term—the quasi-classical Keldysh-Usadel Green function ($\underline{g}_s^K$). To this end, following Eq. (2.38), the electric current density in the dirty limit [86]

$$\begin{aligned}
\boldsymbol{j}(\boldsymbol{R}, T) &= -\frac{eN_0}{4}\int_{-\infty}^{\infty}\mathrm{d}\epsilon\int\frac{d\Omega_p}{4\pi}v_{\mathrm{F}}\hat{\boldsymbol{p}}\,\mathrm{tr}\left[\underline{\tau}_z(\underline{g}_s^K + \hat{\boldsymbol{p}}\cdot\hat{\boldsymbol{g}}_{\boldsymbol{p}}^K)\right], \\
&= \frac{eN_0D}{4}\int_{-\infty}^{\infty}\mathrm{d}\epsilon\,\mathrm{tr}\left[\underline{\tau}_z(\underline{g}_s\partial_{\boldsymbol{R}}\,\underline{g}_s)^K\right] \\
&= \frac{eN_0D}{4}\int_{-\infty}^{\infty}\mathrm{d}\epsilon\,\mathrm{tr}\left[\underline{\tau}_z(\underline{g}_s^R\partial_{\boldsymbol{R}}\,\underline{g}_s^K + \underline{g}_s^K\partial_{\boldsymbol{R}}\,\underline{g}_s^A)\right],
\end{aligned} \tag{2.47}$$

and in the same fashion as Eq. (2.39):

$$\boldsymbol{j}_{\mathrm{th}}(\boldsymbol{R}, T) = \frac{N_0D}{4}\int_{-\infty}^{\infty}\mathrm{d}\epsilon\;\epsilon\,\mathrm{tr}\left[\underline{\tau}_z(\underline{g}_s^R\partial_{\boldsymbol{R}}\,\underline{g}_s^K + \underline{g}_s^K\partial_{\boldsymbol{R}}\,\underline{g}_s^A)\right]. \tag{2.48}$$

2.4 Quantum circuit theory

The so-called circuit theory allows studying quantum transport in arbitrarily complex mesoscopic multi-terminals, where can be described by a set of discrete elements, namely terminals, finite-volume nodes, and barriers [74, 75]. Connections to the external voltage, temperatures, phases, or current sources are taking into account via the help of the terminals, the overall layout is provided via nodes, and barriers represent the connectors between nodes [96]. The nodes will contain matrix Green functions [7], which will include the necessary boundary conditions supplementing the Usadel equation. The transport properties through each connector then will be characterized by a matrix Green function current, where via exploiting a Kirchhoffs' like rule on the node, the sum of all these currents must vanish.

We will write down a drift-diffusion-like equation via expressing the Usadel equa-

[7] with satisfied normalization constraint $\check{\mathcal{G}}_c^2 = \mathbb{1}_4$

tion for a generalized matrix current. Following Eq. (2.46) for the stationary case and via defining $\hat{E} \equiv \epsilon\hat{\tau}_z - \hat{\Delta}$, we get

$$\partial_{\boldsymbol{R}} \cdot \hat{\boldsymbol{j}} - ie^2\nu \left[\hat{E}, \hat{g}_s\right] = 0_4, \tag{2.49}$$

where, $\hat{\boldsymbol{j}} = -\sigma\hat{g}_s\partial_{\boldsymbol{R}}\, \hat{g}_s$ is the matrix current, conductivity $\sigma = e^2\nu D$, and ν the DOS at the Fermi energy [85]. However, at non-zero energies, Eq. (2.49) indicates that matrix current is not fully conserved ($\partial_{\boldsymbol{R}} \cdot \hat{\boldsymbol{j}} \neq 0$). To overcome this problem and be able to use the conservation laws, we can introduce an additional term, namely the density of the "leakage" matrix current: $\hat{j}_L = -ie^2\nu[\hat{E}, \hat{g}_s]$. Keeping this in mind, integrating over Eq. (2.49) will lead [85]

$$0_4 \equiv \hat{\mathcal{I}} + \hat{\mathcal{I}}_L = \int \mathrm{d}S(\hat{\boldsymbol{j}} \cdot \boldsymbol{N}) + \int \mathrm{d}V\hat{j}_L, \tag{2.50}$$

where, the first and second integrals are over a surface of the volume, and the volume, respectively. The unit vector normal to the surface is indicated as N. Employing the discretized Usadel equation (integrals in Eq. (2.50)) will lead to the conservation of the total matrix current on the node which is reflected within the Kirchhoff's rule $0_4 = \sum_k \hat{\mathcal{I}}_{ik} + \hat{\mathcal{I}}_L$ [8] and allows to develop a finite-element circuit theory employing proper boundary conditions.

The next step after having discretized the structure would be to calculate the Green function on the nodes. If we consider two nodes with the Green functions $\hat{G}_{ci}$ and $\hat{G}_{ck}$, the matrix current in the junction between them reads [75]

$$\hat{\mathcal{I}}_{ik} = 2G_Q \sum_n T_n^{(i,k)} \frac{\left[\hat{G}_{ci}, \hat{G}_{ck}\right]}{4 + T_n^{(i,k)}\left(\{\hat{G}_{ci}, \hat{G}_{ck}\} - 2\right)}, \tag{2.51}$$

where this relation is for a spin-independent case and characterized via $T_n^{(i,k)} \in [0,1]$ as a set of transmission eigenvalues with channel index n, and $G_Q = 2e^2/h$ is the conductance quantum. As a simplied version, we can consider a tunnel junction structure where due to the small transmission eigenvalues, the current takes the form $\hat{\mathcal{I}}_{ik} = G_Q \sum_n T_n^{(i,k)} \left[\hat{G}_{ci}, \hat{G}_{ck}\right]/2$ [9]. In general, one could rewrite the boundary condi-

[8] It's summed over all the other nodes (k) which are connected to the node i, and L stands for leakage.

[9] $G_T = G_Q \sum_n T_n^{(i,k)}$ indicates the tunnel conductance of the junction

tion in Eq. (2.51) as $\hat{\mathcal{I}}_{ik} = \left[\hat{G}_{ci}, \hat{L}_{ik}\right]$, so the second Kirchhoff's rule takes the form

$$0_4 \equiv \left[\hat{G}_{ci}, \hat{\mathcal{G}}_i\right] = \left[\hat{G}_{ci}, \sum_k \hat{L}_{ik} + \hat{L}_L\right] = \left[\hat{G}_{ci}, \sum_k g_{ik}\hat{G}_k - ie^2\nu V_i \hat{E}\right], \tag{2.52}$$

here, g_{ik} is the conductance of the terminal (or the barrier), and $\hat{G}_k$ is the corresponding Green function. Since the nodes are finite-element, it means that there is a volume (here, V_i for the node i) associated with them.

In the following chapter 3, we will implement the currents conservation within the finite-element quasi-classical circuit theory and analyze spin-dependent quasiparticle and Cooper-pair transport of a proximity-coupled S/F heterostructure.

Appendix A includes a comprehensive analytical calculation concerning the derivation of the physical observables using this technique.

Chapter 3

Phase-controlled spin and charge currents in S/F hybrids

The provision and manipulation of equal-spin triplet pairs are indispensable for low-power spintronics, but arduous attainable due to their elusive nature. So far, equal-spin triplets have been investigated in equilibrium configurations. We go a step further and propose a setup that facilitates the generation, control, and detection of spin-triplet Cooper pairs under non-equilibrium conditions. In particular, we offer a detection scheme for equal-spin triplet correlations based on charge-current measurements and show how to establish a tunable spin current into the ferromagnets.

Investigating spin-dependent quasiparticle and Cooper-pair transport through a central node interfaced with two superconductors and two ferromagnets, we demonstrate that voltage biasing of the ferromagnetic contacts induces superconducting triplet correlations on the node and reverses the supercurrent flowing between the two superconducting contacts. We further find that such triplet correlations can mediate tunable spin current flow into the ferromagnetic contacts. Our key finding is that unequal spin-mixing conductances for the two interfaces with the ferromagnets result in equal-spin triplet correlations on the node detectable via a net charge current between the two magnets. Our proposed device thus enables generation, control, and detection of the typically elusive equal-spin triplet Cooper pairs [66].

3.1 Introduction

Cooper-pairs from a superconductor (S) placed in the vicinity of a ferromagnet (F) may diffuse into the latter, thus, modifying their electronic properties [17, 26, 97–99].

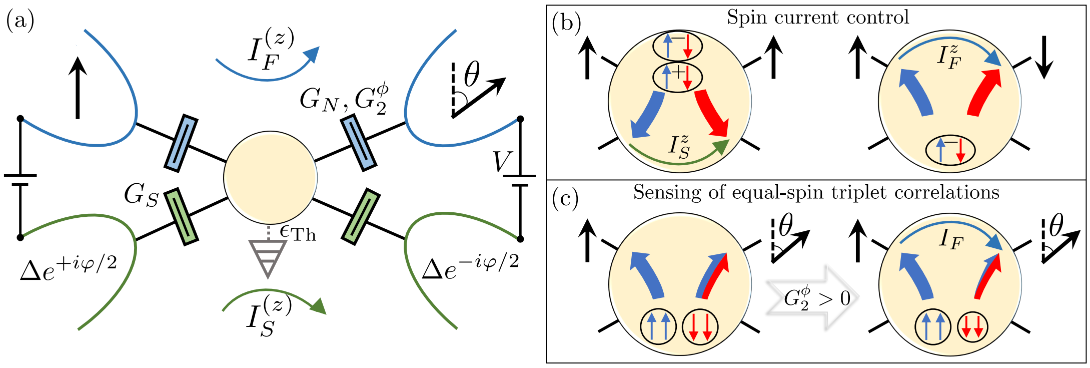

Figure 3.1: (a) 4-terminal circuit consisting of two superconductors with phases $\pm\varphi/2$ and gap Δ (green bordered region), and two ferromagnets with relative magnetization angle θ (blue-bordered region), interfaced with a central node (yellow shaded area). (b) The left (right) sketch of the central node illustrates a net spin current injection into the superconductors (ferromagnets) due to (anti)parallel magnetization of the ferromagnetic leads for $\epsilon_{\mathrm{Th}} \ll \Delta$ and $V \gtrsim \Delta/e$ in absence of the spin-mixing conductances, $G_1^\phi = G_2^\phi = 0$. Singlet (mixed-triplet) correlations are indicated by encircled arrows containing a minus (plus) sign. (c) Finite spin-mixing, $G_2^\phi > 0$, allows the generation and detection of equal-spin triplet correlations (encircled arrows of the same color) by measuring a net charge current between the ferromagnetic leads.

The engineering of this proximity effect in hybrid structures generated over the last few years considerable interest in thermoelectricity [50, 51, 53, 54, 60, 100–105], spin calorics [106, 107], and topological superconductivity [108–113]. Somehow unexpected is that not only ferromagnets [57, 114–116] and antiferromagnetic insulators [64] can cause spin imbalances into superconductors, but also normal metals [117] exploiting that a superconductor itself may serve as a spin filter [118].

By sandwiching a normal metal or ferromagnet between two superconductors, one can realize a Josephson junction featuring a current of Cooper-pairs between them, which is characterized by the junctions' free energy [119–121]. Its global minimum determines the ground state of the system. While a ground state at zero phase difference indicates a Josephson current mediated by singlet Cooper-pairs, a shifted groundstate about π—occurring in magnetic Josephson junctions [42–44, 122–128]—signals triplet superconductivity [24, 129, 130] manifesting in a reversed current phase relation (CPR) [131–133]. Such magnetic Josephson junctions are interesting for quantum computation [134–136] and cryogenic memories [137, 138].

Recently, the generation of equal-spin triplet pairs has been demonstrated in S/F structures utilizing inhomogeneous magnetic fields [65], as initially predicted by Kadigrobov [46] and Bergeret [45, 139] et al. Mixed-spin triplet pairs with a zero spin projection on the z-axis, however, already arise for homogeneous magnetization. It is the immunity of equal-spin triplet correlations against internal magnetic fields, which causes long penetration length into ferromagnets compared to the ones of singlet and mixed-triplet correlations [25, 140, 141]. This property of equal-spin triplet pairs makes them particular attractive for low-power spintronics [35–37].

In this chapter, we study spin and charge transport in a four-terminal S/F system consisting of two s-wave superconducting ($n{=}S_1, S_2$) and two ferromagnetic ($n{=}F_1, F_2$) terminals with a common contact region, Fig. 3.1(a). Our key finding is that a voltage applied to the two ferromagnetic leads in combination with noncollinear magnetizations can induce triplet superconducting correlations in the system and, therewith, transitions in the CPR. We show that the relative magnetization angle θ qualifies for voltages above the superconducting gap as a control knob for spin currents in the ferromagnetic and superconducting contacts [Fig. 3.1(b)]. Finally, we highlight that asymmetric spin-mixing conductances (e.g. $G_1^{\phi} = 0$, $G_2^{\phi} > 0$) can induce for non-collinear magnetization, $0 < \theta < \pi$, but otherwise symmetric configuration, indeed, a finite net charge current between the ferromagnetic contacts [Fig. 3.1(c)]. This effect is attributed to the generation of equal-spin triplet correlations in the central node and allows for their experimental detection and exploitation in a convenient manner.

The outline of the chapter is as follows. In Sec. 3.2, we introduce the schematic of the system under consideration and outline the theoretical framework employed to describe the physical observables. The following Sec. 3.3 concerns with the different energy scalings used in the chapter. Sec. 3.4 presents a discussion regarding 0–π-transitions and tunable spin current, and Sec. 3.5 studies the effect of spin-mixing on the equal-spin triplet correlations. We close the chapter with a summary of the results in Sec. 3.6.

3.2 Method

We study diffusive transport, within a quasi-classical [69, 87, 88, 93, 94] circuit theory [74, 75]. In this framework, hybrid structures are discretized as a network of nodes, terminals, and connectors. Here, we map our system to a layout consisting of a central node that is interfaced with two superconducting and two ferromagnetic terminals via corresponding connectors, see Fig. 3.1(a). An additional *leakage* terminal can account for losses of superconducting correlations. The Green function $\check{\mathcal{G}}_c$ of the node, which is an 8×8-matrix in Keldysh-Nambu-spin space, is here the central

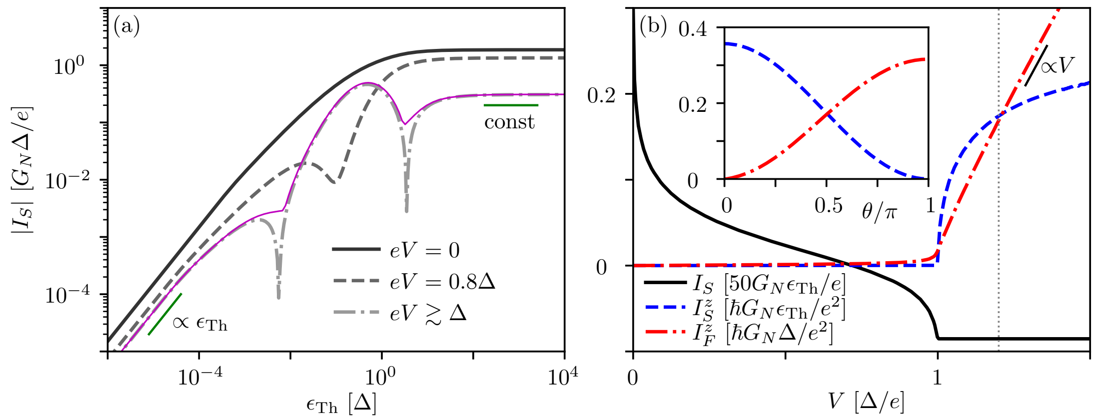

Figure 3.2: (a) Modulus of the net supercurrent I_S as a function of the Thouless energy ϵ_{Th} for different voltages V. The divergencies occurring for large voltages, $V \gtrsim \Delta/e$, indicate 0–π-transitions. Parameters are $\theta = 0$, $\varphi = \pi/2$, and $G_1^\phi = G_2^\phi = 0$. The thin purple line indicates the corresponding critical current $I_C = \max_\varphi I_S$ for $V \gtrsim \Delta/e$. (b) Net supercurrent I_S, and net spin currents I_S^z and I_F^z between the superconducors/ferromagnets for $\theta = \varphi = \pi/2$ and $\epsilon_{\mathrm{Th}} \ll \Delta$. The inset shows the net spin currents as function of θ for $V = 1.2\Delta/e$, as indicated by the horizontal dotted line in panel (b).

quantity characterizing the transport properties. The conservation of the total matrix current at this node, $0_8 = \sum_n \check{\mathcal{I}}_n$, together with the normalization condition $\check{\mathcal{G}}_c^2 = \mathbb{1}_8$ determines $\check{\mathcal{G}}_c$ and, therewith, the individual matrix currents $\check{\mathcal{I}}_n \equiv [\check{\mathcal{M}}_n, \check{\mathcal{G}}_c]$ between terminal $n \in \{S1, S2, F1, F2, \mathrm{Leak}\}$ and the central node [1].

The superconducting contacts are characterized by the BCS bulk Green functions $(2/G_S)\check{\mathcal{M}}_{S\alpha} = \check{\mathcal{G}}_{S\alpha}$ with G_S being the conductance of the corresponding connector to the central node, and $\alpha = 1, 2$. Its retarded/advanced component reads in the spinor basis $\{\Psi_\uparrow^\dagger, \Psi_\downarrow^\dagger, \Psi_\downarrow, -\Psi_\uparrow\}$ [50, 100]

$$\hat{\mathcal{G}}_{S\alpha}^{R,A} = \frac{\pm\,\mathrm{sgn}(\epsilon)}{\sqrt{(\epsilon \pm i\delta)^2 - |\Delta_\alpha|^2}} \begin{pmatrix} \pm i\Gamma + \epsilon & \Delta_\alpha \\ -\Delta_\alpha^* & \mp i\Gamma - \epsilon \end{pmatrix} \otimes \mathbb{1}_2, \tag{3.1}$$

whereby $\Delta_\alpha \equiv \Delta \exp[\pm i\varphi/2]$ denotes the superconducting gap with phase difference φ across the junction. A small imaginary component in the denominator of Eq. (3.1) (here $\delta = 10^{-3}\Delta$) accounts for a finite lifetime $\hbar/\delta$ of the quasiparticles, with energy ϵ, smearing out the superconducting gap [142]. The ferromagnetic contacts are governed

[1]The symbol 0_n ($\mathbb{1}_n$) labels the zero (identity) matrix in $n \times n$ dimensions, and the superscript $\hat{.}$ ($\check{.}$) indicates the Nambu⊗spin (Keldysh⊗Nambu⊗spin) space of dimension 4×4 (8×8).

by $\check{\mathcal{M}}_{F\alpha} = (G_N/2)[(\mathbb{1}_8 + P\check{\kappa}_\alpha)\check{\mathcal{G}}_F - iG^\phi_\alpha \check{\kappa}_\alpha]$ with G_N (G^ϕ_α) being the normal (spin-mixing) conductances of the corresponding connectors. P denotes the contact spin polarization, and $\check{\kappa}_\alpha = \mathbb{1}_2 \otimes \underline{\sigma}_z \otimes (\boldsymbol{m_\alpha} \cdot \underline{\boldsymbol{\sigma}})$ is the spin matrix which is diagonal in Keldysh space. In the last expression, $\underline{\boldsymbol{\sigma}} = \{\underline{\sigma}_x, \underline{\sigma}_y, \underline{\sigma}_z\}$ labels the vector of Pauli matrices and $\boldsymbol{m_\alpha}$ is the magnetization vector corresponding to the ferromagnet $F\alpha$. Here, we fix $\boldsymbol{m_1} = (0, 0, 1)$ in z-direction and consider $\boldsymbol{m_2} = (\sin\theta, 0, \cos\theta)$ tilted by an arbitrary angle θ. We further consider fully polarized ferromagnetic contacts, i.e. $P = 1$. The retarded/advanced component of the ferromagnetic Green function is given by $\hat{\mathcal{G}}^{R,A}_F = \pm\underline{\sigma}_z \otimes \mathbb{1}_2$. Finally, the leakage terminal is described by $\check{\mathcal{M}}_{\text{Leak}} = -iG_S(\epsilon/\epsilon_{\text{Th}})\mathbb{1}_2 \otimes \underline{\sigma}_z \otimes \mathbb{1}_2$ with ϵ_{Th} being the Thouless energy.

The Keldysh component of the S and F Green functions follows from $\hat{\mathcal{G}}^K_n = \hat{\mathcal{G}}^R_n \hat{h}_n - \hat{h}_n \hat{\mathcal{G}}^A_n$ with the distribution function $\hat{h}_n = \text{diag}\big(\tanh[(\epsilon - eV_n)/2k_BT], \tanh[(\epsilon + eV_n)/2k_BT]\big) \otimes \mathbb{1}_2$. Hereafter, we assume all contacts at zero temperature, $k_BT = 0$, the superconductors at zero (reference) voltage, $V_{S1} = V_{S2} = 0$, equally biased ferromagnetic contacts, $V \equiv V_{F1} = V_{F2}$, and equal conductances $G_S = G_N$.

The Keldysh component of the matrix currents $\check{\mathcal{I}}_n$ leads to the charge currents

$$I_n = \frac{1}{8e}\int_{-\infty}^{\infty} d\epsilon \, \text{tr}\left[(\underline{\sigma}_z \otimes \mathbb{1}_2)\hat{\mathcal{I}}^K_n(\epsilon)\right], \tag{3.2}$$

and the z-polarized spin currents

$$I^z_n = \frac{\hbar}{16e^2}\int_{-\infty}^{\infty} d\epsilon \, \text{tr}\left[(\mathbb{1}_2 \otimes \underline{\sigma}_z)\hat{\mathcal{I}}^K_n(\epsilon)\right] \tag{3.3}$$

[75, 143–145], see Fig. A.4 for more technical details. In particular, we will analyze the charge, $I_X = I_{X1} - I_{X2}$, and spin net current, $I^z_X = I^z_{X1} - I^z_{X2}$ between the superconductors/ferromagnets ($X = S, F$). The matrix elements $\underline{f} = \langle\Psi_s|\hat{\mathcal{G}}^K_c|\Psi_{s'}\rangle$

$$\underline{f} = \begin{pmatrix} f_{ss'} & -f_{ss} \\ f_{s's'} & -f_{s's} \end{pmatrix}, \tag{3.4}$$

with $s, s' =\uparrow, \downarrow$ contain the spectral information about spin-pair correlations in nonequilibrium. Here, we consider the integrated quantities over positive energies $\epsilon > 0$ (triplet correlations are odd in ϵ), to quantify singlet, $F_S = \int d\epsilon \; (f_{\uparrow\downarrow} - f_{\downarrow\uparrow})/\sqrt{2}$, mixed-spin triplet, $F_{T0} = \int d\epsilon \; (f_{\uparrow\downarrow} + f_{\downarrow\uparrow})/\sqrt{2}$, and equal-spin triplet correlations $F_{Ts} = \int d\epsilon \; f_{ss}$.

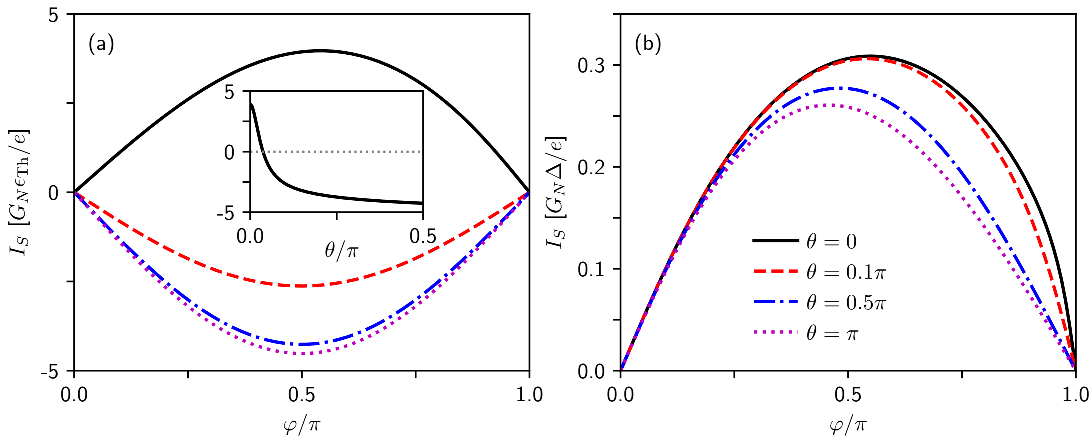

Figure 3.3: The net supercurrent I_S dependence on the relative magnetization angle from parallel ($\theta = 0$) to antiparallel ($\theta = \pi$) alignment. (a) and (b) are calculated fo the large-island $\epsilon_{\mathrm{Th}} \ll \Delta$, and small-island $\epsilon_{\mathrm{Th}} \gg \Delta$ regimes, respectively. The spin-mixing is set to zero $G_1^\phi = G_2^\phi = 0$, and we have considered the voltages exceeding the superconducting gap, $V \gtrsim \Delta/e$. The inset in (a), plotted for $\varphi = \pi/2$, indicates the sharp 0–π-transitions of the CPR appearing for very low θ.

3.3 Scaling

Before analyzing the interplay between Cooper-pair and quasiparticle transport, let us recall that in equilibrium, only a Josephson current $I_S^{\mathrm{eq}} = I_C \sin\varphi$ may flow between both s-wave superconductors, which has a purely sinusoidal CPR—all other currents require quasiparticle excitations. Depending on the effective size L of the central node, the Josephson current scales in the diffusive regime for $\epsilon_{\mathrm{Th}} \ll \Delta$ (large-island) with the Thouless energy $\epsilon_{\mathrm{Th}} \equiv \hbar D/L^2$, where D is the diffusion constant [see black solid line in Fig. 3.2(a)]. For $\epsilon_{\mathrm{Th}} \gg \Delta$ (small-island), however, it is characterized by the superconducting gap Δ [121]. While the superconducting condensate is in equilibrium entirely formed by spin-singlet Cooper-pairs, finite voltages may cause triplet correlations in the system. They can lead for voltages below the gap [dashed line in Fig. 3.2(a)] to a reduction of the Josephson current—here, we consider parallel collinear magnetization. For voltages above the gap and intermediate values of the Thouless energy ϵ_{Th}, such triplet correlations can even induce current reversals in the Josephson current I_S. In Fig. 3.2(a), showing the modulus of I_S on a double logarithmic scale, these zero-crossings (at which the logarithm diverges) result in the two sharp dips of the dash-dotted curve. Also the corresponding critical current $I_C = \max_\varphi I_S$ [purple line in Fig. 3.2(a)] indicates with the kinks the presence of

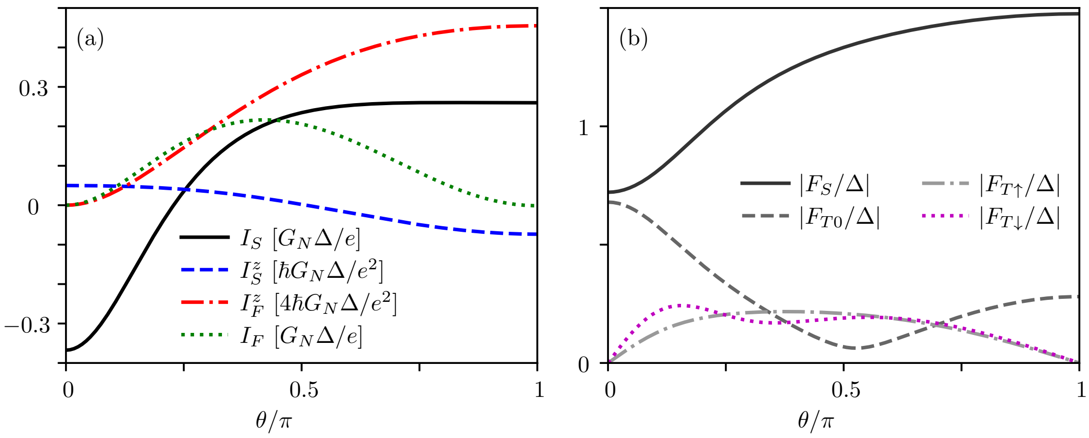

Figure 3.4: (a) Net charge I_S, I_F, and net spin currents I_S^z, I_F^z, as a function of θ for $\epsilon_{\mathrm{Th}} \gg \Delta$, $\varphi = \pi/2$, $V = 2\Delta/e$, $G_1^\phi = 0$, and $G_2^\phi = 2G_N$. (b) Modulus of the spin-singlet, F_S, and the spin-triplet pairing functions F_{T0}, $F_{T\uparrow}$, $F_{T\downarrow}$. Notice, the finite spin-mixing conductance G_2^ϕ induces a net charge current I_F (left panel) following the curve progression of the equal-spin triplet correlations, thus, making them experimentally attainable by standard current measurements [see Fig. 3.1(c) for $\theta \approx 0.3\pi$, where $|F_{T\downarrow}| < |F_{T\uparrow}|$].

triplet correlations.

For Thouless energies much smaller (larger) than the superconducting gap, the corresponding net supercurrent always stays positive and is dominated by spin-singlet correlations. We show in the following that applied voltages in combination with non-parallel magnetization, $\theta \neq 0$, can induce triplet correlations and transitions in the CPR *also* in the regimes $\epsilon_{\mathrm{Th}} \ll \Delta$ and $\epsilon_{\mathrm{Th}} \gg \Delta$.

3.4 Phase transitions and spin current control

First, let us consider the large-island regime, $\epsilon_{\mathrm{Th}} \ll \Delta$, where the Cooper-pair transport is characterized by the Thouless energy, i.e., $I_S, I_S^z \propto \epsilon_{\mathrm{Th}}$. While the net current between both superconductors follows in equilibrium the usual sinusoidal CPR, $I_S \propto \sin\varphi$, non-equilibrium in combination with non-collinear magnetization, $0 < \theta < \pi$, can induce 0–π-transitions for voltages $V > 0$ below the gap Δ, due to mixed-spin triplet correlations [solid black line in Fig. 3.2(b)]. Above the gap, quasiparticle transport sets in, and I_S saturates. In this regime, a finite net spin current $I_F^z \propto V$ (red dot-dashed line) emerges between the ferromagnetic contacts for a nonzero magnetization angle, irrespective of φ. For the chosen symmetric configuration, however, no corresponding net charge current I_F flows.

A special feature of our setup is the occurrence of a finite net spin current I_S^z between both superconductors (blue dashed line) for voltages $V \gtrsim \Delta/e$. This effect is maximal for parallel magnetization, $\theta = 0$, as can be seen in the inset, for which *only* mixed-triplet and singlet correlations are present. It vanishes for antiparallel orientation, $\theta = \pi$, where *no* triplet correlations arise, and when the Josephson phase φ is a multiple of π. Notice that I_S^z is antisymmetric in φ, and I_S^z as well as I_F^z are symmetric in θ. While voltages V above the gap can trigger net spin currents I_F^z and I_S^z, the magnetization angle θ can control their ratio (see inset), making the proposed setup, thus, attractive for future applications. Similarily, θ can control the 0–π-transition of the net supercurrent above the gap, as depicted in Fig. 3.3(a).

3.5 Spin-mixing induced charge current

Let us now turn to the small-island regime, $\epsilon_{\mathrm{Th}} \gg \Delta$, where losses of superconducting coherences become irrelevant, and the Josephson transport is characterized by the gap energy, i.e. $I_S, I_S^z \propto \Delta$. Here, we find that non-antiparallel magnetization $\theta \neq \pi$ can, indeed, induce triplet correlations for sufficiently large voltages, which results in an asymmetric sinusoidal CPR. However, it *cannot* induce 0–π-transitions [see Fig. 3.3(b)]. To cure this circumstance, we consider in the following finite spin-mixing for voltages above the gap. Indeed, figure 3.4(a) indicates that the system is for parallel magnetization, $\theta = 0$, but finite spin-mixing $G_2^\phi > 0$ in a π-phase featuring a negative net supercurrent I_S (solid black line). Roughly at $\theta = \pi/4$, the CPR undergoes a π–0-transition. Similar to the large-island regime [inset of Fig. 3.2(b)], a net spin current I_F^z may flow between the ferromagnets [red dot-dashed line in Fig. 3.4(a)] which is essentially unaffected by spin-mixing. The net spin current I_S^z between the superconductors (dashed blue line), on the contrary, is apart from $\theta = 0$ modified by spin-mixing, and features a current reversal about $\theta = \pi/2$.

Remarkable, the non-collinear magnetization in this system gives rise to equal-spin triplet correlations $|F_{T\uparrow}|$ and $|F_{T\downarrow}|$, see Fig. 3.4(b). A distinctive feature, however, is that these equal-spin triplet correlations can induce a finite net charge current I_F into the ferromagnets [dotted green line in Fig. 3.4(a)] for asymmetric spin-mixing, $G_1^\phi \neq G_2^\phi$. Where this feature is attributed to the creation of an imbalance in the ferromagnetic spin channels, see Fig. 3.1(c). This effect also persists for vanishing Josephson phase φ. Under a mutual exchange of the spin-mixing conductances ($G_1^\phi \leftrightarrow G_2^\phi$), the net charge current I_F just inverts. An experimentally measurable charge current I_F also serves in the large-island regime as a signature of equal-spin triplet correlations. It features in this regime a similar curve progression, but scales instead with ϵ_{Th}.

3.6 Conclusions

Spin-dependent quasiparticle and Cooper-pair transport have been analyzed in a proximity-coupled multi-terminal S/F-heterostructure in non-equilibrium. We have shown that 0–π-transitions can be induced in the CPR by biasing the ferromagnetic contacts and bearing non-collinear magnetic moments, as long as the loss of superconducting coherences is large, $\epsilon_{\mathrm{Th}} \ll \Delta$. In this limit, voltages exceeding the superconducting gap, $V \gtrsim \Delta/e$, trigger net spin currents into the ferromagnets/superconductors, which can be controlled by the relative magnetization angle θ. The small-island regime, however, requires additionally finite spin-mixing to induce CPR. The considered heterostructure qualifies as an ideal platform for the generation of triplet correlations of different spin projection, and as a voltage- and phase-controlled switch for spin and electron currents. In particular, it constitutes a minimal setup for generating, controlling, and detecting equal-spin triplet correlations by current measurements.

Chapter 4

Spin-flip enhanced thermoelectricity in S/F bilayers

The purpose of this chapter is to study the effects of spin-splitting and spin-flip scattering in a superconductor on the thermoelectric properties of a tunneling contact to a metallic ferromagnet using the Green function method. A giant thermopower has been theoretically predicted and experimentally observed in such structures. This is attributed to the spin-dependent particle-hole asymmetry in the tunneling density of states in the superconductor/ferromagnet heterostructure.

In particular, we evaluate the superconductor density of states and thermopower for a range of temperature, Zeeman-splitting, and spin-flip scattering. In contrast to the naive expectation based on the negative effect of spin-flip scattering on Cooper pairing, we find that the spin-flip scattering strongly enhances the thermoelectric performance of the system in the low-field and low-temperature regime due to a complex interplay between the charge and spin conductances caused by the softening of the spin-dependent superconducting gaps. The maximal value of the thermopower exceeds k_B/e by a factor of ≈ 5 and has a nonmonotonic dependence on spin-splitting and spin-flip rate [60].

4.1 Introduction

The field of thermoelectrics has drawn an enormous amount of interest in recent years, driven by the goal of recovering unused heat and converting it into usable electricity. The thermoelectric (TE) or Seebeck effect refers to the conversion of a temperature difference (ΔT) across a system into the usable electrical potential (V). The effect is quantified in terms of the so-called Seebeck coefficient S as $S = -V/\Delta T$. The efficiency of a TE device is determined by the type of materials used in making the device. Thus, a major focus of efforts towards addressing today's energy challenge is to find more efficient TE structures [146].

In a conductor, an applied temperature gradient drives the flow of electrons and holes. These two contributions are in general unequal due to the density of states (DOS) variations vs. quasiparticle energy giving rise to a net charge current. Under open circuit conditions, the requirement of no net current flow leads to a charge buildup that appears as a voltage called thermopower. However, the variation in the DOS is typically very small and, consequently, the TE response is negligibly low. Hence, achieving a strong variation in the DOS near the Fermi energy is the key to enhancing TE effects in a system.

A superconductor (S) with spin-splitting breaks the spin-dependent electron-hole symmetry. Its interface with a ferromagnet (F) results in a spin-polarized conductance leading to a large Seebeck effect. Following this principle, large spin-dependent TE effects have been theoretically proposed [48–52, 100, 147, 148] and experimentally demonstrated [53, 54, 102] in spin-split and -polarized superconducting tunnel contacts [118]. Furthermore, TE properties in a variety of systems have been investigated [149–151]. In most of the cases, the Zeeman-split superconducting heterostructures exhibiting sizeable TE effects require a large externally applied magnetic field [53, 152]. A recent experiment [102] reported a large Seebeck coefficient employing the spin-splitting field provided by the proximity to a ferromagnetic insulator, instead of an applied magnetic field. The spin-orbit interaction has been found to reduce the TE performance of such structures [57]. The usability of these devices as thermal detectors has also been discussed [153].

Similar to the influence of the spin-orbit interaction [57], spin-flip scattering may be expected to reduce the TE performance of such devices. This is because spin-flip scattering is known to have a detrimental effect on superconductivity via breaking of the Cooper pairs resulting in a reduction of the gap. One may assume a different perspective and argue that the reduction of the superconducting gap might enable quasiparticles at lower energies to contribute towards the Seebeck effect, thereby enhancing it. Thus, competing mechanisms are in play, and a clear winner may not be ascertained apriori. Earlier works have investigated the influence of spin-flip on the critical current [154] and critical temperature [154, 155], the spatial and energy dependences of the DOS [20, 88, 156, 157], the anomalous Green function [92], and quasiparticles distribution [158, 159] in a spin-split S.

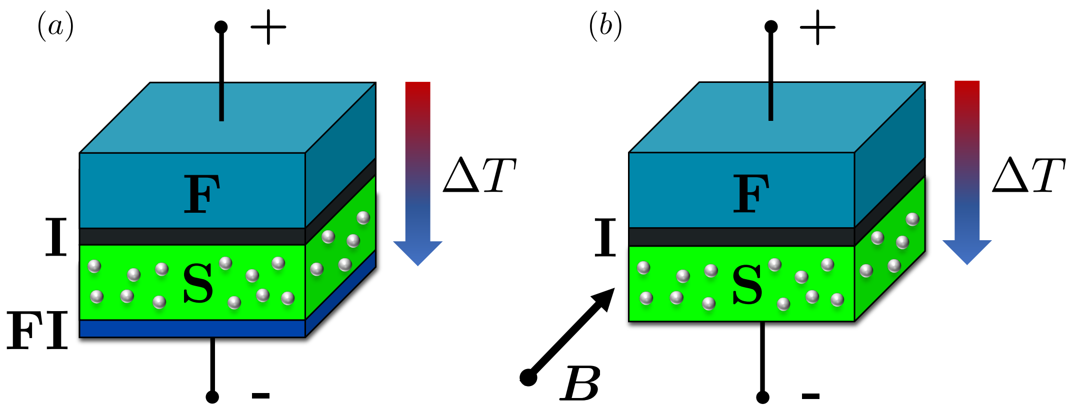

Figure 4.1: Schematic illustration of the investigated thin-film heterostructure. The structure consists of a ferromagnet (top region in light-blue) coupled to a superconductor (the bottom region in green) through a thin insulating barrier (the black region in the center). The quasiparticle DOS in the superconductor is modified by the presence of a spin-splitting field which may be realized via either (a) proximity with a ferromagnetic insulator (FI) film (dark-blue region), or (b) an applied external magnetic field ($\boldsymbol{B}$). ΔT denotes the temperature gradient through the junction. The presence of magnetic impurities inside the superconductor or scattering at its interface with the magnetic insulator results in spin-flip processes.

In this chapter, we theoretically address the effect of spin-flip scattering on the Seebeck coefficient in such S/F bilayers. We find that spin-flip enhances the TE response at low values of the spin-splitting. We incorporate spin-flip scattering and incoherent broadening self-consistently within the quasi-classical description of S [72, 88]. The spin-flip scattering shifts the peaks of the DOS toward zero-energy [see Fig. 4.2] thus reducing the gap and eventually leading to gapless superconductivity. We find that spin-flip scattering suppresses (enhances) the Seebeck coefficient at large (low) spin-splitting of the DOS. We find that the maximal Seebeck coefficient is relatively insensitive to the spin-flip rate, while the value of spin-splitting at which this maximum is achieved is reduced by increasing the spin-flip rate. Thus, for weak spin-flip scattering, its major effect is to decrease the spin-splitting required to achieve a large Seebeck coefficient, corresponding to the reduction of the superconducting gap. However, the TE response is substantial even in the gapless regime and starts to diminish rapidly as an increasing spin-flip rate destroys the superconductivity. We find an analogous behavior of the figure of merit ZT.

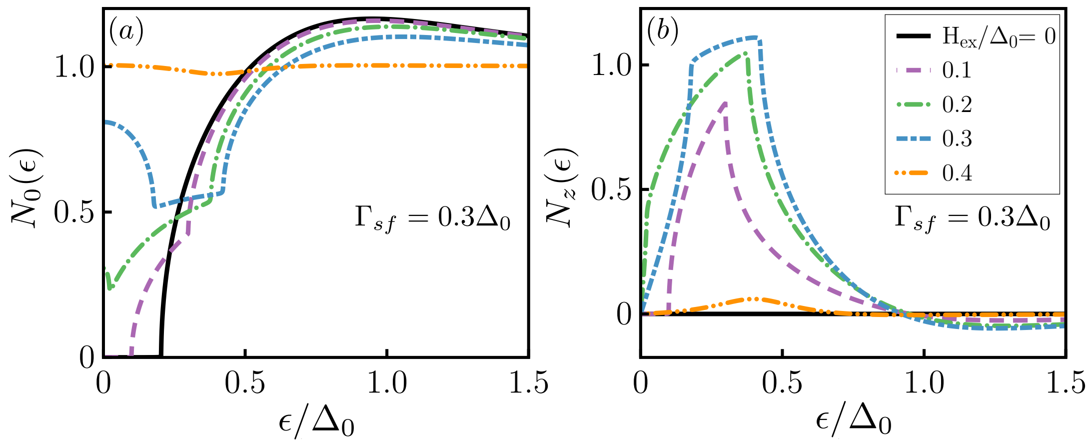

Figure 4.2: Self-consistently determined total [$N_0(\epsilon)$ (left panel)] and spin-polarized [$N_z(\epsilon)$ (right panel)] quasiparticle DOS. We employ $\Gamma_{sf} = 0.3\Delta_0$, $k_BT/\Delta_0 = 0.1$ and incoherent broadening parameter $\delta = 10^{-3}\Delta_0$. Δ_0 is the superconducting pair potential at $T = 0$, $H_{ex} = 0$ and $\Gamma_{sf} = 0$.

We will organize this chapter as follows. In Sec. 4.2, we outline the model and detailed theoretical framework employed to describe the TE effects in an S/F bilayer. In the following Sec. 4.3, we present and discuss our findings regarding the dependence of self-consistently calculated total and spin-polarized DOS (4.3.1), and Seebeck coefficient (4.3.2) on the spin-flip scattering rate. We conclude by summarising our main results in Sec. 4.4.

4.2 Model

We consider a ferromagnetic metal (F) coupled via a spin-polarized tunneling contact to a spin-split superconductor (S) layer. A sketch of the device under consideration has been shown in Fig. 4.1. The spin-splitting in S may be caused by proximity effect with a ferromagnetic insulator [79, 160–163] [Fig. 4.1(a)] or an applied magnetic field ($\boldsymbol{B}$) [152, 164] [Fig. 4.1(b)]. The spin-splitting field is oriented collinear to the magnetization direction of F. The thickness of the superconducting film is assumed smaller than the superconducting coherence length. To not affect each others equilibrium properties, the coupling between the two layers is assumed to be weak enough. The exchange-splitting field H_{ex} in the S layer acts on the spins of the electrons and breaks the particle-hole symmetry in the spin-resolved DOS. Considering the drives to be small, one can write TE coefficients in terms of the linear-response matrices.

Adopting the notation of Ref. [51], we have for the charge I^q and energy I^ϵ currents

$$\begin{pmatrix} I^q \\ I^\epsilon \end{pmatrix} = \begin{pmatrix} G & P\alpha \\ P\alpha & G_{th}T \end{pmatrix} \begin{pmatrix} V \\ \Delta T/T \end{pmatrix}, \tag{4.1}$$

and for spin I^s and 'spin-energy' I^{ϵ_s} currents flowing through the tunnel contact

$$\begin{pmatrix} I^s \\ I^{\epsilon_s} \end{pmatrix} = \begin{pmatrix} PG & \alpha \\ \alpha & PG_{th}T \end{pmatrix} \begin{pmatrix} V \\ \Delta T/T \end{pmatrix}, \tag{4.2}$$

where G, α, P, G_{th}, and T are the charge conductance, TE coefficient, spin-polarization of the tunnel contact, thermal conductance, and the absolute temperature of the system, respectively. The different coefficients may further be expressed in term of the spin-resolved DOS: [51]

$$G = G_T \int_{-\infty}^{\infty} d\epsilon N_0(\epsilon)\delta_T(\epsilon)\,, \tag{4.3a}$$

$$G_{th} = \frac{G_T}{e^2 T} \int_{-\infty}^{\infty} d\epsilon N_0(\epsilon)\epsilon^2\delta_T(\epsilon)\,, \tag{4.3b}$$

$$\alpha = \frac{G_T}{2e} \int_{-\infty}^{\infty} d\epsilon N_z(\epsilon)\epsilon\delta_T(\epsilon)\,. \tag{4.3c}$$

Here, G_T is the conductance of the junction in the normal state, ϵ is the quasiparticle energy, k_B is the Boltzmann constant, and $\delta_T(\epsilon) \equiv \left[4k_BT\cosh^2(\epsilon/2k_BT)\right]^{-1}$ being the normalized difference of the quasiparticle distributions functions of the F and S expanded with respect to a small temperature difference ΔT across the junction. The total and the spin-polarized DOS in S $N_0(\epsilon)$ and $N_z(\epsilon)$, respectively, are defined as

$$N_0(\epsilon) = \big(N_\uparrow(\epsilon) + N_\downarrow(\epsilon)\big)/2\,,$$
$$N_z(\epsilon) = N_\uparrow(\epsilon) - N_\downarrow(\epsilon)\,, \tag{4.4}$$

where $N_{\uparrow(\downarrow)}$ is the DOS for spin-up (down) quasiparticles in S. These are evaluated employing the quasi-classical Green function method, as described below.

The charge Seebeck coefficient describes the buildup of a TE voltage caused by a temperature gradient throughout a material as induced by the Seebeck effect when one opens the circuit and the net electric current vanishes such that $I^q = 0$ [51]. Employing Eqs. (4.3), the charge Seebeck coefficient is obtained as $S = -P\alpha/(GT)$. Thus, an increase of the TE coefficient and a reduction of the charge conductance are required to enhance the thermopower. It is crucial to note that a non-zero spin-

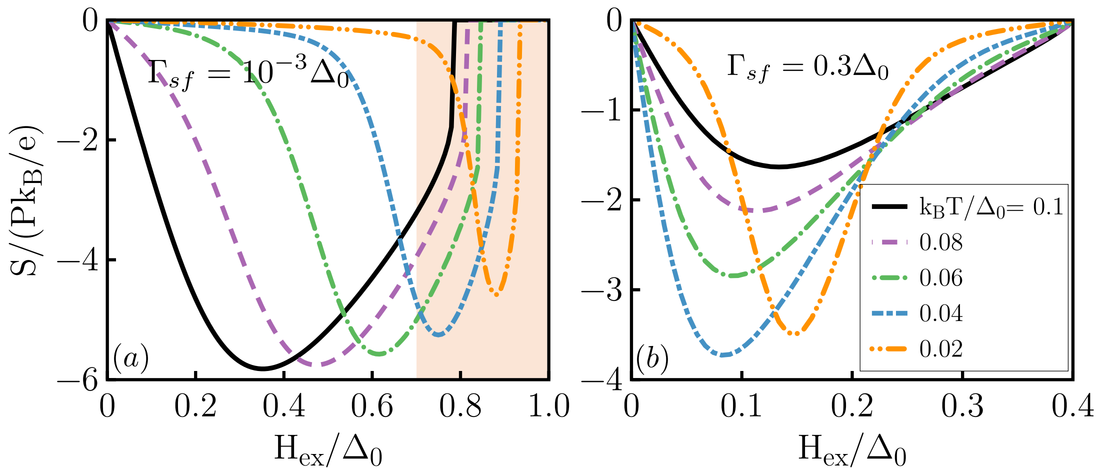

Figure 4.3: Self-consistently determined Seebeck coefficient S normalized to Pk_B/e vs. normalized spin-splitting field H_{ex}/Δ_0. The curves are shown for different temperatures $k_BT/\Delta_0 = 0.1$ (black solid line), 0.08 (purple dashed line), 0.06 (green dash-dot line), 0.04 (blue long dash-dash) and 0.02 (orange dash-dot-dot). $\Gamma_{sf}/\Delta_0 = 10^{-3}$ and 0.3 for left and right panels, respectively. We have set the incoherent broadening $\delta = 10^{-3}\Delta_0$. The Seebeck coefficient increases with increasing the magnetic field, approaches a maximum for a specific field and eventually drops to zero when the exchange field has destroyed the superconductivity. The shaded region depicts the exchange fields above the CC limit. Thus, the plotted results correspond to the metastable superconducting state, while the true stable state is normal and corresponds to vanishing thermoelectric response.

polarization of the interface is necessary to observe the TE phenomena in the present system. From Eqs. (4.3), we see that the charge conductance is proportional to the averaged tunneling DOS $N_0(\epsilon)$. However, the TE coefficient is a function of the spin-polarized DOS $N_z(\epsilon)$. In the system under investigation, the spin Seebeck effect[165–170] is determined by the same parameters as the charge Seebeck coefficient and refers to the creation of spin voltage due to a temperature difference. In other words, the spin thermopower is a way to quantify thermally produced spin voltages [149]. In the linear response regime, the efficiency of the device to produce TE power is quantified by the so-called dimensionless figure of merit ZT: $ZT = S^2GT/\tilde{G}_{th}$, where $\tilde{G}_{th} = G_{th} - (P\alpha)^2/GT$ is the energy conductance at zero current [51]. The larger the value of the ZT, the more efficient is the TE device [171].

The Usadel equation of motion [72] for a homogeneous spin-split superconductor is given by

$$\left[i\epsilon\underline{\tau}_3 - \underline{\Delta} - iH_{ex}\sigma\underline{\tau}_3 - \underline{\Sigma}_{sf}, \underline{g}^{R/A}\right] = 0_2\,, \tag{4.5}$$

where we have set $k_B = \hbar = 1$. $\underline{g}^{R/A}$ is the retarded (advanced) Green function and is obtained by replacing the quasiparticle energy ϵ by $[\epsilon \pm i\delta]$, respectively. The incoherent broadening δ or the so-called 'Dynes' parameter [172], parametrizes inelastic scattering and conserves the analytic structure of the Green functions [88]. Here, $\underline{\Delta}$ is the superconducting pair-potential matrix given as $\underline{\Delta} = \Delta\underline{\tau}_1$, with $\underline{\tau}_i$ the Pauli matrices in Nambu space. For a homogeneous superconducting state, Δ may be chosen as real and corresponds to the superconducting gap. The gap at zero temperature, exchange field, and spin-flip scattering rate is denoted by Δ_0. The exchange field H_{ex} acts on the electron spins and splits the DOS for spin-up and spin-down electrons [24–26]. $\sigma = \{\pm 1\}$ is a spin label referring to the up and down orientations of the spin $(\uparrow, \downarrow)$ with respect to the external spin-splitting field. Since we neglect inhomogeneities, gradient-terms do not appear in Eq. (4.5). The self-energy corresponding to spin-flip scattering via magnetic impurities [25, 67, 86, 173, 174] corresponds to $\underline{\Sigma}_{sf} = (1/2\tau_{sf})\underline{\tau}_3\underline{g}^{R/A}\underline{\tau}_3$. Alternatively, such a self-energy can originate from a magnetic interface with strong spin-dependent scattering [71, 83, 175, 176] thereby influencing the DOS [177, 178]. The spin-flip scattering time τ_{sf} is the average time between changes of the spin state of an electron [67].

We employ the following ϑ parametrization for the quasi-classical Green function matrix

$$\underline{g}^R_\pm = \cos\vartheta_\pm\underline{\tau}_3 + \sin\vartheta_\pm\underline{\tau}_1 = \begin{pmatrix} \cos\vartheta_\pm & \sin\vartheta_\pm \\ \sin\vartheta_\pm & -\cos\vartheta_\pm \end{pmatrix}, \tag{4.6}$$

which automatically ensures the normalization $\underline{g}^R_\pm \underline{g}^R_\pm = \mathbb{1}_2$ [1]. In these expressions $\cos\vartheta_\pm$ and $\sin\vartheta_\pm$ are normal and anomalous Green functions, respectively, and $\vartheta_\pm$ is, in general, a complex quantity. Using equations (4.5) and (4.6), the equation of motion simplifies to a set of nonlinear equations for up and down-spins as

$$-2\big((\epsilon + i\delta) + \sigma H_{ex}\big)\sin\vartheta_\pm + 2i\Delta\cos\vartheta_\pm - i\Gamma_{sf}\sin 2\vartheta_\pm = 0\,, \tag{4.7}$$

with $\Gamma_{sf} = 1/\tau_{sf}$ being the spin-flip scattering rate. The superconducting pair potential is determined self-consistently from the imaginary component of the anomalous

[1] Here, the superscript $\underline{\ldots}$ indicates the Nambu (particle-hole) space of dimension 2×2.

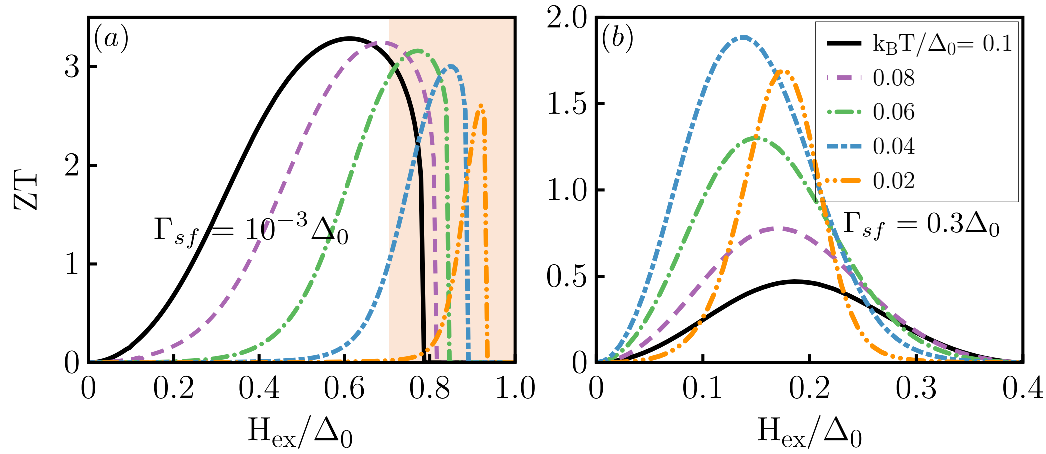

Figure 4.4: Self-consistently determined thermoelectric figure of merit ZT as a function of the normalized spin-splitting field H_{ex}/Δ_0. The parameter values employed are $\Gamma_{sf}/\Delta_0 = 10^{-3}$ and 0.3 for left and right panels, respectively. We have set polarization of the interface conductance to $P = 0.9$ and the incoherent broadening is $\delta = 10^{-3}\Delta_0$. The shaded region represents fields larger than the CC limit, and the curves displayed pertain to the metastable superconducting state.

part (pair amplitude) of the retarded Green function [see Fig. A.5 for more technical details]:

$$\Delta = \frac{\lambda}{2}\int_0^{\Omega_c} d\epsilon \tanh\frac{\epsilon}{2k_BT}\,\mathrm{Im}\left[\sin\vartheta_+ + \sin\vartheta_-\right], \tag{4.8}$$

where λ is the dimensionless Bardeen-Cooper-Schrieffer (BCS) [179] interaction constant. Ω_c is a suitably chosen cut-off. Once we have an expression for the self-consistent superconducting pair potential, we can calculate the DOS from the real part of the normal component of the retarded Green function. Within this framework, the DOS for spin up ($\uparrow$) and down ($\downarrow$) are

$$\begin{aligned} N_\uparrow(\epsilon) &= N(0)\,\mathrm{Re}[\cos\vartheta_+]\,, \\ N_\downarrow(\epsilon) &= N(0)\,\mathrm{Re}[\cos\vartheta_-]\,, \end{aligned} \tag{4.9}$$

with $N(0)$ the DOS in the normal state at the Fermi level.

Our approach outlined above determines the relevant properties of the superconducting state via the self-consistency equation and then employs the knowledge

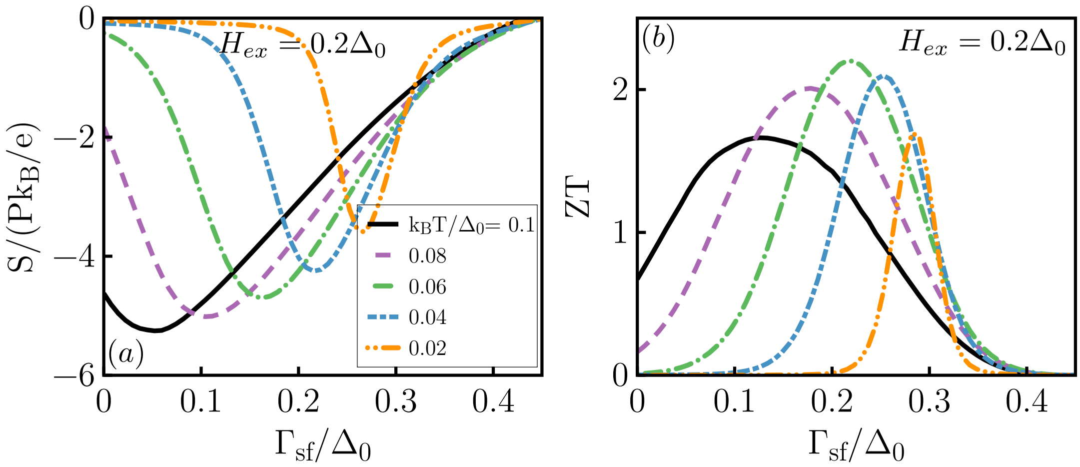

Figure 4.5: Spin-flip scattering rate (Γ_{sf}/Δ_0) dependence of the self-consistently determined normalized thermopower $S/(Pk_B/e)$ (left panel) and thermoelectric figure of merit ZT (right panel) for $H_{ex}/\Delta_0 = 0.2$ at different temperatures. Polarization of the interface conductance and incoherent broadening are assumed $P = 0.9$ and $\delta = 10^{-3}\Delta_0$.

of these equilibrium properties to evaluate the system response. However, this approach does not guarantee that the equilibrium state thus obtained is lowest in energy, and hence, the 'true' ground state. For a superconductor in a magnetic field, the superconducting ground state, disregarding spin-flip scattering, is stable, i.e., lowest in energy, only below the so-called 'Chandrasekhar-Clogston' (CC) limit $H_{ex} = \Delta_0/\sqrt{2}$ [180, 181], which is lower than the Pauli limit $H_{ex} = \Delta_0$ captured by our approach. Hence, our calculations pertain to the superconducting solution of the self-consistency equation, which becomes metastable in a certain range of exchange splitting, i.e., its energy is larger than the system being in normal state. The true ground state in this range is the normal state, the thermoelectric response in which is vanishingly small. We determine this range from the literature [57, 182] and mark it via shaded regions in Figs. 4.3, 4.4 and 4.6.

4.3 Results and discussions

4.3.1 Density of states

In Fig. 4.2, we show the total [$N_0(\epsilon)$] and spin-polarized [$N_z(\epsilon)$] DOS obtained within a self-consistent evaluation for different exchange fields and spin-flip scattering rates as described above. For the total DOS, the gap slowly closes as the spin-splitting is increased, and via the gapless state, it eventually results in the destruction of the superconducting state. The spin-polarized DOS vanishes for no spin-splitting as well as at-large spin-splittings when the superconducting state is destroyed. Hence, we see that the optimal situation for the TE coefficient α, corresponding to the largest $N_z(\epsilon)$ [Eqs. (4.3)], is achieved for a certain value of the spin-splitting.

4.3.2 Thermopower and figure of merit

Employing the obtained DOS and Eqs. (4.3), we evaluate the Seebeck coefficient and plot it vs. spin-splitting in Fig. 4.3. As expected from our earlier discussion on spin-polarized DOS, the Seebeck coefficient vanishes at zero and large spin-splittings yielding a maximum in between. We see that the value of spin-splitting at which the maximal Seebeck coefficient is achieved varies strongly with the temperature. Remarkably, in the absence of spin-flip scattering (left panel of Fig. 4.3) at the lowest temperatures, the Seebeck coefficient is drastically reduced due to the CC restriction on the field range.

Now comparing the cases of low and sizeable spin-flip rate in Fig. 4.3, we observe that a large spin-flip rate reduces the temperature sensitivity of the spin-splitting ($H_{\max}$) corresponding to a maximal Seebeck effect. Overall, putting the CC physics aside for the moment, $H_{\max}$ lowers with an increasing spin-flip rate, but this lowering does not have a one-to-one correspondence with the reduction of the superconducting gap, as evidenced by the relative insensitivity of $H_{\max}$ to the temperature. The TE performance of the device is particularly insensitive to the temperature for large spin-flip, as long as the system is not in the gapless state. Fig. 4.4 presents the analogous results for ZT corroborating the similar dependence of TE efficiency on the relevant physical variables. We emphasize that ZT values large than one can only be achieved in the low-temperature and low-exchange field regime by introducing a spin-flip rate, as shown in the lower right panel of Fig. 4.4. In the absence of spin-flip scattering, ZT is dramatically reduced at very low temperatures $T \lesssim 0.05\Delta_0$ due to the instability of the superconducting state in the CC regime. We now focus on the dependence of the TE response on the spin-flip rate. To this end, we plot the Seebeck coefficient (left panel) and ZT (right panel) vs. spin-flip rate at a fixed value of spin-splitting in Fig. 4.5. The Seebeck coefficient exhibits an initial enhancement with the spin-flip rate followed by an eventual reduction. This behavior may be understood in terms of spin-flip mediated lowering of the superconducting gap. The first-order effect of spin-flip in this regime is to reduce the superconducting gap, thereby shifting the quasiparticle

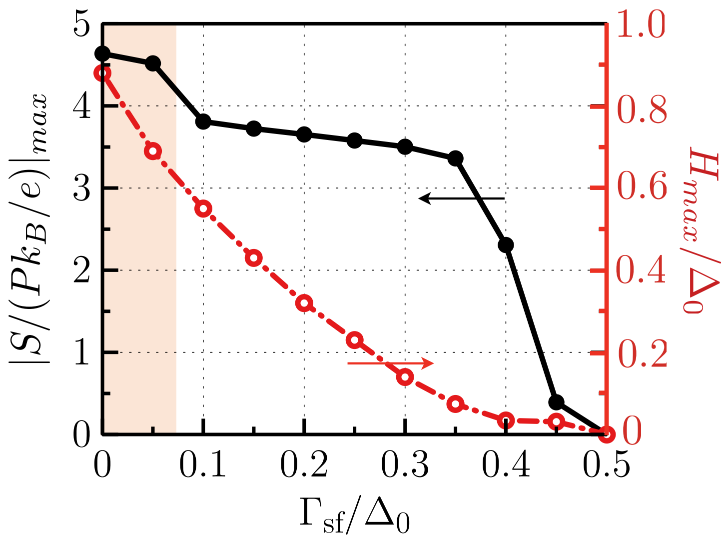

Figure 4.6: Maximal thermopower $|S/(Pk_B/e)|_{max}$ vs spin-flip rate Γ_{sf} at $k_BT/\Delta_0 = 0.02$ [the orange curves in Fig. 4.3]. For a small incoherent broadening $\delta = 10^{-3}\Delta_0$, the maximum Seebeck coefficient for medium spin-flip scattering rates is relatively insensitive to the spin-flip rate. The shaded region represents the destruction of the superconducting state due to CC physics. The upper spin-flip rate limit of the shaded region has not been calculated exactly and represents an approximate value. At low spin-flip rates, the true Seebeck-coefficient vanishes due to the required exchange field being above the CC limit. On the other hand, large spin-flip rates destroy superconductivity via the usual depairing channel.

dynamics to lower energies without significantly damaging the superconducting state. At more significant spin-flip rates, the Cooper pair depairing effect dominates the modification of the system TE response. ZT essentially mirrors the dependence of the Seebeck coefficient with minor differences.

We finally study the dependence of the maximum Seebeck coefficient (as a function of spin-splitting) on the spin-flip rate in Fig. 4.6. The maximum Seebeck coefficient is relatively insensitive to the spin-flip. However, the value of spin-splitting (not shown explicitly) at which this maximum value is achieved decreases monotonically with the spin-flip rate.

Based on the analysis presented above, we are now in a position to summarise the effect of spin-flip on the TE response of the system under investigation. At small values of the spin-flip scattering rate, its predominant effect is reducing the superconducting gap. This provides quasiparticles at lower energies and results in the enhancement of Seebeck effect at low values of the spin-splitting exchange field. After the superconductor enters the gapless regime, there is no such gain in further increasing the spin-flip. On the contrary, the depairing effect only tends to make the DOS flatter resulting in a decreasing TE response with increasing spin-flip. We note that spin-orbit scattering will mix the spin-split energy bands [182] and hence reduce the overall TE performance. Thus, the optimal value of spin-flip is around the point when the superconductor is about to enter the gapless regime.

Moreover, by studying the effect of a more sizeable incoherent broadening parameter in appendix B for the same parameter space as here, we find that it leads to a drastic reduction of the achievable maximum thermoelectric performance of our system.

4.4 Conclusions

We have investigated the thermoelectric response of a tunnel structure formed by a thin insulating layer sandwiched between a spin-split superconducting film and a ferromagnetic metal. We take into account the important roles of the spin-flip scattering and Chandrasekhar-Clogston (CC) paramagnetic limit [180–182]. The relevant physical quantities have been obtained self-consistently within the quasi-classical Green function technique. We have shown that increasing the spin-flip scattering rate leads to a strong enhancement of the thermoelectric performance in the low-field and low-temperature regime. Specifically, the maximum of the thermopower exceeds k_B/e by a factor of ≈ 5 down to temperatures of the order of $k_BT/\Delta_0 = 0.02$ and has a nonmonotonic behavior with respect to the spin-splitting and the spin-flip scattering rate in the superconductor. We also predict a sizeable thermoelectric figure of merit of $ZT \gtrsim 1$ in the low-temperature regime. Although the spin-flip rate does affect the maximum of the thermopower only weakly, it shifts the peak from a higher to a lower field. This appears to be of crucial importance since we find that the high fields required in the absence of spin-flip scattering are larger than the CC limit, and thus do not support superconductivity. Our results constitute a promising prediction for reducing the necessary spin-splitting in these structures, which might also be useful to avoid other detrimental effects related to external magnetic fields, magnetization control, or the CC-limited ultra-low-temperature degradation of the TE response.

Chapter 5

Spin effects in proximity-coupled MI/S systems

Inspired by recent feats in exchange coupling antiferromagnets to an adjacent material, in this chapter we demonstrate the possibility of employing them for inducing spin-splitting in a superconductor, thereby avoiding the detrimental, parasitic effects of ferromagnets employed to this end.

We derive the Gor'kov equation for the matrix Green function in the superconducting layer, considering a microscopic model for its disordered interface with a two-sublattice magnetic insulator. We find that an antiferromagnetic insulator with effectively uncompensated interface induces a large, disorder-resistant spin-splitting in the adjacent superconductor. In addition, we find contributions to the self-energy stemming from the interfacial disorder. Within our model, these mimic impurity and spin-flip scattering, while another breaks the symmetries in particle-hole and spin spaces. The latter contribution, however, drops out in the quasi-classical approximation and thus, does not significantly affect the superconducting state [64].

5.1 Introduction

Conventional Bardeen-Cooper-Schrieffer (BCS) superconductors [179] are incompatible with magnetic interactions as the latter tend to break the Cooper pairing [183] between the opposite-spin electrons. Nevertheless, the so-called Pauli contribution, associated with the energy splitting of the two spin states, leads to interesting new phenomena when the spin-splitting is comparable to the 'unperturbed' superconducting gap [184]. These include spatially inhomogeneous order parameter in an otherwise homogeneous superconductor [29, 30], gapless superconductivity [185, 186], and a first-order phase transition between superconducting and normal states [180, 181], all of which have been experimentally observed [26, 187]. Furthermore, hybrids incorporating such spin-split superconductors were recently predicted [50, 51, 100], and found [53, 102], to exhibit large thermoelectric effects. The spin-splitting in the superconducting layer may be induced by a magnetic field or via exchange coupling to a magnetic layer [102, 177] and leads to intriguing transport properties reviewed in [57].

The success of 'exchange biasing' a ferromagnet (F) layer via its coupling to an adjacent antiferromagnet (AF) has been instrumental in the contemporary memory technology [61, 62, 188]. A simplified picture of exchange biasing in F/AF bilayers requires the AF interface to be uncompensated, i.e., possess finite surface magnetization [61–63]. Several theoretical models [63], most of which assume the AF surface to be uncompensated, have been employed to understand the experiments. Recent progress in surface characterization methods [189] and epitaxial sample growth [190] has enabled to resolve [190, 191] several previously open questions [61]. Numerous experiments [192–200] have succeeded in direct observation and quantification of uncompensated spins at interfaces thereby improving the understanding of their role in exchange bias and the control of the effect.

Recently, the presence of surface magnetization, stemming from broken translational symmetry at interfaces, in magnetoelectric AFs has been predicted [201]. This has also been observed experimentally and exploited in achieving electrically switchable exchange bias [202] and magnetic memory [203] using α-Cr_2O_3. Furthermore, uncompensated AF interfaces have been theoretically predicted to amplify the transfer of magnonic spin from a magnetic insulator to an adjacent non-magnetic conductor [204, 205].

In this chapter, we suggest employing insulating AFs, with their uncompensated surfaces, to induce an effective exchange field in an adjacent superconducting layer. To the best of our knowledge, only Fs have been employed to this end so far. AFs offer several advantages over Fs in this regard [206–208]. These include the minimization of stray magnetic fields, the possibility of electrical tunability [202, 203, 209], avoiding parasitic negative effects of low-energy magnon excitations [58, 59] and so on. The proximity effect due to metallic antiferromagnets has been investigated experimentally [210] and theoretically [211]. Antiferromagnetically ordered impurity chains may also give rise to Majorana state [212].

Considering a two-sublattice magnetic insulator (MI)/superconductor (S) bilayer structure, we derive the Gor'kov equation for the matrix Green function in S from a microscopic Hamiltonian including the interface [67]. Our model for MI encompasses the full range of single-domain magnets from ferro- to antiferro- via ferrimagnets [59, 204]. We explicitly include interfacial disorder in our model and find that the induced exchange field is resistant to it, within the Born approximation. We find that the effect of the MI layer is captured by a self-energy, which includes interfacial disorder-mediated terms, in addition to the spin-splitting term. The latter is found to be large for an uncompensated interface with an AF. For the system considered here, with the Hamiltonian diagonal in spin space [1], the interfacial disorder-mediated terms take a form identical to spin-independent impurity and spin-flip scattering. A third disorder contribution breaks the particle-hole and spin symmetries, but predominantly renormalizes the normal state properties leaving the superconducting state essentially unaffected.

This chapter is organized as follows. In Sec. 5.2, we illustrate the possible interface microstructures for bilayers under investigation and introduce the model Hamiltonian. Sec. 5.3 derives and solves the Gor'kov equation, and the interfacial self-energy for the matrix Green function is evaluated in Sec. 5.4. We discuss our findings in Sec. 5.5, and finally, we draw our conclusions in Sec.5.6.

5.2 Model and Hamiltonian

We consider a MI/S bilayer (Fig. (5.1)) with the S thickness d_{S} much smaller than the superconducting coherence length. MI is comprised of a single-domain two-sublattice magnetic insulator where sublattice magnetizations are considered static and collinear to the z-axis. We consider S to be a BCS superconductor in the weak coupling regime such that the Hamiltonian in the grand canonical ensemble reads [67]:

$$\tilde{\mathcal{H}} = \int d^3r \left[\sum_{\alpha} \tilde{\psi}^{\dagger}_{\alpha}(\boldsymbol{r}) \left[-\partial^2 + V_s(\boldsymbol{r}) \left(\delta_{\alpha\uparrow} - \delta_{\alpha\downarrow} \right) \right] \tilde{\psi}_{\alpha}(\boldsymbol{r}) + \sum_{\alpha,\beta} \frac{g}{2} \tilde{\psi}^{\dagger}_{\beta}(\boldsymbol{r}) \tilde{\psi}^{\dagger}_{\alpha}(\boldsymbol{r}) \tilde{\psi}_{\alpha}(\boldsymbol{r}) \tilde{\psi}_{\beta}(\boldsymbol{r}) \right] \tag{5.1}$$

here, $\tilde{\psi}_{\alpha}(\boldsymbol{r})$ is the electron annihilation operator for z-projected spin α at position $\boldsymbol{r}$, $\partial^2 \equiv \nabla^2/2m + \mu - V_i(\boldsymbol{r})$, μ is the chemical potential, m is electron effective mass, $V_i(\boldsymbol{r})[V_s(\boldsymbol{r})]$ represents the spin-independent (dependent) potential energy, g (< 0)

[1] Contributions, non-diagonal in spin space, to the Hamiltonian may arise owing to different physical origins. These include inhomogeneous magnetization, coupling to multiple magnets with non-collinear magnetizations, and unconventional pairing between same-spin electrons in the superconductor.

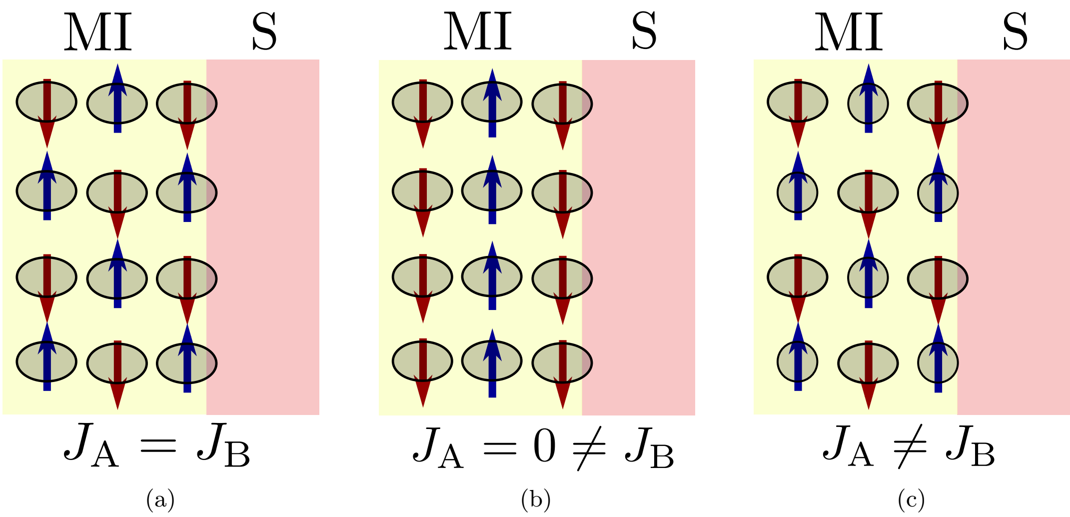

Figure 5.1: Possible interface microstructures for magnetic insulator (MI)/superconductor (S) bilayers. Sublattices A and B are depicted in blue and red, respectively. Cases (a) and (b), respectively, represent antiferromagnets with compensated and fully uncompensated interfaces with S. Case (c) depicts a ferrimagnet with a compensated interface. In this case, the symmetry of interfacial coupling between S and the two sublattices is broken [213, 214] by, for example, different wavefunction clouds associated with the localized moments that comprise the sublattice. Interfacial disorder accounted for in our model, is not depicted explicitly here.

parametrizes the electron-electron attraction, and we have set $\hbar$ to 1. All operators are in the Heisenberg picture and are decorated by a $.\tilde{.}.$ above. The interface with MI results in the potential energy terms $V_{i,s}(\boldsymbol{r})$. For simplicity, we do not explicitly include bulk contributions to the potential energy here.

The MI/conductor interface is typically modeled as an effective exchange interaction between the spin densities on the two sides [204]:

$$\tilde{\mathcal{H}}_{\text{int}} = -\int d^2 s \sum_{\mathcal{G}=\text{A,B}} \left[J_{\mathcal{G}} \tilde{\boldsymbol{S}}_{\mathcal{G}}(\boldsymbol{s}) \cdot \tilde{\boldsymbol{S}}(\boldsymbol{s}) \right]. \tag{5.2}$$

Here, $\boldsymbol{s}$ is the two-dimensional position vector in the interfacial plane defined by $y = 0$, $\tilde{\boldsymbol{S}}$ is the electronic spin density operator in S, and $\tilde{\boldsymbol{S}}_{\text{A(B)}}$ is MI sublattice A (B) spin density operator. $J_{\text{A(B)}}$ parametrizes the exchange strength between the MI sublattice A (B) and the S electrons, and depends upon the details of the interface such as its microstructure (Fig. (5.1)). The magnetic spin densities are related to the corresponding magnetizations via the sublattice gyromagnetic ratios $\gamma_{\mathrm{A,B}}$, assumed negative, $\tilde{\boldsymbol{M}}_{\mathrm{A,B}} = -|\gamma_{\mathrm{A,B}}|\tilde{\boldsymbol{S}}_{\mathrm{A,B}}$. We consider sublattice A (B) to be saturated along positive (negative) z-direction with saturation magnetization $M_{\mathrm{A0(B0)}}$.

Augmenting the interfacial interaction above [Eq. (5.2)] with a spin-independent contribution and disorder, the net interfacial Hamiltonian may be expressed as:

$$\tilde{\mathcal{H}}_{\mathrm{int}} = \int d^3r \sum_{\alpha} \tilde{\psi}_{\alpha}^{\dagger}(\boldsymbol{r}) U(\boldsymbol{s}) \delta(y) \left[a + b\left(\delta_{\alpha\uparrow} - \delta_{\alpha\downarrow}\right)\right] \tilde{\psi}_{\alpha}(\boldsymbol{r}), \tag{5.3}$$

where a parametrizes the spin-independent contribution of the interfacial interaction and $b = J_{\mathrm{A}} M_{\mathrm{A0}}/2|\gamma_{\mathrm{A}}| - J_{\mathrm{B}} M_{\mathrm{B0}}/2|\gamma_{\mathrm{B}}|$. $U(\boldsymbol{s})$ accounts for the interfacial disorder which is modeled in a manner analogous to the treatment of impurities-mediated disorder in a bulk conductor [67, 215]:

$$U(\boldsymbol{s}) = 1 + \sum_{\boldsymbol{s}_i} u(\boldsymbol{s} - \boldsymbol{s}_i), \tag{5.4}$$

with $u(\boldsymbol{s} - \boldsymbol{s}_i)$ representing the fluctuation in potential energy associated with a 'disorder center' located at $\boldsymbol{s}_i$, and we assume $\int d^2s \ u(\boldsymbol{s}) = 0$. Employing Eq. (5.3), the potential energy contribution to the total Hamiltonian [Eq. (5.1)] corresponds to $V_{i[s]}(\boldsymbol{r}) = U(\boldsymbol{s})\delta(y)a[b]$.

5.3 Gor'kov equation

We now formulate the problem at hand in terms of imaginary-time Green functions in Nambu-spin space. Decorating four-dimensional entities (vectors and matrices) by a $\hat{\ldots}$ above and two-dimensional by a $\underline{\ldots}$ subscript, we define $\hat{\Psi}^{\dagger} \equiv \left[\tilde{\psi}_{\uparrow}^{\dagger}, \tilde{\psi}_{\downarrow}^{\dagger}, \tilde{\psi}_{\downarrow}, \tilde{\psi}_{\uparrow}\right]$. We further define the matrix, imaginary-time Green function as [67]:

$$\hat{G}(x_1, x_2) \equiv -\underline{\tau}_z \otimes \underline{\sigma}_0 \left\langle \mathrm{T}_{\tau} \hat{\Psi}(x_1) \hat{\Psi}^{\dagger}(x_2) \right\rangle = \begin{pmatrix} G_{\uparrow\uparrow} & G_{\uparrow\downarrow} & F_{\uparrow\downarrow} & F_{\uparrow\uparrow} \\ G_{\downarrow\uparrow} & G_{\downarrow\downarrow} & F_{\downarrow\downarrow} & F_{\downarrow\uparrow} \\ -\bar{F}_{\downarrow\uparrow} & -\bar{F}_{\downarrow\downarrow} & \bar{G}_{\downarrow\downarrow} & \bar{G}_{\downarrow\uparrow} \\ -\bar{F}_{\uparrow\uparrow} & -\bar{F}_{\uparrow\downarrow} & \bar{G}_{\uparrow\downarrow} & \bar{G}_{\uparrow\uparrow} \end{pmatrix}, \tag{5.5}$$

where $\tau = it$ is the imaginary time, $x_1 \equiv (\boldsymbol{r}_1, \tau_1)$, $\underline{\tau}_{0,x,y,z}$ and $\underline{\sigma}_{0,x,y,z}$ are the 2×2 identity and Pauli matrices in, respectively, Nambu and spin spaces, and the outer product is expanded as:

$$\underline{\tau}_z \otimes \underline{\sigma}_0 = \begin{pmatrix} \underline{\sigma}_0 & 0 \\ 0 & -\underline{\sigma}_0 \end{pmatrix}.$$

Employing Heisenberg equation of motion for $\tilde{\psi}_\alpha(x_1)$ with the Hamiltonian given by Eq. (5.1), we obtain the dynamical equation for $G_{\alpha\beta}(x_1, x_2)$:

$$\frac{\partial G_{\alpha\beta}(x_1,x_2)}{\partial \tau_1} = -\delta_{\alpha\beta}\delta(x_1 - x_2) + \left[\partial_1^2 - V_s(\boldsymbol{r}_1)\left(\delta_{\alpha\uparrow} - \delta_{\alpha\downarrow}\right)\right] G_{\alpha\beta}(x_1, x_2) \tag{5.6}$$

$$- i \sum_\gamma \Delta_{\alpha\gamma}(x_1)\bar{F}_{\gamma\beta}(x_1, x_2), \tag{5.7}$$

where $\Delta_{\alpha\beta}(x) \equiv i|g|F_{\alpha\beta}(x,x)$. In simplifying the four-point correlator above, we have employed Wick's theorem [2] and disregarded terms which lead to a mere renormalization of the chemical potential [67]. Dynamical equations for the other components of the matrix Green function can be derived in an analogous manner [67]. All these equations may be expressed as a single Gor'kov equation for the matrix Green function:

$$\hat{\mathcal{G}}^{-1}(x_1)\hat{G}(x_1, x_2) = \delta(x_1 - x_2)\underline{\tau}_0 \otimes \underline{\sigma}_0, \tag{5.8}$$

where

$$\hat{\mathcal{G}}^{-1}(x_1) = -\frac{\partial}{\partial \tau_1}\underline{\tau}_z \otimes \underline{\sigma}_0 + \partial_1^2 \underline{\tau}_0 \otimes \underline{\sigma}_0 - V_s(\boldsymbol{r}_1)\underline{\tau}_z \otimes \underline{\sigma}_z - \hat{\Delta}(\boldsymbol{r}_1).$$

For a homogeneous superconducting state, the pair potential matrix may be chosen as $\hat{\Delta}(\boldsymbol{r}) = -i\Delta\underline{\tau}_y \otimes \underline{\sigma}_z$ with real Δ [67, 88].

5.4 Interfacial self-energy

Since the Gor'kov equation can rarely be solved exactly, we resort to perturbation theory within the Green function method [215] and obtain the self-energy arising from the interfacial contribution to the Hamiltonian [Eq. (5.3)]. To this end, we express $\hat{\mathcal{G}}^{-1}(x_1) = \hat{\mathcal{G}}_0^{-1}(x_1) - \hat{\mathcal{H}}_{\text{int}}(x_1)$ as the sum of the clean superconducting layer plus the

[2] Strictly speaking, Wick's theorem is not valid here since our Hamiltonian contains terms fourth-order in the ladder operators. However, the application of Wick's theorem here is equivalent to the mean-field approximation made in the BCS theory. Please refer to the discussion by Kopnin for further details [67].

interfacial contribution, which assumes the form [using Eqs. (5.3) and (5.9)]:

$$\hat{\mathcal{H}}_{\text{int}}(x_1) = \; U(\boldsymbol{s}_1)\delta(y_1)\,[a\; \underline{\tau}_0 \otimes \underline{\sigma}_0 + b\; \underline{\tau}_z \otimes \underline{\sigma}_z] \equiv \; U(\boldsymbol{s}_1)\delta(y_1)\; \hat{t}. \tag{5.9}$$

The evaluation of the corresponding self-energy follows the method analogous to the case of impurities-mediated disorder in a bulk conductor [67, 215], and is detailed in the appendix C. Within this method, the so-called cross-diagram technique [67, 215], the following assumptions are made. (i) The perturbation is assumed small, thus making the Born approximation. (ii) We average over the positions $\boldsymbol{s}_i$ of the disorder centers. (iii) All diagrams with intersecting impurity scattering lines may be disregarded. (iv) We further neglect diagrams with more than two scattering events. In addition, we employ the quasi-classical approximation in treating the homogeneous superconducting state. With these assumptions, diagrams of all orders can be summed [67, 215] and we obtain the main result of this Letter:

$$\hat{\Sigma}_{\text{int}}(\omega_n, \boldsymbol{p}) = \frac{1}{d_S}\left[\hat{t} + N_{\text{dis}} \int \frac{d^3 p_1}{(2\pi)^3}\; |u(\boldsymbol{\kappa} - \boldsymbol{\kappa}_1)|^2\; \hat{t}\; \hat{G}(\omega_n, \boldsymbol{p}_1)\; \hat{t}\right], \tag{5.10}$$

where the result is expressed concisely in the frequency ω_n and momentum $\boldsymbol{p}$ representation [3]. Here, $u(\boldsymbol{\kappa}) \equiv \int d^2 s\; u(\boldsymbol{s})\; \exp(-i\boldsymbol{\kappa}\cdot\boldsymbol{s})$, N_{dis} is the areal density of disorder centers, $\boldsymbol{\kappa}$ is the in-plane component of the momentum $\boldsymbol{p}$, and d_S is the thickness of S layer assumed to be much smaller than the superconducting coherence length. The Green function for the proximity-coupled superconducting layer is given by $\hat{G}^{-1}(\omega_n, \boldsymbol{p}) = \hat{G}_0^{-1}(\omega_n, \boldsymbol{p}) - \hat{\Sigma}_{\text{int}}(\omega_n, \boldsymbol{p})$, in terms of the unperturbed Green function $\hat{G}_0(\omega_n, \boldsymbol{p})$ and the self-energy evaluated above.

5.5 Discussion

The self-energy [Eq. (5.10)], stemming from the interface with MI, comprises a contribution independent of, and thus resistant to, interfacial disorder and a term proportional to the areal density of disorder centers N_{dis}. Apart from a small renormalization of the chemical potential, the former contribution is simply the effective exchange field, $\propto b = J_{\text{A}} M_{\text{A0}}/2|\gamma_{\text{A}}| - J_{\text{B}} M_{\text{B0}}/2|\gamma_{\text{B}}|$, induced in S. Thus, an AF with uncompensated surface, for which $J_{\text{A}} \neq J_{\text{B}}$, $M_{\text{A0}} = M_{\text{B0}}$, and $\gamma_{\text{A}} = \gamma_{\text{B}}$, induces spin-splitting in the adjacent S layer.

[3] See appendix C for a detailed derivation of the interfacial self-energy, the conventions followed for expressing the Green functions in the frequency and momentum representation, and a discussion on some relevant materials along with the corresponding expected spin-splittings.

The interfacial disorder-mediated contribution to the self-energy can be further divided into three terms with the integrands in Eq. (5.10) respectively proportional to (i) $a^2\big(\underline{\tau}_0\otimes\underline{\sigma}_0\hat{G}\underline{\tau}_0\otimes\underline{\sigma}_0\big)$, (ii) $b^2\big(\underline{\tau}_z\otimes\underline{\sigma}_z\hat{G}\underline{\tau}_z\otimes\underline{\sigma}_z\big)$, and the third term (iii) $ab\big(\underline{\tau}_0\otimes\underline{\sigma}_0\hat{G}\underline{\tau}_z\otimes\underline{\sigma}_z + \underline{\tau}_z \otimes \underline{\sigma}_z\hat{G}\underline{\tau}_0 \otimes \underline{\sigma}_0\big)$. The term (i) looks like the self-energy due to non-magnetic impurities [67]. Assuming isotropic scattering, this contribution drops out of the superconducting gap as well as the Eilenberger equations for s-wave superconductors, in consistence with the Anderson theorem [216]. Assuming that $\hat{\mathcal{G}}^{-1}(x_1)$ is diagonal in spin space, which is the case here [Eq. (5.9)], the total matrix Green function is also diagonal in spin space. Taking this into consideration, term (ii) may be rewritten as $\propto \underline{\tau}_z\otimes\underline{\sigma}_0\hat{G}\underline{\tau}_z\otimes\underline{\sigma}_0$, which has the same form as the self-energy contribution due to spin-flip scattering via magnetic impurities [67]. The effect of such a term has been studied and is known to result in phenomena such as gapless superconductivity [186]. It also has consequences for the density of states [83, 176, 178] and leads to an enhancement of the Seebeck effect in magnet/superconductor heterostructures [60].

Again, accounting for the diagonal in the spin space structure of the total Green function, the contribution to the self-energy corresponding to the term (iii) assumes the matrix structure $\propto \underline{\tau}_0 \otimes \underline{\sigma}_z$, thereby breaking the symmetries in both Nambu and spin spaces. An explicit evaluation of the quasi-classical Green function matrix shows that this term drops out on integrating over the excitation energy. Thus, this term renormalizes the normal-state properties of the S layer while dropping out in the quasi-classical description of the superconducting state. The analogous term in the self-energy evaluated beyond the Born approximation for magnetic impurities in a bulk superconductor, which does not lead to any spin splitting, was found to break the particle-hole symmetry [147]. Its key manifestation was asymmetric scattering with Yu-Shiba-Rusinov states [217–219] resulting in a large thermoelectric effect [147].

In general, the Hamiltonian, and thus the total matrix Green function, may be non-diagonal in spin space when, for example, the magnetization is spatially inhomogeneous, or the superconductor exhibits unconventional same-spin electron pairing. Under those circumstances, terms (ii) and (iii) may not be interpreted as discussed above.

Here, we have considered a superconducting layer much thinner than the coherence length. For a thick superconductor, the evaluated self-energy may be incorporated in the boundary conditions for the Gor'kov equation in the bulk. Thus, our theory also provides a microscopic derivation of the boundary conditions describing the interface of a superconductor with a magnetic insulator, complementary to the corresponding evaluations within a scattering theory approach [71, 79, 160]. Furthermore, we have considered a single-domain magnet leaving possible generalizations to textured and multi-domain interfaces for future work [220]. Reference [191] reviews exchange bias and magnetic proximity effect together, thereby delineating the connection between the two phenomena further and providing directions for generalizing our results.

From the experimental point of view, it is considered difficult to grow metals on insulators due to lattice mismatch. Such interfaces are inevitably disordered. Nevertheless, a strong interfacial exchange coupling has been observed in a wide range of such structures [167, 213, 214, 221–225]. This is consistent with our result which demonstrates that interfacial disorder does not lead to any qualitative changes in physics and the induced exchange field is resistant to this disorder. It, however, leads to additional spin-flip scattering like contributions which, in some cases [60, 92, 147, 150], may be desirable.

As elaborated in the appendix C, the existing literature on exchange bias [61] and spin-mixing conductance [78, 221, 225, 226] provides valuable guidance regarding materials and corresponding expected spin-splittings. Several AFs, such as CoO, FeF_2, and FeS, may induce fields greater than 100 mT in a 10 nm thick superconducting layer [61]. Furthermore, multilayers incorporating one or more ferromagnetic seed layers are expected to be particularly effective [61], while still circumventing the disadvantages of spin-splitting induced via a ferromagnetic layer.

5.6 Summary

We have derived and solved the Gor'kov equation for two-sublattice magnetic insulator/thin superconductor bilayer structures. Starting with a microscopic description of the interface, we have evaluated the interfacial self-energy for the matrix (Nambu-spin space) Green function in the superconducting layer. Our findings show that an antiferromagnet with an uncompensated surface, in addition to ferrimagnets, induces interfacial disorder-resistant spin-splitting in the adjacent superconductor. Additional contributions mimicking non-magnetic impurities and spin-flip scattering result due to the interfacial disorder. Our findings, in conjunction with related experiments [61, 202, 203, 225], pave the way for employing antiferromagnetic insulators in inducing exchange field in an adjacent superconductor, thereby addressing the feasibility of a wide range of concepts and devices involving spin-split superconductors.

Chapter 6

Summary and perspective

As the key ingredients of this Ph.D. thesis, we have considered three theoretically exciting but still not experimentally explored fundamental transport phenomena related to non-equilibrium and spin properties of nanoscale heterostructures consisting of conventional (s-wave) superconductors in proximity to magnetic materials. This chapter will summarize the main results and perspectives of our detailed investigations presented previously in the chapters 3, 4, and 5, which might be the starting point for further understandings of the questions that are still open within the field of superconducting spintronics.

To give the main motivation, in chapter 1, we have shortly introduced the physics of superconductivity and afterward elaborated on its coexistence with ferromagnetism via the proximity phenomena. A brief overview of the fundamental theoretical concepts, including the derivation of Gor'kov equation for matrix Green functions [employed in chapter 5] and its generalization to out of thermal equilibrium via Keldysh technique, the quasi-classical Green function approximation, and its extension to the dirty limit, namely the Usadel equation of motion [utilized in chapter 4] have been presented in chapter 2. As the next step, we have introduced the Nazarov's circuit theory [applied in chapter 3] and its derivation from the discretized version of the diffusion-like Usadel equation. The quantum circuit theory offers a relatively easy analytical (numerical) description, which is notably helpful in dealing with the interfaces of complex multi-terminal mesoscopic heterostructures.

In chapter 3, we study spin-dependent quasiparticle and Cooper-pair transport in a novel type of an experimentally achievable four-terminal superconductor/ferromagnet system. One highlight is that as long as the loss of superconducting coherences is large, biasing the ferromagnetic leads in combination with non-collinear magnetic moments, can induce triplet superconducting correlations and, therewith, 0–π-transitions can be induced in the current phase relation. In this limit, the relative magnetization angle θ qualifies for voltages above the superconducting gap, as a control knob for charge and spin currents in the ferromagnetic and superconducting contacts. Our key finding is that our proposed setup allows for triggering a finite net charge current between the two magnets by non-collinear magnetization alignment and general spin-mixing conductances. This pumping effect is directly linked to the appearance of the equal-spin triplet Cooper pairs on the central node, and hence, might have a direct impact on generating, controlling, and detecting spin-polarized Cooper pairs by conventional current measurements and be an essential ingredient for technological applications in future superconducting spintronics devices. This work is of interest to a broad community in superconductivity and magnetism. One might even imagine realizing our proposal with heterostructures of two-dimensional crystals, in which superconductivity and magnetism have already been found. The proposed minimal setup might solve the problem to detect spin-triplet Cooper pairs by controlling charge currents and be the key to a non-equilibrium control of spin-polarized current flows in superconductors.

Chapter 4 addresses the effect of spin-flip scattering and Chandrasekhar-Clogston (CC) paramagnetic limit on the thermoelectric response of a tunnel contact between a spin-split superconducting layer and a ferromagnetic metal. We show that increasing the spin-flip scattering rate leads to a substantial enhancement of the thermoelectric performance in the experimentally relevant low-field and low-temperature regime; i.e., (i) the maximum of the thermopower $|S|_{max}$ exceeds k_B/e by a factor of ≈ 5 and (ii) improvement in figure of merit. Since $|S|_{max}$ gets shifted from a higher to a lower field due to the spin-flip rate, a sizeable thermoelectric effect in the ultra-low temperature regime can only be gained via including spin-flip scattering. The reason for this is that the unphysically high values of the exchange fields at which $|S|_{max}$ is achieved in the absence of spin-flip scattering, are larger than the CC limit, and thus lead to a vanishing thermoelectric response by not supporting superconductivity. By establishing a promising prediction towards the reduction of the necessary spin-splitting, which might also be beneficial in avoiding other detrimental effects related to external magnetic fields, magnetization control, or the CC-limited ultra-low-temperature degradation of the thermoelectric response, our proposal has a strong impact on the emerging field of energy control in superconducting nanostructures.

For the first time, to the best of our knowledge, in the chapter 5 we propose and accordingly demonstrate the possibility of employing insulating antiferromagnets in inducing the exchange field in an adjacent superconductor, thereby addressing the feasibility of a wide range of concepts and devices involving spin-split superconductors. Considering a two-sublattice magnetic insulator/superconductor bilayer, we have derived and solved the Gor'kov equation for the matrix (Nambu-spin space) Green function in the superconducting layer. We find that the effect of the magnetic insulator layer is captured by the interfacial self-energy, which includes disorder-mediated terms—identical to spin-independent impurity and spin-flip scattering—in addition to the spin-splitting term. In the hitherto published literature, such effects have usually been achieved via ferromagnets, both in experiment and theory. Antiferromagnets, however, offer several improvements, e.g., the minimization of stray magnetic fields, the possibility of electrical tunability, avoiding negative effects of low-energy magnonic excitations, etc., over the detrimental and parasitic effects arising due to the ferromagnets. Our proposal, in conjunction with related experiments, strongly enhances the practical feasibility of this design and constitutes a significant step towards realizing several concepts and devices involving spin-split superconductors. In addition to its fundamental physics importance, our work has high relevance for the recently invigorated fields of antiferromagnetic spintronics as well as spin-triplet superconductors.

Appendix A

Illustration of the quantum circuit theory

The materials presented here allow for a better understanding of the quasi-classical circuit theory introduced in chapter 2, we apply it in this appendix on two heterostructures and afterward give some notes on its numerical implementation. Introducing the physical observables of interest, e.g., DOS, charge, and spin currents in Sec. A.1, Sec. A.1.1 presents the first example of a special case of the four-terminal system discussed in chapter 3. To make the results analytically attainable, we consider the limiting case of $\epsilon \ll \Delta$, where the decoherence (leakage of coherence) has been disregarded, i.e., $\epsilon_{\mathrm{Th}} \to \infty$ with ϵ_{Th} the Thouless energy. The aim here is to provide an expression for the density of states, which is already described by the equilibrium properties of the system. In the second example, we treat a more straightforward but equally important heterostructure in Sec. A.1.2 formed by two ferromagnets connected to a central node, allowing analytical access to its nonequilibrium properties. It serves as a model system to better understand the role of spin-polarized leads. Before doing so, let us recall how the basic transport properties are encoded in the Green function of the node.

A.1 Central transport quantities

Once we have an expression for the Green function of the node $\check{\mathcal{G}}_c(\varepsilon)$, we can calculate the DOS in the central contact region from the real part of the normal component of the retarded Green function as

$$N(\epsilon) = \frac{1}{4}\,\mathrm{Re}\,\mathrm{tr}\left[(\underline{\tau}_3 \otimes \underline{\sigma}_0)\,\hat{\mathcal{G}}_c^R(\epsilon)\right]. \tag{A.1}$$

Moreover, knowing $\check{\mathcal{G}}_c(\varepsilon)$ we can derive the resulting matrix current $\check{\mathcal{I}}_n$ and thereby one can obtain the currents in the system via its Keldysh component $\hat{\mathcal{I}}_n^K(\epsilon)$ as

$$I_n^{\blacksquare} = \frac{1}{8e^2}\int_{-\infty}^{\infty} d\epsilon\,\mathrm{tr}\left[\hat{\Lambda}^{\blacksquare}\,\hat{\mathcal{I}}_n^K(\epsilon)\right], \tag{A.2}$$

with the currents coefficients

$$\hat{\Lambda}^{\blacksquare} = \begin{cases} e(\underline{\tau}_z \otimes \underline{\sigma}_0), & \blacksquare = q \text{ (charge)} \\ \epsilon(\underline{\tau}_0 \otimes \underline{\sigma}_0), & \blacksquare = \epsilon \text{ (energy)} \\ (\hbar/2)(\underline{\tau}_0 \otimes \underline{\sigma}_\nu), & \blacksquare = s \text{ (spin)} \\ (\epsilon\hbar/2)(\underline{\tau}_z \otimes \underline{\sigma}_\nu), & \blacksquare = \epsilon_s \text{ (spin-energy)} \end{cases} \tag{A.3}$$

given in the spin-Nambu basis as $\Psi^\dagger = \{\Psi_\uparrow^\dagger, \Psi_\downarrow^\dagger, \Psi_\downarrow, -\Psi_\uparrow\}$. The spin current, for instance, becomes

$$I_n^\nu = \frac{\hbar}{16e^2}\int_{-\infty}^{\infty} d\epsilon\,\mathrm{tr}\left[(\underline{\tau}_0 \otimes \underline{\sigma}_\nu)\hat{\mathcal{I}}_n^K(\epsilon)\right]. \tag{A.4}$$

The Pauli matrices in spin (Nambu) space for each direction $\nu = \{x, y, z\}$ are $\underline{\sigma}_\nu(\underline{\tau}_\nu)$, where $\underline{\sigma}_0(\underline{\tau}_0)$ denotes the corresponding identity matrix [1]. However, to make the formalism in chapter 3 easier to follow, we have implemented the symbol 0_n ($\mathbb{1}_n$) which labels the zero (identity) matrix in $n \times n$ dimensions and the superscript $\check{\ldots}$ indicating the Keldysh⊗Nambu⊗spin space of dimension 8×8 and simplified the z-polarized spin current to the one presented in Eq. (3.3).

A.1.1 Application: Four-terminal S/F-heterostructures

In this section, we present the results for a regime in a four-terminal S/F system consisting of two s-wave superconducting and two ferromagnetic terminals with a common contact region, Fig. 3.1, in which we can solve analytically the corresponding generalized Kirchhof equations $0_8 = \sum_n \check{\mathcal{I}}_n$, with $\check{\mathcal{I}}_n \equiv [\check{\mathcal{M}}_n, \check{\mathcal{G}}_c]$ linear in the central Green function $\check{\mathcal{G}}_c$ of the node. From its retarded component, one obtains straightforwardly an expression for the local DOS, see Eq. (A.1).

[1] In the notation we have utilized here, "$\underline{\ldots}$", "$\hat{\ldots}$", and "$\check{\ldots}$" denote the vector of 2×2, 4×4, and 8×8 matrices, respectively, in spin (Nambu), Nambu⊗spin, and Keldysh⊗Nambu⊗spin spaces.

Within the node, the Green function obeys the normalization constraint

$$\check{\mathcal{G}}_c^2 = \begin{pmatrix} \hat{\mathcal{G}}_c^R & \hat{\mathcal{G}}_c^K \\ 0 & \hat{\mathcal{G}}_c^A \end{pmatrix} \begin{pmatrix} \hat{\mathcal{G}}_c^R & \hat{\mathcal{G}}_c^K \\ 0 & \hat{\mathcal{G}}_c^A \end{pmatrix} = \begin{pmatrix} \hat{\mathcal{G}}_c^{R^2} & \hat{\mathcal{G}}_c^R \hat{\mathcal{G}}_c^K + \hat{\mathcal{G}}_c^K \hat{\mathcal{G}}_c^A \\ 0 & \hat{\mathcal{G}}_c^{A^2} \end{pmatrix} \stackrel{!}{=} \begin{pmatrix} \mathbb{1}_4 & 0_4 \\ 0_4 & \mathbb{1}_4 \end{pmatrix} \tag{A.5}$$

resulting in the following equations,

$$\begin{aligned} \hat{\mathcal{G}}_c^R \hat{\mathcal{G}}_c^R &= \hat{\mathcal{G}}_c^A \hat{\mathcal{G}}_c^A = \mathbb{1}_4, \\ \hat{\mathcal{G}}_c^R \hat{\mathcal{G}}_c^K &+ \hat{\mathcal{G}}_c^K \hat{\mathcal{G}}_c^A = 0_4. \end{aligned} \tag{A.6}$$

While the equilibrium physics is incorporated in retarded (advanced) $\hat{\mathcal{G}}^{R(A)}(\epsilon)$, the non-equilibrium properties are contained in the Keldysh component, $\hat{\mathcal{G}}_c^K(\epsilon)$, which satisfies $\hat{\mathcal{G}}_c^K(\epsilon) = \hat{\mathcal{G}}_c^R(\epsilon)\hat{h}(\epsilon) - \hat{h}(\epsilon)\hat{\mathcal{G}}_c^A(\epsilon)$ via Eq. (A.6). Here, $\hat{h}(\epsilon) = f_L(\underline{\tau}_0 \otimes \underline{\sigma}_0) + f_T(\underline{\tau}_z \otimes \underline{\sigma}_0)$ is the non-equilibrium distribution matrix in a terminal biased at voltage V, and $f_{L(T)} = (\tanh[(\epsilon + eV)/2k_B T] + (-)\tanh[(\epsilon - eV)/2k_B T])/2$ refers to longitudinal (transverse) changes of the order parameter associated with the respective non-equilibrium distributions [88].

On the other hand, by expanding

$$\check{\mathcal{M}} = \begin{pmatrix} \hat{\mathcal{M}}^R & \hat{\mathcal{M}}^K \\ 0_4 & \hat{\mathcal{M}}^A \end{pmatrix} \tag{A.7}$$

into its components in Keldysh space and applying the current conservation rule for the net matrix current,

$$[\check{\mathcal{M}}, \check{\mathcal{G}}_c] = \begin{pmatrix} [\hat{\mathcal{M}}^R, \hat{\mathcal{G}}_c^R] & \hat{\mathcal{M}}^R \hat{\mathcal{G}}_c^K + \hat{\mathcal{M}}^K \hat{\mathcal{G}}_c^A - \hat{\mathcal{G}}_c^R \hat{\mathcal{M}}^K - \hat{\mathcal{G}}_c^K \hat{\mathcal{M}}^A \\ 0_4 & [\hat{\mathcal{M}}^A, \hat{\mathcal{G}}_c^A] \end{pmatrix} = 0_8, \tag{A.8}$$

we obtain

$$\begin{aligned} &\left[\hat{\mathcal{M}}^R, \hat{\mathcal{G}}_c^R\right] = 0_4, \\ &\left[\hat{\mathcal{M}}^A, \hat{\mathcal{G}}_c^A\right] = 0_4, \\ &\hat{\mathcal{M}}^R \hat{\mathcal{G}}_c^K + \hat{\mathcal{M}}^K \hat{\mathcal{G}}_c^A - \hat{\mathcal{G}}_c^R \hat{\mathcal{M}}^K - \hat{\mathcal{G}}_c^K \hat{\mathcal{M}}^A = 0_4. \end{aligned} \tag{A.9}$$

Now, implementing the boundary conditions for the superconducting and ferromagnetic matrix currents according to chapter 3, the total $\check{\mathcal{M}}$ becomes

$$\check{\mathcal{M}} = \sum_\alpha \left[\frac{G_{S\alpha}}{2} \check{\mathcal{G}}_{S\alpha} + \frac{G_\alpha}{2} \check{\mathcal{G}}_{F\alpha} + G_\alpha^P \check{\kappa}_\alpha \check{\mathcal{G}}_{F\alpha} - i \frac{G_\alpha^\phi}{2} \check{\kappa}_\alpha \right], \tag{A.10}$$

where we restrict to the case $\epsilon_{\mathrm{Th}} \to \infty$ (a BCS lookalike DOS), in which the leackage term vanishes. Here, $G^P = PG_\alpha/2$ accounts for the spin polarization of both leads with $\alpha = 1, 2$.

The retarded (advanced) and Keldysh components of the matrix Eq. (A.10) read

$$\begin{aligned}\hat{\mathcal{M}}^{R(A)} &= \frac{G_{S1}}{2}\hat{\mathcal{G}}_{S1}^{R(A)} + \frac{G_{S2}}{2}\hat{\mathcal{G}}_{S2}^{R(A)} + \frac{(G_1+G_2)}{2}\hat{\mathcal{G}}_F^{R(A)} \\ &+ \big(\underline{\tau}_z \otimes (\boldsymbol{m}_1 \cdot \underline{\boldsymbol{\sigma}})\big)\Big(G_1^P\hat{\mathcal{G}}_F^{R(A)} - i\frac{G_1^\phi}{2}\Big) + \big(\underline{\tau}_z \otimes (\boldsymbol{m}_2 \cdot \underline{\boldsymbol{\sigma}})\big)\Big(G_2^P\hat{\mathcal{G}}_F^{R(A)} - i\frac{G_2^\phi}{2}\Big), \qquad \text{(A.11)}\end{aligned}$$

and

$$\begin{aligned}\hat{\mathcal{M}}^K &= \frac{G_{S1}}{2}\hat{\mathcal{G}}_{S1}^K + \frac{G_{S2}}{2}\hat{\mathcal{G}}_{S2}^K + \frac{G_1}{2}\hat{\mathcal{G}}_1^K + G_1^P\big(\underline{\tau}_z \otimes (\boldsymbol{m}_1 \cdot \underline{\boldsymbol{\sigma}})\big)\hat{\mathcal{G}}_1^K + \frac{G_2}{2}\hat{\mathcal{G}}_{F2}^K \\ &+ G_2^P\big(\underline{\tau}_z \otimes (\boldsymbol{m}_2 \cdot \underline{\boldsymbol{\sigma}})\big)\hat{\mathcal{G}}_{F2}^K. \qquad \text{(A.12)}\end{aligned}$$

By inserting the S and FM retarded Green function $\big[$Eq. (3.1) and $\hat{\mathcal{G}}_F^{R/A} = \pm(\underline{\tau}_z \otimes \underline{\sigma}_0)\big]$ in Eq. (A.11), employing $(\Delta_\alpha - \Delta_\alpha^*)/2 = i\Delta \sin\varphi_\alpha$, $i(\Delta_\alpha + \Delta_\alpha^*)/2 = i\Delta\cos\varphi_\alpha$, and the property of the tensor product

$$\big(\underline{\tau}_z \otimes (\boldsymbol{m}_\alpha \cdot \underline{\boldsymbol{\sigma}})\big)\big(\underline{\tau}_z \otimes \underline{\sigma}_0\big) = \underline{\tau}_z^2 \otimes (\boldsymbol{m}_\alpha \cdot \underline{\boldsymbol{\sigma}}) = \underline{\tau}_0 \otimes (\boldsymbol{m}_\alpha \cdot \underline{\boldsymbol{\sigma}}),$$

one can simplify $\hat{\mathcal{M}}^R$ to

$$\begin{aligned}\hat{\mathcal{M}}^R &= \frac{\mathrm{sgn}(\epsilon)}{2\sqrt{(\varepsilon+i\delta)^2 - \Delta^2}}\Big[(\epsilon+i\delta)(G_{S1}+G_{S2})(\underline{\tau}_z \otimes \underline{\sigma}_0) \\ &+ i\Delta\big(G_{S1}\sin\varphi_1 + G_{S2}\sin\varphi_2\big)(\underline{\tau}_x \otimes \underline{\sigma}_0) + i\Delta\big(G_{S1}\cos\varphi_1 + G_{S2}\cos\varphi_2\big)(\underline{\tau}_y \otimes \underline{\sigma}_0)\Big] \\ &+ \frac{(G_1+G_2)}{2}(\underline{\tau}_z \otimes \underline{\sigma}_0) + G_1^P\big(\underline{\tau}_0 \otimes (\boldsymbol{m}_1 \cdot \underline{\boldsymbol{\sigma}})\big) - i\frac{G_1^\phi}{2}\big(\underline{\tau}_z \otimes (\boldsymbol{m}_1 \cdot \underline{\boldsymbol{\sigma}})\big) \\ &+ G_2^P\big(\underline{\tau}_0 \otimes (\boldsymbol{m}_2 \cdot \underline{\boldsymbol{\sigma}})\big) - i\frac{G_2^\phi}{2}\big(\underline{\tau}_z \otimes (\boldsymbol{m}_2 \cdot \underline{\boldsymbol{\sigma}})\big). \qquad \text{(A.13)}\end{aligned}$$

One can simplify $\hat{\mathcal{M}}^{A/K}$ in the same way, but at the moment since we are determining the DOS, the retarded part is adequate. In order to further simplify the retarded component, we assume a symmetric configuration (i.e., $G_{S1} = G_{S2} = G_S$, $G_1 = G_2 = G$, $G_1^P = G_2^P = G^P$, $G_1^\phi = G_2^\phi = G^\phi$ and for the magnetization vectors $\boldsymbol{m}_1 = \boldsymbol{m}_2 = \boldsymbol{m}$)

leading to

$$\hat{\mathcal{M}}^R = \frac{\mathrm{sgn}(\epsilon)G_S}{\sqrt{(\varepsilon+i\delta)^2-\Delta^2}}\Bigg[(\epsilon+i\delta)(\underline{\tau}_z\otimes\underline{\sigma}_0) + i\Delta\cos\frac{\varphi_1-\varphi_2}{2}\Big(\sin\frac{\varphi_1+\varphi_2}{2}(\underline{\tau}_x\otimes\underline{\sigma}_0)$$
$$+\cos\frac{\varphi_1+\varphi_2}{2}(\underline{\tau}_y\otimes\underline{\sigma}_0)\Big)\Bigg] + G(\underline{\tau}_z\otimes\underline{\sigma}_0) + 2G^P\big(\underline{\tau}_0\otimes(\boldsymbol{m}\cdot\underline{\boldsymbol{\sigma}})\big) - iG^\phi\big(\underline{\tau}_z\otimes(\boldsymbol{m}\cdot\underline{\boldsymbol{\sigma}})\big). \tag{A.14}$$

Employing the properties of the superconducting phase (i.e., $0=\varphi_1+\varphi_2$ and $\varphi = \varphi_1-\varphi_2$), Eq. (A.14) reduces to

$$\hat{\mathcal{M}}^R = \left(\frac{\mathrm{sgn}(\epsilon)(\epsilon+i\delta)}{\sqrt{(\epsilon+i\delta)^2-\Delta^2}}G_S + G\right)(\underline{\tau}_z\otimes\underline{\sigma}_0) + i\Delta\frac{\mathrm{sgn}(\epsilon)G_S}{\sqrt{(\epsilon+i\delta)^2-\Delta^2}}\cos\frac{\varphi}{2}(\underline{\tau}_y\otimes\underline{\sigma}_0)$$
$$+2G^P\big(\underline{\tau}_0\otimes(\boldsymbol{m}\cdot\underline{\boldsymbol{\sigma}})\big) - iG^\phi\big(\underline{\tau}_z\otimes(\boldsymbol{m}\cdot\underline{\boldsymbol{\sigma}})\big). \tag{A.15}$$

In the limit $\epsilon \ll \Delta$, we get

$$\hat{\mathcal{M}}^R = \left(-i\frac{(\varepsilon+i\delta)}{\Delta}G_S + G\right)(\underline{\tau}_z\otimes\underline{\sigma}_0) + G_S\cos\frac{\varphi}{2}\left(\underline{\tau}_y\otimes\underline{\sigma}_0\right) + 2G^P\big(\underline{\tau}_0\otimes(\boldsymbol{m}\cdot\underline{\boldsymbol{\sigma}})\big)$$
$$- iG^\phi\big(\underline{\tau}_z\otimes(\boldsymbol{m}\cdot\underline{\boldsymbol{\sigma}})\big). \tag{A.16}$$

For the magnetization unit vector $\boldsymbol{m}$, we consider a parallel collinear alignment (i.e., $\theta=0$); $\boldsymbol{m}=(0,0,1)$ is pinned along the spin quantization (z-) axis which results in $(\boldsymbol{m}\cdot\underline{\boldsymbol{\sigma}}) = \underline{\sigma}_z$. One last step would be to use the existing symmetry in the 4×4 tensor products defined above and seprate them to two 2×2 blocks. To this end, we define $\sigma=\pm1$ as a spin label referring to the $\uparrow,\downarrow$ orientation of spin, and Eq. (A.16) may written as

$$\underline{\mathcal{M}}^R = \frac{G_S}{\Delta}\left[\left(-i(\epsilon+i\delta) + \frac{G}{G_S}\Delta - i\sigma\frac{G^\phi}{G_S}\Delta\right)\underline{\tau}_z + \Delta\cos\frac{\varphi}{2}\underline{\tau}_y + 2\sigma\frac{G^P}{G_S^2}\Delta\underline{\tau}_0\right], \tag{A.17}$$

where, the last term will drop out from the commutator relation $\left[\hat{\mathcal{M}}^R, \hat{\mathcal{G}}_c^R\right] = 0_4$.

Furthermore, we can write down $\hat{\mathcal{M}}^R$ and the resulting DOS in terms of an effective spin-splitting $\tilde{h} = (G^\phi/G_S)\Delta$, inhomogeneous broadening $\tilde{\Gamma} = (G/G_S)\Delta$ and phase dependent gap $\tilde{\Delta} = \Delta\cos(\varphi/2)$ via

$$\underline{\mathcal{M}}^R = \frac{G_S}{\Delta}\left[\left(-i\big(\epsilon+i\delta+\sigma\tilde{h}\big)+\tilde{\Gamma}\right)\underline{\tau}_z + \tilde{\Delta}\underline{\tau}_y\right]. \tag{A.18}$$

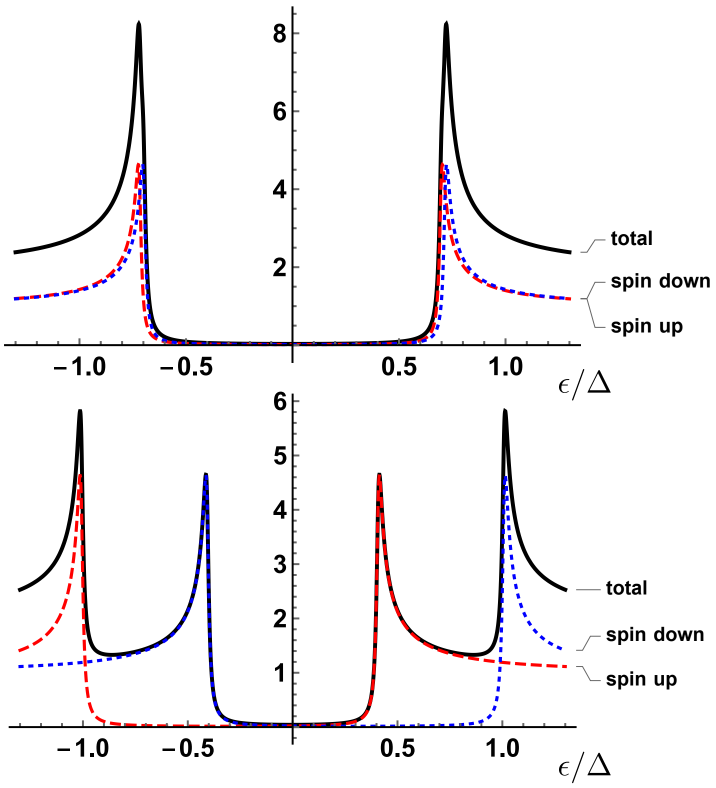

Figure A.1: The total (solid black line), spin up (red dashed line), and spin down (blue dotted line) DOS calculated for $G^{\phi}/G_S = 0.01$ (upper panel) and $G^{\phi}/G_S = 0.3$ (lower panel). We have set a finite phase difference across the junction $\varphi = \pi/2$, $G/G_S = 0.01$, and $\delta = 10^{-3}\Delta$.

The normalization condition for a general matrix of the form $\underline{\mathcal{M}}^R = A\underline{\tau}_x + B\underline{\tau}_y + C\underline{\tau}_z$ will be automatically ensured via $\underline{\mathcal{G}}_c^R = \left(A\underline{\tau}_x + B\underline{\tau}_y + C\underline{\tau}_z\right)/\sqrt{A^2+B^2+C^2}$, which results in the retarded Green function of the node,

$$\underline{\mathcal{G}}_c^R = \frac{\left(-i\left(\epsilon + i\delta + \sigma\tilde{h}\right) + \tilde{\Gamma}\right)\underline{\tau}_z + \tilde{\Delta}\underline{\tau}_y}{\sqrt{-(\epsilon+i\delta)^2 + 2\left(-i\left(\varepsilon + i\delta + \sigma\tilde{h}\right) + \tilde{\Gamma}/2\right)\tilde{\Gamma} - 2\left(\sigma(\epsilon+i\delta) + \tilde{h}/2\right)\tilde{h} + \tilde{\Delta}^2}}. \tag{A.19}$$

In summary, we presented a detailed analysis of the central island's retarded (advanced) Green function. The last step here towards the calculation of the DOS [see Fig. A.1], would be to insert this relation in Eq. (A.1).

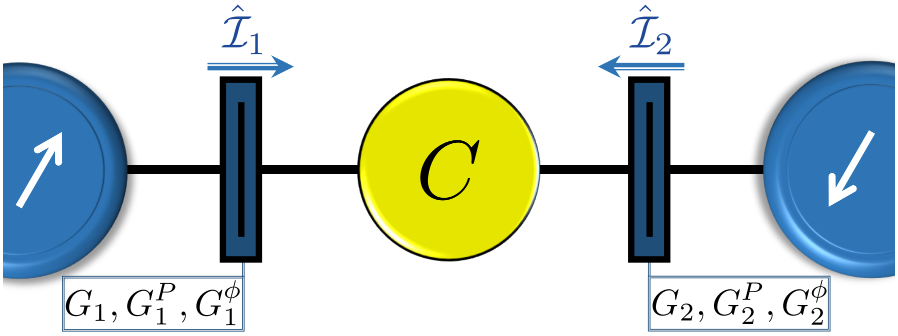

Figure A.2: Circuit diagram of the system under investigation, consisting of two ferromagnetic leads (blue regions) coupled through a normal central node (yellow area). The arbitrary alignments of the magnetizations are shown via the white arrows.

A.1.2 Application: Two-terminal FNF contact

In this section, we want to study the diffusive transport of the two-terminal system depicted in Fig. A.2.

The total $\hat{\mathcal{M}}$ considering the boundary conditions of both ferromagnets reads

$$\hat{\mathcal{M}} = \sum_\alpha \left[\frac{G_\alpha}{2} \hat{\mathcal{G}}_{F\alpha} + G_\alpha^P \hat{\kappa}_\alpha \hat{\mathcal{G}}_{F\alpha} - i \frac{G_\alpha^\phi}{2} \hat{\kappa}_\alpha \right], \tag{A.20}$$

where the index $\alpha = 1, 2$ labels each FM lead with the sub-matrices $\hat{\mathcal{M}}_\alpha = G_\alpha \hat{\mathcal{G}}_{F\alpha}/2 + G_\alpha^P \hat{\kappa}_\alpha \hat{\mathcal{G}}_{F\alpha} - i G_\alpha^\phi \hat{\kappa}_\alpha / 2$. Following recipes of the previous application, Sec. A.1.1, and implementing the relation for the Keldysh component $\underline{\mathcal{G}}^K(\epsilon) = \underline{\mathcal{G}}^R(\epsilon)\underline{h}(\epsilon) - \underline{h}(\epsilon)\underline{\mathcal{G}}^A(\epsilon)$, the 4×4 ferromagnetic Green functions read in Keldysh-spin space [2]

$$\hat{\mathcal{G}}_\alpha = \begin{pmatrix} \underline{\mathcal{G}}^R & \underline{\mathcal{G}}_\alpha^K \\ 0 & \underline{\mathcal{G}}^A \end{pmatrix} = \begin{pmatrix} \underline{\sigma}_0 & 2\underline{\sigma}_0 \underline{h}_\alpha \\ 0_2 & -\underline{\sigma}_0 \end{pmatrix}, \tag{A.21}$$

where, in thermal equilibrium

$$\underline{h}_\alpha(\epsilon) = \begin{pmatrix} \tanh \frac{\epsilon + eV_\alpha}{2T_\alpha} & 0 \\ 0 & \tanh \frac{\epsilon - eV_\alpha}{2T_\alpha} \end{pmatrix} = \begin{pmatrix} 1 - 2f_e & 0 \\ 0 & 2f_h - 1 \end{pmatrix} = \begin{pmatrix} 1 - 2f(\epsilon) & 0 \\ 0 & 2f(-\epsilon) - 1 \end{pmatrix} \tag{A.22}$$

[2]For the current case since there is no S included in this picture, the Nambu space is excluded and thereby the 2×2 retarded (advanced) Green functions of the F in spin space are $\underline{\mathcal{G}}_F^{R/A} = \pm \underline{\sigma}_0$.

in Nambu (particle-hole) space. By implementing the spin matrix and $\hat{\kappa}_\alpha = \mathbb{1}_2 \otimes (\boldsymbol{m}_\alpha \cdot \underline{\boldsymbol{\sigma}})$, the retarded (advanced) component of $\hat{\mathcal{M}}$ becomes

$$\begin{aligned}\underline{\mathcal{M}}^{R(A)} &= \pm\frac{(G_1+G_2)}{2}\underline{\sigma}_0 \pm G_1^P(\boldsymbol{m}_1\cdot\underline{\boldsymbol{\sigma}})\underline{\sigma}_0 - i\frac{G_1^\phi}{2}(\boldsymbol{m}_1\cdot\underline{\boldsymbol{\sigma}}) \pm G_2^P(\boldsymbol{m}_2\cdot\underline{\boldsymbol{\sigma}})\underline{\sigma}_0 \\ &\quad - i\frac{G_2^\phi}{2}(\boldsymbol{m}_2\cdot\underline{\boldsymbol{\sigma}}), \qquad \text{(A.23)}\end{aligned}$$

leading to $\underline{\mathcal{G}}_c^{R(A)}(\epsilon) = \pm\underline{\sigma}_0$ of the node as an outcome of the normalization constraint and the commutation relation introduced in Eq. (A.9). Moreover, the Keldysh component of $\hat{\mathcal{M}}$ may be expressed as

$$\underline{\mathcal{M}}^K = G_1\underline{h}_1 + G_2\underline{h}_2 + 2G_1^P(\boldsymbol{m}_1\cdot\underline{\boldsymbol{\sigma}})\underline{h}_1 + 2G_2^P(\boldsymbol{m}_2\cdot\underline{\boldsymbol{\sigma}})\underline{h}_2, \qquad \text{(A.24)}$$

where, via inserting it in the commutation relation, stated in Eq. (A.9), we get

$$\underline{\mathcal{M}}^R\underline{h}_c - \underline{\mathcal{M}}^K - \underline{h}_c\underline{\mathcal{M}}^A = 0_2. \qquad \text{(A.25)}$$

The main objective is here to calculate the distribution function on the node, $\underline{h}_c$, which will allow us to describe non-equilibrium transport properties. This distribution function has the matrix form of $\underline{h}_c = h_c^0 + (\boldsymbol{h_T}\cdot\underline{\boldsymbol{\sigma}})$, where the subscripts refer to the longitudinal (h_c^0) and transverse (h_T) contribution. The terms $(\boldsymbol{m}_\alpha\cdot\underline{\boldsymbol{\sigma}})$ commute with the diagonal one h_c^0 but not with the term $(\boldsymbol{h_T}\cdot\underline{\boldsymbol{\sigma}})$. Implementing the distribution function in spin space by $\underline{h}_\alpha = \big(1-2f_\alpha(\epsilon)\big)\underline{\sigma}_0$, Eq. (A.25) can be written after some simplifications as

$$\begin{aligned}&(G_1+G_2)\big(h_c^0 + (\boldsymbol{h_T}\cdot\underline{\boldsymbol{\sigma}})\big)\underline{\sigma}_0 - G_1\big(1-2f_1(\epsilon)\big) - G_2\big(1-2f_2(\epsilon)\big) \\ &+ 2\underline{\sigma}_0 h_c^0\big(G_1^P\boldsymbol{m}_1 + G_2^P\boldsymbol{m}_2\big)\cdot\underline{\boldsymbol{\sigma}} + 2\underline{\sigma}_0\big(G_1^P\boldsymbol{m}_1 + G_2^P\boldsymbol{m}_2\big)\cdot\boldsymbol{h_T} \\ &+ \underline{\boldsymbol{\sigma}}\cdot\big(G_1^\phi\boldsymbol{m}_1 + G_2^\phi\boldsymbol{m}_2\big)\times\boldsymbol{h_T} - 2\big(G_1^P\boldsymbol{m}_1 + G_2^P\boldsymbol{m}_2\big)\cdot\underline{\boldsymbol{\sigma}} \\ &+ 4\big(G_1^P\boldsymbol{m}_1 f_1(\epsilon) + G_2^P\boldsymbol{m}_2 f_2(\epsilon)\big)\cdot\underline{\boldsymbol{\sigma}} = 0_2. \qquad \text{(A.26)}\end{aligned}$$

Here, we have employed the property of matrix product, $\boldsymbol{m}_\alpha\underline{\boldsymbol{\sigma}}\cdot\boldsymbol{h_T}\underline{\boldsymbol{\sigma}} = \boldsymbol{m}_\alpha\cdot\boldsymbol{h_T} + i\underline{\boldsymbol{\sigma}}\cdot(\boldsymbol{m}_\alpha\times\boldsymbol{h_T})$. So, after close inspection, we find that we can cast Eq.(A.26) into the compact matrix form. $\boldsymbol{A}\underline{\sigma}_0 + \boldsymbol{B}\cdot\underline{\boldsymbol{\sigma}} = 0_2$ with the matrices

$$\boldsymbol{A} = (G_1+G_2)h_c^0 - G_1\big(1-2f_1(\epsilon)\big) - G_2\big(1-2f_2(\epsilon)\big) + 2\big(G_1^P\boldsymbol{m}_1 + G_2^P\boldsymbol{m}_2\big)\cdot\boldsymbol{h_T}, \qquad \text{(A.27)}$$

and

$$\boldsymbol{B} = (G_1 + G_2)\boldsymbol{h_T} + \big(G_1^\phi \boldsymbol{m}_1 + G_2^\phi \boldsymbol{m}_2\big) \times \boldsymbol{h_T} + 2(h_c^0 - 1)\big(G_1^P \boldsymbol{m}_1 + G_2^P \boldsymbol{m}_2\big)$$
$$+ 4\big(G_1^P \boldsymbol{m}_1 f_1(\epsilon) + G_2^P \boldsymbol{m}_2 f_2(\epsilon)\big). \tag{A.28}$$

Now, adopting the notation in Ref. [227], we can expand $\boldsymbol{h_T}$ into a convenient basis of the vectors $\boldsymbol{m}_1$, $\boldsymbol{m}_2$, and $\boldsymbol{m}_1 \times \boldsymbol{m}_2$ as $\boldsymbol{h_T} = h_u \boldsymbol{u} + h_v \boldsymbol{v} + h_w \boldsymbol{w}$. The orthogonal vectors read

$$\boldsymbol{u} = \frac{\boldsymbol{m}_1 + \boldsymbol{m}_2}{\sqrt{2(1+m)}}, \qquad \boldsymbol{v} = \frac{\boldsymbol{m}_1 - \boldsymbol{m}_2}{\sqrt{2(1-m)}}, \qquad \boldsymbol{w} = \frac{\boldsymbol{m}_1 \times \boldsymbol{m}_2}{\sqrt{1-m^2}}. \tag{A.29}$$

where $m = \boldsymbol{m}_1 \cdot \boldsymbol{m}_2 = \cos\theta$. Next, we express $\boldsymbol{m}_1$ and $\boldsymbol{m}_2$ in terms of the functions $\boldsymbol{u}$ and $\boldsymbol{v}$ [227]

$$\boldsymbol{m}_1 = \sqrt{\frac{1+m}{2}}\boldsymbol{u} + \sqrt{\frac{1-m}{2}}\boldsymbol{v}, \tag{A.30}$$
$$\boldsymbol{m}_2 = \sqrt{\frac{1+m}{2}}\boldsymbol{u} - \sqrt{\frac{1-m}{2}}\boldsymbol{v}, \tag{A.31}$$

in order to simplify $\boldsymbol{A}$ and $\boldsymbol{B}$ given in Eqs. (A.27) and (A.28):

$$\boldsymbol{A} = (G_1 + G_2)\big(h_c^0 - 1\big) + 2\big(G_1 f_1(\epsilon) + G_2 f_2(\epsilon)\big) + \sqrt{2(1+m)}\big(G_1^P + G_2^P\big)h_u$$
$$+ \sqrt{2(1-m)}\big(G_1^P - G_2^P\big)h_v, \tag{A.32}$$

and

$$\boldsymbol{B} = \Bigg[(G_1 + G_2)h_u + \big(h_c^0 - 1\big)\sqrt{2(1+m)}\big(G_1^P + G_2^P\big) + 2\sqrt{2(1+m)}\big(G_1^P f_1(\epsilon) +$$
$$G_2^P f_2(\epsilon)\big) + \big(G_1^\phi - G_2^\phi\big)\sqrt{\frac{1-m}{2}}h_w\Bigg]\boldsymbol{u}$$
$$+ \Bigg[(G_1 + G_2)h_v + \big(h_c^0 - 1\big)\sqrt{2(1-m)}\big(G_1^P - G_2^P\big) + 2\sqrt{2(1-m)}\big(G_1^P f_1(\epsilon) -$$
$$G_2^P f_2(\epsilon)\big) - \big(G_1^\phi + G_2^\phi\big)\sqrt{\frac{1+m}{2}}h_w\Bigg]\boldsymbol{v}$$
$$+ \Bigg[(G_1 + G_2)h_w + \big(G_1^\phi + G_2^\phi\big)\sqrt{\frac{1+m}{2}}h_v - \big(G_1^\phi - G_2^\phi\big)\sqrt{\frac{1-m}{2}}h_u\Bigg]\boldsymbol{w}. \tag{A.33}$$

The matrix equation $\boldsymbol{A}\underline{\sigma}_0 + \boldsymbol{B}\cdot\underline{\boldsymbol{\sigma}} = 0_2$, reduces in the symmetric case—$G_1 = G_2 = G$, $G_1^P = G_2^P = G^P$, and $G_1^\phi = G_2^\phi = G^\phi$—to the linear system of equations

$$G\big(h_c^0 - 1 + f_1(\epsilon) + f_2(\epsilon)\big) + \sqrt{2(1+m)}G^P h_u = 0, \tag{A.34}$$
$$Gh_u + \sqrt{2(1+m)}G^P\big(h_c^0 - 1 + f_1(\epsilon) + f_2(\epsilon)\big) = 0, \tag{A.35}$$
$$Gh_v + \sqrt{2(1-m)}G^P\big(f_1(\epsilon) - f_2(\epsilon)\big) - \sqrt{\frac{1+m}{2}}G^\phi h_w = 0, \tag{A.36}$$
$$Gh_w + \sqrt{\frac{1+m}{2}}G^\phi h_v = 0. \tag{A.37}$$

Solving these equations results in the following expressions for the elements of the distribution function

$$h_c^0 = 1 - f_1(\epsilon) - f_2(\epsilon), \tag{A.38}$$
$$h_u = 0, \tag{A.39}$$
$$h_v = \Big(-\sqrt{2(1-m)}GG^P\big(f_1(\epsilon) - f_2(\epsilon)\big)\Big) \Big/ \Big(G^2 + (\frac{1+m}{2})G^{\phi^2}\Big), \tag{A.40}$$
$$h_w = \Big(\sqrt{1-m^2}G^\phi G^P\big(f_1(\epsilon) - f_2(\epsilon)\big)\Big) \Big/ \Big(G^2 + (\frac{1+m}{2})G^{\phi^2}\Big). \tag{A.41}$$

Now, there is only one step left before we can construct the distribution function $\underline{h}_c$ of the node. The vector of Pauli matrices ($\underline{\boldsymbol{\sigma}}$) need to be defined in the basis of $\boldsymbol{m}_1$, $\boldsymbol{m}_2$, and $\boldsymbol{m}_1 \times \boldsymbol{m}_2$ which is accomplished by $\underline{\boldsymbol{\sigma}} = \underline{\sigma}_u \boldsymbol{u} + \underline{\sigma}_v \boldsymbol{v} + \underline{\sigma}_w \boldsymbol{w}$ with $\underline{\sigma}_u$, $\underline{\sigma}_v$, and $\underline{\sigma}_w$ defined as

$$\underline{\sigma}_u = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \underline{\sigma}_v = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \underline{\sigma}_w = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{A.42}$$

following the usual commutation relations. Therewith, the distribution function of the central island $\underline{h}_c = h_c^0 + (\boldsymbol{h_T}\cdot\underline{\boldsymbol{\sigma}})$ reads

$$\underline{h}_c = \begin{pmatrix} h_c^0 + h_w & -ih_v \\ ih_v & h_c^0 - h_w \end{pmatrix}. \tag{A.43}$$

A.1.3 Total conductance

The ferromagnetic matrix currents $\hat{\mathcal{I}}_\alpha \equiv \Big[\hat{\mathcal{M}}_\alpha, \hat{\mathcal{G}}_c\Big]$ can be now calculated with the aid of the distribution function, Eq. (A.43). To this end, we write the components of total $\hat{\mathcal{M}}$ (Eqs. (A.23) and (A.24)) in the basis $\{\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}\}$ given in Eq. (A.29) (recall

$\underline{\boldsymbol{\sigma}} = \underline{\sigma}_u \boldsymbol{u} + \underline{\sigma}_v \boldsymbol{v} + \underline{\sigma}_w \boldsymbol{w})$,

$$\underline{\mathcal{M}}^{R(A)} = \pm\frac{(G_1+G_2)}{2}\underline{\sigma}_0 \pm \sqrt{\frac{1+m}{2}}\Big(G_1^P + G_2^P\Big)\underline{\sigma}_u \pm \sqrt{\frac{1-m}{2}}\Big(G_1^P - G_2^P\Big)\underline{\sigma}_v$$
$$- \frac{i}{2}\sqrt{\frac{1+m}{2}}\Big(G_1^\phi + G_2^\phi\Big)\underline{\sigma}_u - \frac{i}{2}\sqrt{\frac{1-m}{2}}\Big(G_1^\phi - G_2^\phi\Big)\underline{\sigma}_v, \tag{A.44}$$

and

$$\underline{\mathcal{M}}^{K} = (G_1+G_2)\underline{\sigma}_0 - 2\big(G_1 f_1(\epsilon) + G_2 f_2(\epsilon)\big)\underline{\sigma}_0 + 2\sqrt{\frac{1+m}{2}}\Big(G_1^P + G_2^P\Big)\underline{\sigma}_u$$
$$+ 2\sqrt{\frac{1-m}{2}}\Big(G_1^P - G_2^P\Big)\underline{\sigma}_v - 4\sqrt{\frac{1+m}{2}}\Big(G_1^P f_1(\epsilon) + G_2^P f_2(\epsilon)\Big)\underline{\sigma}_u$$
$$- 4\sqrt{\frac{1-m}{2}}\Big(G_1^P f_1(\epsilon) - G_2^P f_2(\epsilon)\Big)\underline{\sigma}_v. \tag{A.45}$$

In order to calculate the current through the left ferromagnet ($\alpha = 1$) and the corresponding differential conductance, the Keldysh component of the matrix current $\underline{\mathcal{I}}_1^K(\epsilon)$ needs to be determined, see Eq. (A.2), which follows from the Keldysh component $\underline{\mathcal{G}}_1^K = 2\underline{\sigma}_0\underline{h}_1$ of the spectral matrix current $[\hat{\mathcal{M}}_1, \hat{\mathcal{G}}_c]$. Thus, we get

$$\underline{\mathcal{I}}_1^K(\epsilon) = 2\underline{\mathcal{M}}_1^R\underline{h}_c - 2\underline{\mathcal{M}}_1^K - 2\underline{h}_c\underline{\mathcal{M}}_1^A. \tag{A.46}$$

The retarded (advanced) and Keldysh components of $\underline{\mathcal{M}}_1$ for the symmetric case can be readoff from Eqs. (A.44) and (A.45):

$$\underline{\mathcal{M}}_1^{R(A)} = \pm\frac{G}{2}\underline{\sigma}_0 \pm \left(G^P \mp \frac{i}{2}G^\phi\right)\left(\sqrt{\frac{1+m}{2}}\underline{\sigma}_u + \sqrt{\frac{1-m}{2}}\underline{\sigma}_v\right), \tag{A.47}$$

and

$$\underline{\mathcal{M}}_1^K = \left[G\underline{\sigma}_0 + 2G^P\left(\sqrt{\frac{1+m}{2}}\underline{\sigma}_u + \sqrt{\frac{1-m}{2}}\underline{\sigma}_v\right)\right]\big(1 - 2f_1(\epsilon)\big). \tag{A.48}$$

Inserting Eqs. (A.47) and (A.48) into Eq. (A.46) finally yields the Keldysh component of the matrix current for the left ferromagnetic lead,

$$\begin{aligned}\underline{\mathcal{I}}_1^K(\epsilon)=&2\Big(G\big(f_1(\epsilon)-f_2(\epsilon)\big)+2G^P\sqrt{\frac{1-m}{2}}h_v\Big)\underline{\sigma}_0+2\Big(2G^P\sqrt{\frac{1+m}{2}}\big(f_1(\epsilon)-f_2(\epsilon)\big)+\\&G^\phi\sqrt{\frac{1-m}{2}}h_w\Big)\underline{\sigma}_u+2\Big(Gh_v+2G^P\sqrt{\frac{1-m}{2}}\big(f_1(\epsilon)-f_2(\epsilon)\big)-G^\phi\sqrt{\frac{1+m}{2}}h_w\Big)\underline{\sigma}_v\\&+2\Big(Gh_w+G^\phi\sqrt{\frac{1+m}{2}}h_v\Big)\underline{\sigma}_w.\end{aligned} \tag{A.49}$$

Employing h_v from Eq. (A.40), the trace of the matrix current becomes

$$\operatorname{tr}\left[\underline{\mathcal{I}}_1^K(\epsilon)\right]=4G\left(1-\frac{2(1-m)G^{P^2}}{G^2+(\frac{1+m}{2})G^{\phi^2}}\right)\big(f_1(\epsilon)-f_2(\epsilon)\big), \tag{A.50}$$

with $f_{1,2}(\epsilon)$ denoting the Fermi functions corresponding to the left/right FM lead. Note that, for $m=\pm1$—collinear (parallel and antiparallel) alignment of the relative magnetization angle—the G^ϕ contribution vanishes.

We know from Eq. (A.2) that the charge current follows from integration over the quasi-energies,

$$I_1^q\equiv\frac{1}{8e}\int_{-\infty}^{\infty}d\epsilon\operatorname{tr}\left[\underline{\sigma}_0\,\underline{\mathcal{I}}_1^K(\epsilon)\right]=G_T(\theta)\int_{-\infty}^{\infty}d\epsilon\big(f_1(\epsilon)-f_2(\epsilon)\big), \tag{A.51}$$

with θ being the relative magnetization angle with $m=\cos\theta$, and $G_T(\theta)$ representing the normal state total conductance of the tunnel junction,

$$G_T(\theta)=\frac{G}{2}\left(1-\frac{2(1-\cos\theta)\left(G^P\right)^2}{G^2+(\frac{1+\cos\theta}{2})\left(G^\phi\right)^2}\right). \tag{A.52}$$

By noticing, that the integral in Eq. (A.51) just yields the voltage difference between both FMs, the charge current through the left FM lead reduces to $I_1^q=G^T(\theta)V$. In the limiting case of a normal metal ($G^P=0$) it becomes, as expected, $I_1^q=(G/2)V$.

Since, $G^P=(G_\uparrow-G_\downarrow)/2$ and polarization of the junction is $P=(G_\uparrow-G_\downarrow)/(G_\uparrow+G_\downarrow)=(G_\uparrow-G_\downarrow)/G$, the differential conductance $G_T(\theta)$ given in Eq. (A.52) can be finally expressed as

$$G_T(\theta)=\frac{G}{2}\left(1-\frac{P^2}{2}\frac{(1-\cos\theta)G^2}{G^2+(\frac{1+\cos\theta}{2})\left(G^\phi\right)^2}\right)=\frac{G}{2}\left(1-P^2\frac{\tan^2(\theta/2)}{\tan^2(\theta/2)+1+(\frac{G^\phi}{G})^2}\right). \tag{A.53}$$

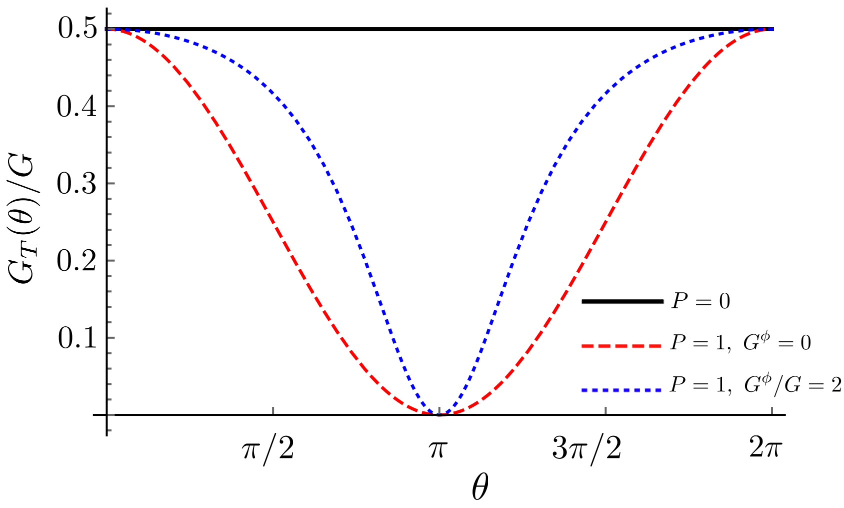

Figure A.3: The total conductance $G_T(\theta)$ vs. the relative magnetization alignment θ. The red dashed and blue dotted line presents the case of the zero and finite spin mixing conductances G^{ϕ}, respectively, for $P = 1$. Increasing G^{ϕ}, sharpens the dip for non-collinear θ and thereby leads to an increase in the total conductance.

where we have plotted it as a function of θ in Fig. A.1.3.

With $1 + (G^{\phi}/G)^2 = |\eta|^2/\,\mathrm{Re}(\eta)$ where η is the complex relative mixing conductance defined in Ref. [227], Eq. (A.53) is in fact, the total conductance of a diffusive FNF system for the symmetric contacts when $l_{sf} \gg L$ (the size of system is much smaller than the spin diffusion length), the magnetic field applied to N is turned off ($\boldsymbol{B} = 0$), and the resistance of N is negligible compared to the ferromagnetic contacts ($G_N \to \infty$), for more details see Ref. [227].

A.2 A note on the numerical implementation

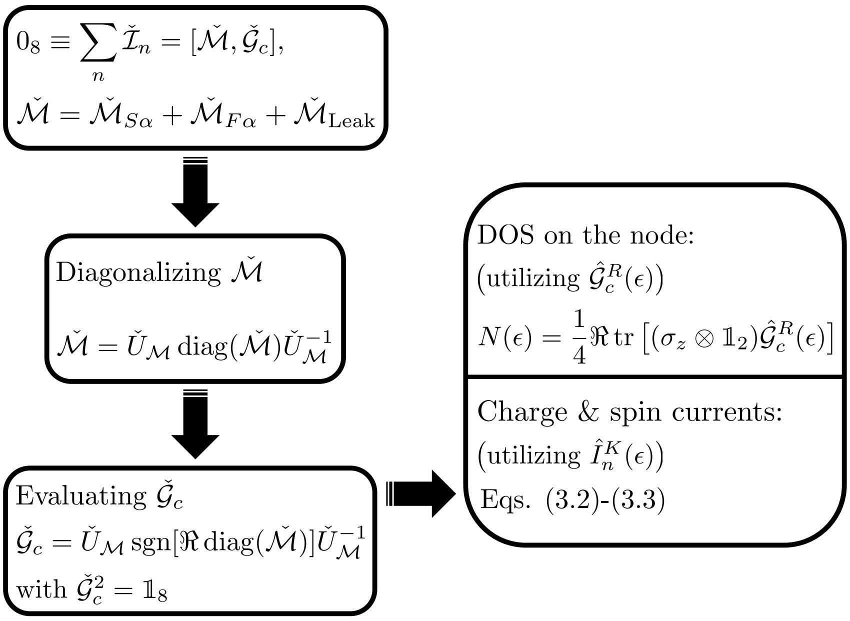

Figure A.4: Scheme to determine transport properties, e.g., electronic DOS in the central region, and the charge (and z-polarized spin) currents between each terminal and the node of the system presented in chapter 3. The main objective here is to calculate the Green function of the node $\check{\mathcal{G}}_c$ which fulfills the normalization constraint $\check{\mathcal{G}}_c^2 = \mathbb{1}_8$ [87]. To this end, we start via employing the discretized boundary conditions in the form of the "matrix currents" refecting the Kirchhoff's current conservation; the net matrix current must become zero at each energy [75]: $0_8 \equiv \sum_n \check{\mathcal{I}}_n$. One can achieve this, adopting the notation of [50], by diagonalizing $\check{\mathcal{M}}$ as $\check{\mathcal{M}} = \check{U}_{\mathcal{M}} \text{diag}(\check{\mathcal{M}}) \check{U}_{\mathcal{M}}^{-1}$. This will allows us to evaluate $\check{\mathcal{G}}_c = \check{U}_{\mathcal{M}} \text{sgn}[\text{Re diag}(\check{\mathcal{M}})] \check{U}_{\mathcal{M}}^{-1}$ with an ensured normalization condition, where $\check{U}_{\mathcal{M}}$ and $\text{diag}(\check{\mathcal{M}})$ denote the eigenvectors and eigenvalues of $\check{\mathcal{M}}$, respectively. Once we find an expression for $\check{\mathcal{G}}_c(\varepsilon)$, DOS and the currents can be calculated implementing the retarded element of $\check{\mathcal{G}}_c$, and the Keldysh component of the matrix currents $\hat{I}_n^K(\varepsilon)$, respectively.

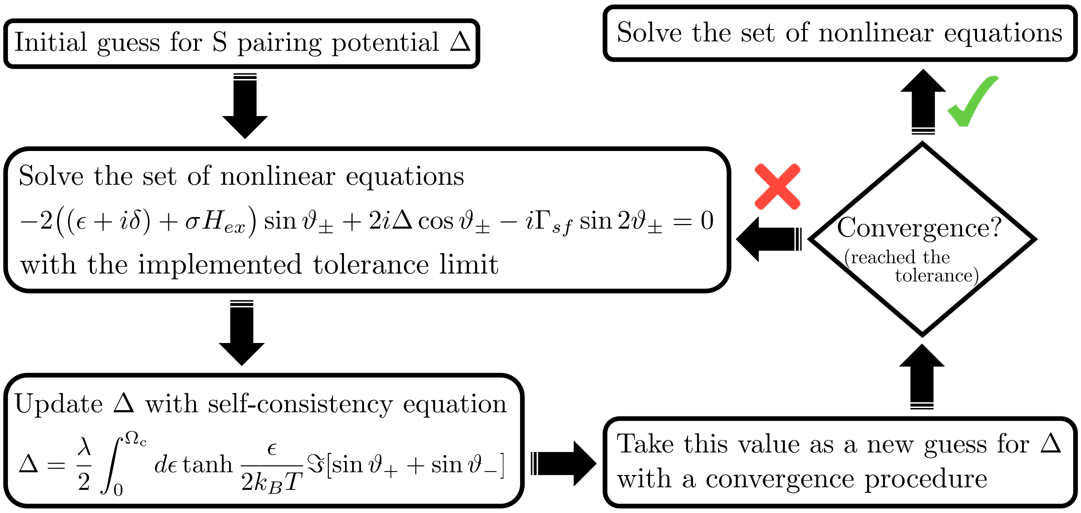

Figure A.5: We are showcasing the self-consistent determination of the S pairing potential employed in chapter 4. There is no exact analytical solution that will give a fully self-consistent pair potential for our system, and a numerical approach must be implemented.

For bulk superconductors, as discussed in this appendix, the DOS and the currents can be also numerically evaluated following the same steps as employed in Secs. A.1 and A.1.2. The basic steps are summarized in Fig. A.4. For thin film superconductors, as discussed in the main text in chapter 4, the gap will be renormalized for instance due to the effect of spin-splitting. Its width has to be found iteratively as outlined in Fig. A.5.

Appendix B

Thermoelectricity: Effect of a sizeable incoherent broadening

The effect of spin-flip scattering on the TE response of a tunnel structure formed by a thin insulating layer sandwiched between a spin-split S and an F has been studied in chapter 4 for a small inhomogeneous broadening. Here, we go a step further and demonstrate that while for a small broadening the achievable maximum Seebeck coefficient is relatively insensitive to the spin-flip rate, in the opposite case of a large broadening, the spin-flip rate consistently has a detrimental effect on it [Fig. B.5]. The results of this appendix are obtained for the case of high incoherent broadening $\delta = 0.1\Delta_0$.

Fig. B.1 presents the self-consistently determined total and spin-polarized DOS, where because of the sizeable spin-flip rate considered in these plots, the system exhibits gapless superconductivity even at zero spin-splitting. As we seen in Fig. 4.3 (a), the value of spin-splitting at which the maximal Seebeck coefficient is achieved varies strongly with the temperature for small incoherent broadening, however, this value is relatively insensitive to the temperature when incoherent broadening is appreciable, see Fig B.2 (a) [ZT mirrors the same, Fig B.3]. This indicates that for large δ, the main contribution to the TE response does not come from the DOS peaks close to the gap energy. Rather, the full quasiparticle spectrum contributes due to the finite DOS in theotherwise gapped region. Based on this, and the DOS dependence, we conclude: (i) the behavior of the system depends qualitatively on the strength of incoherent broadening, and (ii) the TE response of the system is finite even in the gapless regime.

The TE performance of the device, presented in Fig. B.4 is particularly insensitive

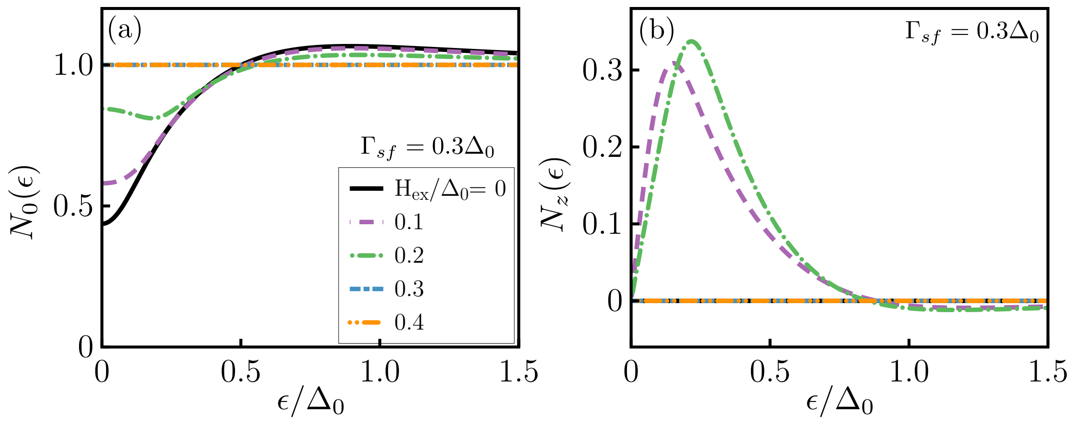

Figure B.1: (a) The total $N_0(\epsilon)$ and (b) spin-polarized $N_z(\epsilon)$ quasiparticle DOS calculated for $\Gamma_{sf} = 0.3\Delta_0$, $k_BT/\Delta_0 = 0.1$. Δ_0 denotes the superconducting pair potential at $T = 0$, $H_{ex} = 0$ and $\Gamma_{sf} = 0$.

to the temperature for large spin-flip and incoherent broadening δ, for which the system is always in the gapless state. The system is less sensitive to the temperature since it is in the gapless state for the larger part of the parameter space, and even when there is a gap, it is not hard.

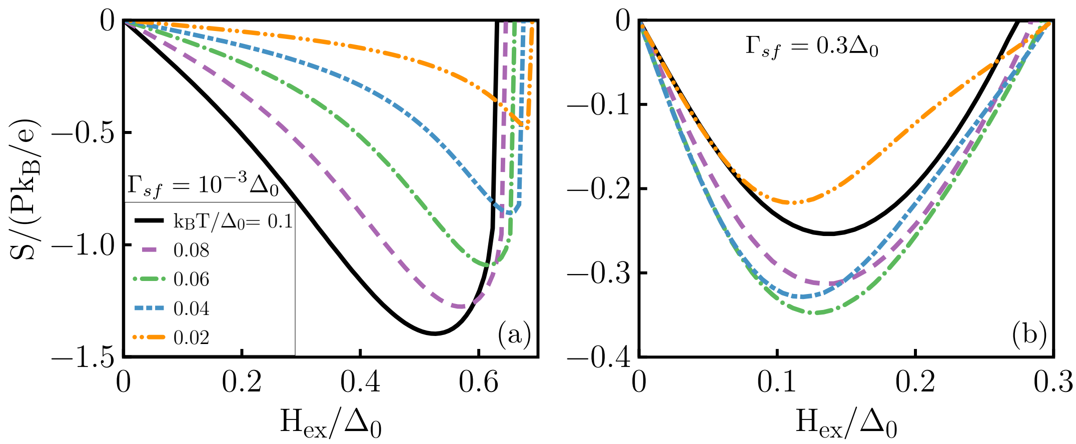

Figure B.2: Seebeck coefficient S vs. spin-splitting field H_{ex}/Δ_0 calculated for (a) and (b) corresponding to $\Gamma_{sf}/\Delta_0 = 10^{-3}$ and 0.3, respectively. The curves indicate different temperatures $k_BT/\Delta_0 = 0.1$ (black solid line), 0.08 (purple dashed line), 0.06 (green dash-dot line), 0.04 (blue long dash-dash) and 0.02 (orange dash-dot-dot).

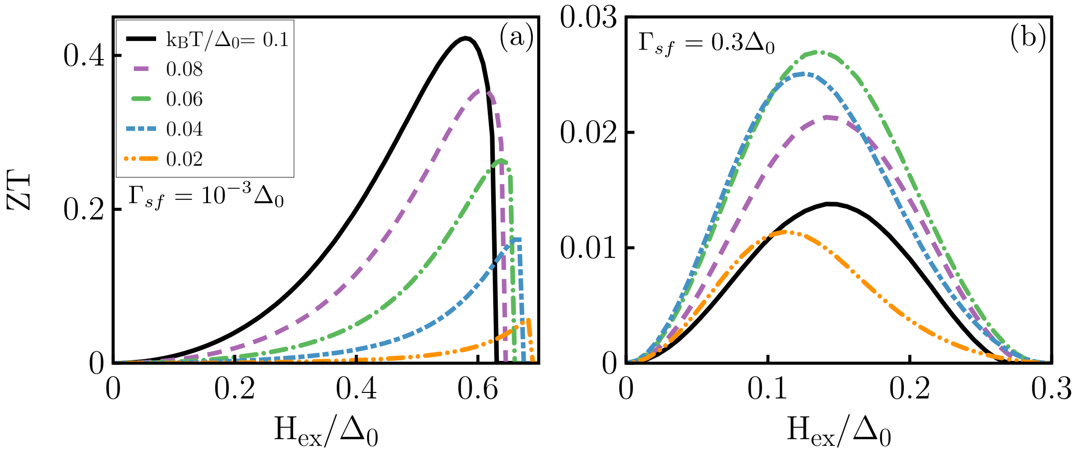

Figure B.3: TE figure of merit ZT as a function of the normalized spin-splitting field H_{ex}/Δ_0. The parameter values employed are $\Gamma_{sf}/\Delta_0 = 10^{-3}$ and 0.3 for (a) and (b), respectively. The polarization of the interface conductance is fixed to $P = 0.9$.

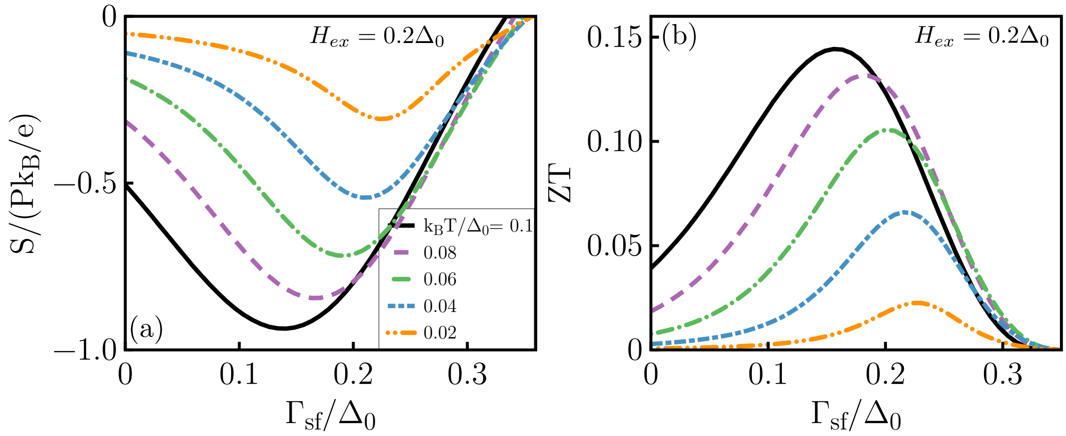

Figure B.4: Spin-flip scattering rate (Γ_{sf}/Δ_0) dependence of (a) normalized thermopower $S/(Pk_B/e)$ and (b) thermoelectric figure of merit ZT for $H_{ex}/\Delta_0 = 0.2$ at different temperatures. Polarization of the interface conductance is $P = 0.9$.

We finally study the dependence of the maximum Seebeck coefficient (as a function of spin-splitting) on the spin-flip rate in Fig. B.5. For small δ [see chapter 4 for more details], the maximum Seebeck coefficient is relatively insensitive to the spin-flip. However, the value of spin-splitting (not shown explicitly) at which this maximum value is achieved decreases monotonically with the spin-flip rate. In the opposite case

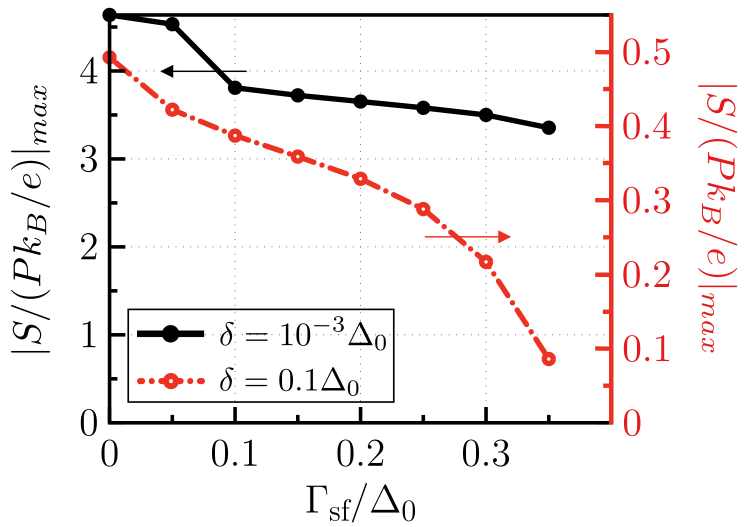

Figure B.5: Maximal thermopower $|S/(Pk_B/e)|_{max}$ vs spin-flip rate Γ_{sf} at $k_BT/\Delta_0 = 0.02$ [the orange curves in Figs. 4.3 and B.2]. For a small incoherent broadening (black curve), the maximum Seebeck coefficient is relatively insensitive to the spin-flip rate while it reduces drastically with spin-flip rate for a larger incoherent broadening (red curve).

of large inhomogeneous broadening, the spin-flip rate consistently has adetrimental effect on the maximum Seebeck coefficient achievable in the system.

Appendix C

Self-energy of proximity-coupled MI/S systems

This appendix includes the conventions followed for expressing the Green functions in the frequency and momentum representation in Sec. C.1, and a detailed derivation of the interfacial self-energy in Sec. C.2, which are significantly essential to follow the results of chapter 5. Moreover, a discussion on some relevant materials, along with the corresponding expected spin-splittings for the MI/S systems is presented in Sec. C.3.

C.1 Frequency-momentum representation

Adapting Mahan's convention and assuming a time-invariant system [215], we express a function $f(x_1, x_2)$ in Matsubara frequency representation as follows:

$$f(x_1, x_2) \equiv f(\tau, \boldsymbol{r}_1, \boldsymbol{r}_2) = \frac{1}{\beta} \sum_n e^{-i\omega_n \tau} F(\omega_n, \boldsymbol{r}_1, \boldsymbol{r}_2), \tag{C.1}$$

$$F(\omega_n, \boldsymbol{r}_1, \boldsymbol{r}_2) = \int_0^\beta e^{i\omega_n \tau} f(\tau, \boldsymbol{r}_1, \boldsymbol{r}_2)\, d\tau, \tag{C.2}$$

where $\beta \equiv 1/k_B T$ with k_B the Boltzmann constant and T the temperature, $x_1 \equiv \{\boldsymbol{r}_1, \tau_1\}$ and so on, $\tau \equiv \tau_1 - \tau_2$, $\omega_n \equiv (2n+1)\pi/\beta$ are the Matsubara frequencies for a Fermionic system. In the following, we drop the explicit distinction between the functions $f()$ and $F()$. The function being referred to is deemed understood based on its arguments. For example, $f(\omega_n, \boldsymbol{r}_1, \boldsymbol{r}_2)$ from this point on represents what we have called $F(\omega_n, \boldsymbol{r}_1, \boldsymbol{r}_2)$ above. Furthermore, the argument ω_n is assumed to be implicit in the following discussion.

A general function $f(\boldsymbol{r}_1,\boldsymbol{r}_2)$ is expressed in the momentum representation:

$$f(\boldsymbol{r}_1,\boldsymbol{r}_2) = \int \frac{d^3k_1}{(2\pi)^3}\frac{d^3k_2}{(2\pi)^3}\; F(\boldsymbol{k}_1,\boldsymbol{k}_2)\; e^{i\boldsymbol{k}_1\cdot\boldsymbol{r}_1}e^{i\boldsymbol{k}_2\cdot\boldsymbol{r}_2}, \tag{C.3}$$

$$f\left(\boldsymbol{r}'+\frac{\boldsymbol{r}}{2},\boldsymbol{r}'-\frac{\boldsymbol{r}}{2}\right) = \int \frac{d^3p}{(2\pi)^3}\frac{d^3k}{(2\pi)^3}\; F\left(\boldsymbol{p}+\frac{\boldsymbol{k}}{2},-\boldsymbol{p}+\frac{\boldsymbol{k}}{2}\right)\; e^{i\boldsymbol{p}\cdot\boldsymbol{r}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}, \tag{C.4}$$

where $\boldsymbol{p} \equiv (\boldsymbol{k}_1-\boldsymbol{k}_2)/2$, $\boldsymbol{k} \equiv \boldsymbol{k}_1+\boldsymbol{k}_2$, $\boldsymbol{r} \equiv \boldsymbol{r}_1-\boldsymbol{r}_2$, and $\boldsymbol{r}' \equiv (\boldsymbol{r}_1+\boldsymbol{r}_2)/2$. With the definitions:

$$f_1\left(\boldsymbol{r},\boldsymbol{r}'\right) \equiv f\left(\boldsymbol{r}'+\frac{\boldsymbol{r}}{2},\boldsymbol{r}'-\frac{\boldsymbol{r}}{2}\right), \tag{C.5}$$

$$F_1\left(\boldsymbol{p},\boldsymbol{k}\right) \equiv F\left(\boldsymbol{p}+\frac{\boldsymbol{k}}{2},-\boldsymbol{p}+\frac{\boldsymbol{k}}{2}\right), \tag{C.6}$$

we can describe the function in the relative and center of mass coordinates representation:

$$f_1\left(\boldsymbol{r},\boldsymbol{r}'\right) = \int \frac{d^3p}{(2\pi)^3}\frac{d^3k}{(2\pi)^3}\; F_1\left(\boldsymbol{p},\boldsymbol{k}\right)\; e^{i\boldsymbol{p}\cdot\boldsymbol{r}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}, \tag{C.7}$$

$$F_1\left(\boldsymbol{p},\boldsymbol{k}\right) = \int d^3\boldsymbol{r}d^3\boldsymbol{r}'\; f_1\left(\boldsymbol{r},\boldsymbol{r}'\right)\; e^{-i\boldsymbol{p}\cdot\boldsymbol{r}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}'}. \tag{C.8}$$

This representation allows us to treat the variations in functions on small (inverse Fermi momentum) and large (superconducting coherence) length scales effectively [67]. In particular, the description of a spatially homogeneous system can be treated as independent of $\boldsymbol{r}'$ and may be developed in terms of $\boldsymbol{p}$ alone, while disregarding $\boldsymbol{k}$. Once again, in the following, and in the main text, we do not explicitly distinguish between the different functions (e.g. $f()$, $f_1()$, $F()$ and so on). We employ the same letters to represent the function appropriate for that particular representation, which, in turn, becomes evident from the arguments specifying the function. For example, $f(\boldsymbol{p},\boldsymbol{k})$ is understood to represent the following expression in terms of the real-space function $f(x_1,x_2) \equiv f(\tau,\boldsymbol{r}_1,\boldsymbol{r}_2)$:

$$f\left(\boldsymbol{p},\boldsymbol{k}\right) = \int_0^\beta d\tau \int d^3\boldsymbol{r}d^3\boldsymbol{r}'\; f\left(\tau,\boldsymbol{r}'+\frac{\boldsymbol{r}}{2},\boldsymbol{r}'-\frac{\boldsymbol{r}}{2}\right)\; e^{i\omega_n\tau}e^{-i\boldsymbol{p}\cdot\boldsymbol{r}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}'}. \tag{C.9}$$

C.2 Perturbative evaluation of Green function

Expressing the problem to be solved as the sum of an unperturbed and a perturbation contributions $\hat{\mathcal{G}}^{-1}(x_1) = \hat{\mathcal{G}}_0^{-1}(x_1) - \hat{\mathcal{H}}_{\rm int}(x_1)$, the total Green's function matrix can be expanded as a sum of contributions to increasing degrees in the perturbation:

$$\hat{G}\left(x_1, x_2\right) = \sum_{n=0}^{\infty} \hat{G}^{(n)}\left(x_1, x_2\right), \tag{C.10}$$

where $\hat{G}^{(0)}\left(x_1, x_2\right)$ is the Green's function matrix for the unperturbed problem. For the case at hand, this corresponds to a superconducting film without the magnet. Substituting above form of the Green's function into the Gor'kov equation and switching to frequency representation, we obtain the following recursive relations [67, 215]:

$$\hat{G}^{(n)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right) = \int d^3 r_3 \; \hat{G}^{(0)}\left(\boldsymbol{r}_1, \boldsymbol{r}_3\right) \hat{\mathcal{H}}_{\text{int}}(\boldsymbol{r}_3) \hat{G}^{(n-1)}\left(\boldsymbol{r}_3, \boldsymbol{r}_2\right), \tag{C.11}$$

with $n \geq 1$. Since we work within the quasi-classical approximation for superconductivity assuming the unperturbed solution to represent a homogeneous superconducting state, the corresponding Green function matrix can be represented as:

$$\hat{G}^{(0)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right) = \int \frac{d^3 p}{(2\pi)^3} \hat{G}^{(0)}\left(\boldsymbol{p}\right) e^{i\boldsymbol{p}\cdot(\boldsymbol{r}_1 - \boldsymbol{r}_2)}. \tag{C.12}$$

This representation will be used repeatedly in the following analysis.

The first order correction can be simplified to:

$$\hat{G}^{(1)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right) = \int \frac{d^3 p_1}{(2\pi)^3} \frac{d^3 p_2}{(2\pi)^3} \; e^{i(\boldsymbol{p}_1\cdot\boldsymbol{r}_1 - \boldsymbol{p}_2\cdot\boldsymbol{r}_2)} \; \hat{G}^{(0)}\left(\boldsymbol{p}_1\right) \hat{\mathcal{H}}_{\text{int}}(\boldsymbol{p}_1 - \boldsymbol{p}_2) \hat{G}^{(0)}\left(\boldsymbol{p}_2\right), \tag{C.13}$$

where

$$\hat{\mathcal{H}}_{\text{int}}(\boldsymbol{p}) \equiv \int d^3 r \; \hat{\mathcal{H}}_{\text{int}}(\boldsymbol{r}) \; e^{-i\boldsymbol{p}\cdot\boldsymbol{r}} = \left[(2\pi)^2 \delta(\boldsymbol{\kappa}) + u(\boldsymbol{\kappa}) \sum_{\boldsymbol{s}_i} e^{-i\boldsymbol{\kappa}\cdot\boldsymbol{s}_i}\right] \; \hat{t}, \tag{C.14}$$

with $\boldsymbol{p} \equiv \{\boldsymbol{\kappa}, \xi\}$. In evaluating the above expression, we have employed the conventions and definitions introduced in the main text. Employing Eq. (C.14) in Eq. (C.13). the first order correction reduces to:

$$\hat{G}^{(1)}(\boldsymbol{r}_1,\boldsymbol{r}_2) = \int \frac{d^3p_1}{(2\pi)^3}\frac{d^3p_2}{(2\pi)^3}\; e^{i(\boldsymbol{p}_1\cdot\boldsymbol{r}_1-\boldsymbol{p}_2\cdot\boldsymbol{r}_2)}\;\hat{G}^{(0)}(\boldsymbol{p}_1)\,(2\pi)^2\delta(\boldsymbol{\kappa}_1-\boldsymbol{\kappa}_2)\;\hat{t}\;\hat{G}^{(0)}(\boldsymbol{p}_2)$$
$$+\int \frac{d^3p_1}{(2\pi)^3}\frac{d^3p_2}{(2\pi)^3}\; e^{i(\boldsymbol{p}_1\cdot\boldsymbol{r}_1-\boldsymbol{p}_2\cdot\boldsymbol{r}_2)}\;\sum_{\boldsymbol{s}_i} e^{-i(\boldsymbol{\kappa}_1-\boldsymbol{\kappa}_2)\cdot\boldsymbol{s}_i}\;\hat{G}^{(0)}(\boldsymbol{p}_1)\,u(\boldsymbol{\kappa}_1-\boldsymbol{\kappa}_2)\;\hat{t}\;\hat{G}^{(0)}(\boldsymbol{p}_2)\,, \tag{C.15}$$

$$=\int \frac{d^3p_1}{(2\pi)^3}\frac{d\xi_2}{2\pi}\; e^{i(\xi_1y_1-\xi_2y_2)}\; e^{i\boldsymbol{\kappa}_1\cdot(\boldsymbol{s}_1-\boldsymbol{s}_2)}\;\hat{G}^{(0)}(\boldsymbol{p}_1)\;\hat{t}\;\hat{G}^{(0)}(\boldsymbol{\kappa}_1,\xi_2)\quad +$$
$$N_{\rm dis}\int \frac{d^3p_1}{(2\pi)^3}\frac{d\xi_2}{2\pi}\; e^{i(\xi_1y_1-\xi_2y_2)}\; e^{i\boldsymbol{\kappa}_1\cdot(\boldsymbol{s}_1-\boldsymbol{s}_2)}\;u(0)\;\hat{G}^{(0)}(\boldsymbol{p}_1)\;\hat{t}\;\hat{G}^{(0)}(\boldsymbol{\kappa}_1,\xi_2)\,, \tag{C.16}$$

where we have averaged over the disorder center locations via the replacement $\sum_{\boldsymbol{s}_i}\to N_{\rm dis}\int d^2s_i$, and $u(0)=\int d^2s\; u(s)=0$ results in the second term vanishing:

$$\hat{G}^{(1)}(\boldsymbol{r}_1,\boldsymbol{r}_2) = \int \frac{d^3p_1}{(2\pi)^3}\frac{d\xi_2}{2\pi}\; e^{i(\xi_1y_1-\xi_2y_2)}\; e^{i\boldsymbol{\kappa}_1\cdot(\boldsymbol{s}_1-\boldsymbol{s}_2)}\;\hat{G}^{(0)}(\boldsymbol{p}_1)\;\hat{t}\;\hat{G}^{(0)}(\boldsymbol{\kappa}_1,\xi_2)\,. \tag{C.17}$$

The expression obtained above describes an inhomogeneous system due to the breaking of translational invariance by the interface. However, within the quasi-classical approximation, we expect a homogeneous superconducting state. Thus the expression above goes beyond the quasi-classical limit. We obtain the contribution relevant for describing superconductivity, stemming from a narrow phase-space around the Fermi energy, by averaging over the thickness d_S of the superconductor:

$$\left\langle \hat{G}^{(1)}(\boldsymbol{r}_1,\boldsymbol{r}_2)\right\rangle = \frac{1}{d_S}\int dy'\;\hat{G}^{(1)}(\boldsymbol{r}_1,\boldsymbol{r}_2)\,, \tag{C.18}$$
$$=\frac{1}{d_S}\int \frac{d^3p_1}{(2\pi)^3}\frac{d\xi_2}{2\pi}dy'\; e^{iy'(\xi_1-\xi_2)}\; e^{iy\frac{(\xi_1+\xi_2)}{2}}\; e^{i\boldsymbol{\kappa}_1\cdot(\boldsymbol{s}_1-\boldsymbol{s}_2)}\;\hat{G}^{(0)}(\boldsymbol{p}_1)\;\hat{t}\;\hat{G}^{(0)}(\boldsymbol{\kappa}_1,\xi_2)\,, \tag{C.19}$$

where $y'=(y_1+y_2)/2$ and $y=y_1-y_2$. This leads us to our result for the first order correction:

$$\left\langle \hat{G}^{(1)}(\boldsymbol{r}_1,\boldsymbol{r}_2)\right\rangle = \int \frac{d^3p}{(2\pi)^3}\; e^{i\boldsymbol{p}\cdot(\boldsymbol{r}_1-\boldsymbol{r}_2)}\;\hat{G}^{(0)}(\boldsymbol{p})\;\frac{\hat{t}}{d_S}\;\hat{G}^{(0)}(\boldsymbol{p})\,. \tag{C.20}$$

The evaluation of the second order correction follows an analysis similar to the above. We wish to evaluate $\left\langle \hat{G}^{(2)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right)\right\rangle$ with

$$\hat{G}^{(2)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right) = \int d^3r_3 \; \hat{G}^{(0)}\left(\boldsymbol{r}_1, \boldsymbol{r}_3\right) \hat{\mathcal{H}}_{\text{int}}(\boldsymbol{r}_3) \hat{G}^{(1)}\left(\boldsymbol{r}_3, \boldsymbol{r}_2\right). \tag{C.21}$$

In order to obtain the desired result within our approximation, we make the following replacement:

$$\begin{aligned}&\hat{G}^{(1)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right) \to \left\langle \hat{G}^{(1)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right)\right\rangle + \\ &\int \frac{d^3p_1}{(2\pi)^3} \frac{d^3p_2}{(2\pi)^3} \; e^{i(\boldsymbol{p}_1 \cdot \boldsymbol{r}_1 - \boldsymbol{p}_2 \cdot \boldsymbol{r}_2)} \sum_{\boldsymbol{s}_i} e^{-i(\boldsymbol{\kappa}_1 - \boldsymbol{\kappa}_2)\cdot \boldsymbol{s}_i} \; \hat{G}^{(0)}\left(\boldsymbol{p}_1\right) \; u(\boldsymbol{\kappa}_1 - \boldsymbol{\kappa}_2) \; \hat{t} \; \hat{G}^{(0)}\left(\boldsymbol{p}_2\right).\end{aligned}$$

The disorder has to be treated separately from the $\left\langle \hat{G}^{(1)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right)\right\rangle$ term since the pre-averaging procedure fails to capture the disorder-mediated scrambling of momenta. Employing the above replacement, Eq. (C.21) can be simplified into several contributions. All contributions stemming from scattering by single and multiple but distinct disorder-centers vanish on account of $u(0) = 0$. The term due to two scattering events from the same disorder-centers leads to a finite result. Combined with the other assumptions of the cross-diagram technique [67, 215], mentioned in the main text, the required second order correction becomes:

$$\left\langle \hat{G}^{(2)}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right)\right\rangle = \int \frac{d^3p}{(2\pi)^3} \; e^{i\boldsymbol{p}\cdot(\boldsymbol{r}_1 - \boldsymbol{r}_2)} \; \hat{G}^{(0)}\left(\boldsymbol{p}\right) \; \hat{\Sigma}^{(1)}_{\text{int}}\left(\boldsymbol{p}\right) \; \hat{G}^{(0)}\left(\boldsymbol{p}\right), \tag{C.22}$$

where

$$\hat{\Sigma}^{(1)}_{\text{int}}\left(\boldsymbol{p}\right) = \frac{N_{\text{dis}}}{d_S} \int \frac{d^3p_1}{(2\pi)^3} \; |u(\boldsymbol{\kappa} - \boldsymbol{\kappa}_1)|^2 \; \hat{t} \; \hat{G}^{(0)}\left(\boldsymbol{p}_1\right) \; \hat{t}. \tag{C.23}$$

Proceeding along similar lines employing the approximations introduced above and evaluating the higher order terms, we can sum all terms in a manner analogous to the treatment of bulk impurity scattering within the cross diagram technique [67, 215]. The final result is obtained as

$$\left\langle \hat{G}\left(\boldsymbol{r}_1, \boldsymbol{r}_2\right)\right\rangle = \int \frac{d^3p}{(2\pi)^3} \; e^{i\boldsymbol{p}\cdot(\boldsymbol{r}_1 - \boldsymbol{r}_2)} \; \hat{G}\left(\boldsymbol{p}\right), \tag{C.24}$$

with the expression for $\hat{G}\left(\boldsymbol{p}\right) \equiv \hat{G}\left(\omega_n, \boldsymbol{p}\right)$ as given in the main text.

Material	**Temperature (K)**	**Texture**	**Effective field (T)**	**Comments**
CoO	150	poly	10^{-1}	–
CoO	100	poly-multi	10^{0}	Ferromagnet in mutlilayer
CoO	77	(1 1 1)	10^{-1}	–
FeS	10	poly	10^{-1}	–
FeF_2	10	(1 1 0)	10^{0}	–
FeF_2	10	(0 0 1)	10^{-3}	–
CrN	10	poly	10^{-1}	–
NiO	RT	poly	$10^{-3} - 10^{-2}$	–
NiO	RT	(1 1 1)	$10^{-3} - 10^{-2}$	Enhancement at low temperatures
NiO	RT	(1 0 0)	$10^{-2} - 10^{-1}$	–
Cr_2O_3	RT	poly	10^{-3}	Comparison with recent studies

Table C.1: Materials and expected induced effective magnetic fields evaluated via the interfacial energy density data compiled in Ref. [61].

C.3 Realistic materials and expected effective fields

In the present section, we discuss some concrete materials along with the spin splittings, or equivalently effective magnetic fields, expected to be induced by them in an adjacent conductor. We first present these estimates on the basis of the experimental data available from exchange bias studies [61]. The compiled data demonstrates the variation in the induced fields with different materials, textures, and fabrication techniques [see table C.1]. In particular, incorporation of a ferromagnetic seed layer can be very effective in achieving a large induced field without necessitating epitaxial growth. Second, we estimate the expected spin splitting from the spin-mixing conductance measured via various experimental studies in ferromagnet/metal bilayers [221, 225, 226]. One can expect a perfectly uncompensated interface to induce a comparable field. The resulting estimate is consistent with the highest values expected from the exchange bias data thus suggesting that several systems exhibit nearly perfect uncompensation at the interface. Detailed spin-mixing conductance studies for antiferromagnets, analogous to exchange bias data, are not available at this point.

C.3.1 Estimates from exchange bias experiments

Enlisting a few examples from Ref. [61] here, we refer the readers to this review article for a more extensive analysis and further references. The effective fields have been estimated employing the reported interfacial energy densities [61] assuming a nominal magnetization of 2×10^5 A/m and a 10 nm thick superconductor. The 'texture' corresponds to the magnet being polycrystalline ('poly') or one of its specific crystal planes exposed at its interface with the superconductor. 'RT' stands for room temperature. A further discussion of the 'comments' is also presented below. Some of the discussed materials are not antiferromagnetic at room temperatures, which does not hinder their use with superconductors at low temperatures.

Discussion of comments:

- *Ferromagnet in multilayer.* – Incorporating one or more ferromagnetic layers into a magnetic multilayer terminated with an antiferromagnetic layer allows for a strong sublattice-asymmetry at the exposed surface. Such an arrangement does not depend on having an epitaxial growth. Furthermore, the presence of a ferromagnetic (seed) layer far away from the exposed antiferromagnet surface, at which the superconductor is deposited, does not influence the conductor. The underlying physics is explained by the field cooling effect [61]. The heterostructure is heated above the Neel temperature of the antiferromagnetic layer. It is then allowed to cool gradually in the presence of an applied magnetic field, which keeps the ferromagnet fully aligned. As the antiferromagnet begins to order below its Neel temperature, the atomic layer next to the ferromagnet is fully ordered due to interfacial exchange interaction with the ordered ferromagnet. The subsequent atomic layers in the antiferromagnet follow appropriate ordering, consistent with the antiferromagnetic interaction with the previous atomic layer, resulting in an atomically layered configuration throughout, which gives a strong sublattice symmetry-breaking at the other end of the antiferromagnet, where the conductor is deposited.

- *Enhancement at low temperatures.* – Reference [61] also documents an isolated observation of an effective field larger by three orders of magnitude for the same system at 10 K.

- *Comparison with recent studies.* – The estimated field here is consistent with the recent experimental observation of exchange bias via epitaxially grown Cr_2O_3 with a reasonably rough surface [202]. Their demonstration of electric switching allows for yet another functionality, and is directly applicable to our proposal as well.

C.3.2 Estimates from spin-mixing conductance experiments

Kajiwara and coworkers [226] have estimated an exchange coupling energy of 10 meV per 'bond' employing the spin transfer studies across yttrium iron garnet (YIG)-platinum interface. This corresponds to a spin-mixing conductance $\sim 10^{19}$ m^{-2}. Considering the lattice constant of around 1 nm for YIG, we estimate the interfacial energy density of $\sim 10^{-3}$ J/m^2. Following an approach analogous to the exchange bias evaluation above, we estimate an effective field of $1-10$ T in the adjacent conductor with thickness 10 nm. There is a slight ambiguity in this estimation stemming from a difference in the lattice constants of YIG and a typical conductor. However, since we only estimate the order of magnitude, this ambiguity is of little consequence. Since similar spin-mixing conductances have been measured across a range of magnet/metal bilayers [225], we expect the induced fields by the corresponding uncompensated interfaces to be of similar magnitude.

Appendix D

Scientific contribution

This appendix presents a short review of the author's main scientific contributions during his time as a Ph.D. student in the Quantum Transport Group at the University of Konstanz.

Poster presentations

- Ph.D. SpinCaT Workshop: New Perspectives for Spin-caloritronics, JGU - SPICE, Mainz, Germany, 18-19 August 2016
- The International Conference and School on Superconducting Hybrid Nanostructures: Physics and Application, Moscow Institute of Physics and Technology (MIPT), Dolgoprudnyy, Russia, 19-26 September 2016
- 628. Wilhelm and Else Heraeus Seminar: Trends in Mesoscopic Superconductivity, Physikzentrum in Bad Honnef, Germany, 14-18 November 2016
- DPG-Frühjahrstagung 2017, Spring Meeting of the Condensed Matter Section, Dresden, Germany, 19-24 March 2017
- Spincaloritronics 8 Funded by the DFG via SPP 1538 SpinCat, Regensburg, Germany, 12-15 June 2017
- LT28, 28th International Conference on Low-temperature Physics, Gothenburg, Sweden, 9-16 August 2017
- DPG-Frühjahrstagung 2019, Spring Meeting of the Condensed Matter Section, Regensburg, Germany, 1-5 April 2019

- 697. Wilhelm and Else Heraeus Seminar: Superconductivity in Low-Dimensional and Interacting Systems, Physikzentrum in Bad Honnef, Germany, 2-6 June 2019

Talks

- DPG-Frühjahrstagung 2018, Spring Meeting of the Condensed Matter Section, Berlin, Germany, 11-16 March 2018
- DPG-Frühjahrstagung 2019, Spring Meeting of the Condensed Matter Section, Regensburg, Germany, 1-5 April 2019

Publications and preprints

The work presented in this thesis were published or submitted for publication in the following peer-reviewed journals, which are listed below in chronological order:

I. **A. Rezaei**, A. Kamra, P. Machon, and W. Belzig, "Spin-flip Enhanced Thermoelectricity in Superconductor-Ferromagnet Bilayers," New Journal of Physics **20**, 073034 (2018).

II. Akashdeep Kamra, **Ali Rezaei**, and Wolfgang Belzig, "Spin Splitting Induced in a Superconductor by an Antiferromagnetic Insulator," Phys. Rev. Lett. **121**, 247702 (2018).

III. **Ali Rezaei**, Robert Hussein, Akashdeep Kamra, and Wolfgang Belzig, "Phase-controlled Spin and Charge Currents in Superconductor-Ferromagnet Hybrids," ArXiv e-prints (2019), arXiv:1908.09610v1 [cond-mat.supr-con].

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